

Chapter 6 :

Transistor Equivalent Circuits and AC Analysis

6.1 Introduction

We have already discussed the construction, device operation, and characteristics of transistor in Chapter 3. We have also discussed different DC biasing techniques of transistor in CB, CE and CC configuration. Before going to transistor amplifier, we will describe how a transistor can be represented by an equivalent circuit. The equivalent circuit of transistor is called its *model* and the process of representing any device by its equivalent circuit is called *modeling*. The equivalent circuit of a transistor is determined for two types of operation. One is for small signal operation, and the other is for large signal operation. Although there is no strict dividing line between the two, we can consider a transistor is working in small signal mode if the AC swing use only 10% of the maximum possible swing span. On the other hand, a large signal amplifier uses almost 100% swing span. There are many types of small signal transistor models. Among them, in this chapter we will describe three most common models of transistor. These are r_e model, hybrid equivalent model and hybrid π model. r_e model and hybrid equivalent models are used for low frequency analysis and hybrid π model is used for high frequency analysis.

6.2 Small Signal Low Frequency Models of Transistor

In Chapter 2, we have seen how a PN junction diode can approximately be represented by its equivalent circuit. The DC equivalent circuit was consisted of an ideal diode, a barrier voltage and a resistance. If we can represent a device by its equivalent circuit, that is, using electrical components and voltage and/or current source, it becomes easier to understand the operation of the circuit and so does the analysis. Similarly, for better understanding and analysis of a transistor circuit we need to represent a transistor by its equivalent circuit. This equivalent circuit of the transistor is called its *model*. In every model, the transistor is represented by some parameters, e.g. impedance (Z) parameters, admittance (Y) parameters and their combination hybrid parameters. As already mentioned, for small signal operation of a transistor there are many types of models, like-impedance model or Z-parameters, admittance model or Y-parameters, hybrid model or h-parameters, r_e modeling, hybrid π modeling etc. However, among them r_e model, hybrid model and hybrid π model are most commonly used. r_e model and hybrid models are used to analyze the transistor circuit when the operating frequencies are low, i.e., the junction and inter-electrode capacitances are negligible. On the other hand, hybrid π model is used for high frequency analysis where the junction and inter-electrode capacitances are not negligible.

Since long, electronics industries and educational institutions are using the hybrid-parameters because they produce more accurate results in the analysis of amplifier circuits. In fact, hybrid-parameter equivalent circuit is popular even today. But their uses are being limited by the following difficulties:

1. The values of h-parameters are not so easily available. Most of the datasheets do not mention all of them.
2. Their values vary considerably from transistor to transistor even of the same type number.
3. Their values give good results for a particular operating condition of the transistor.

On the other hand, r_e model is gaining popularity nowadays for the following advantages.

1. The model is very simple, i.e., it consists of a few parameters.
2. The values of the parameters are easily available.
3. The procedure to find the parameters is simple and easy to understand.
4. The r_e model produces reasonably accurate results.

As r_e model is easy to understand, we will first describe this model and then the others.

6.3 The r_e Model of BJT

Already we have known that r_e model is a small signal model for bipolar junction transistor (BJT). The advantages of this model are also discussed. There are three types of configuration of BJT amplifier circuits. These are common-base (CB), common-emitter (CE), and common-collector (CC). Here, we will discuss r_e model of each of them.

6.3.1 r_e Model for CE Circuit

An NPN transistor circuit in common-emitter configuration is shown in Fig.6-1(a). The transistor is biased in the active region. The directions of the transistor terminal currents are as before. We may consider this circuit as a four-terminal network, keeping the emitter common to both the input and output port. Therefore, B and E will be in the input port, and the C and E will be in the output port. Generally, in a four terminal network the directions of currents (i_i , and i_o) and the polarities of voltages (v_i , and v_o) are always as shown in the figure. The direction of the output current is towards the port. If actual polarity and directions do not match with the assumed directions, they are represented as negative values. Here, the parameters (e.g. i_e , i_b) are written using lower-case letters to represent the AC values.

As we have seen in Chapter 4, in a CE circuit the input current (i_b) flows through a forward biased B - E junction, and due to the transistor action almost all of the input current is transferred to the collector terminal. The value of the collector current is βi_b . We can consider the input side as a forward biased diode and the output side as a dependent current source whose value is $i_c = \beta i_b$. Thus, the transistor can be represented by a diode and a current source as in Fig.6-1(b). In a forward biased diode after the knee point, its behavior is like a resistor. So, the diode can be replaced by a resistor, which results in the circuit of Fig.6-1(c). Generally, these equivalent circuits are used for AC analysis. So, r_e is the AC resistance (or dynamic resistance) of the forward biased B - E junction as determined in Chapter 1. It can be calculated as

$$r_e = \frac{26 \text{ mV}}{I_E} + r_B \approx \frac{26 \text{ mV}}{I_E} \quad (6-1)$$

Here, r_B is the base spreading resistance. As the value of r_B is very small, it can be neglected. In calculating the value of r_e , the quiescent emitter current (I_{EQ}) is used.

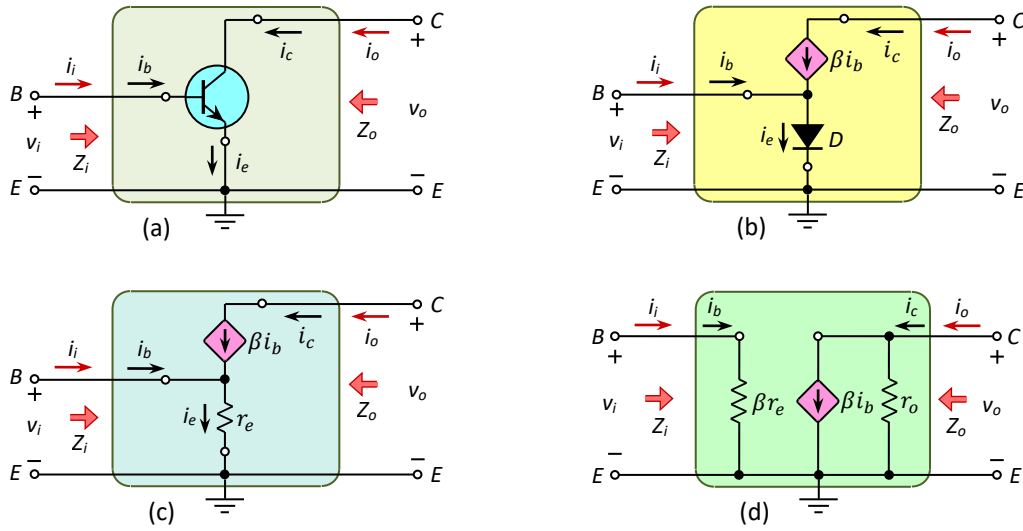


Fig.6-1:(a) NPN transistor in CEC, (b) Diode equivalent circuit, (c) Diode is replaced by r_e , and (d) r_e equivalent circuit.

Now, for the simplicity of calculation, we want to separate the input circuit from the output circuit. To do that let us find the relation between the input current and input voltage. From the Fig.6-1(c), it can be written as,

$$v_i = i_e r_e = i_b (\beta + 1) r_e \approx i_b \beta r_e \quad (6-2)$$

From Equ.(6-2), we find that the emitter resistance r_e is reflected to the base circuit being multiplied by a factor $(\beta + 1) \approx \beta$. With this concept the circuit of Fig.5.1(c) can be drawn separately as in Fig.6-1(d). One more thing is the output impedance. If the output impedance were infinite, the collector circuit could have been represented by a single current source. As shown in Fig.6-2, the output characteristic curves have a finite slope. So the output impedance has a finite value and can be determined using the Early voltage as

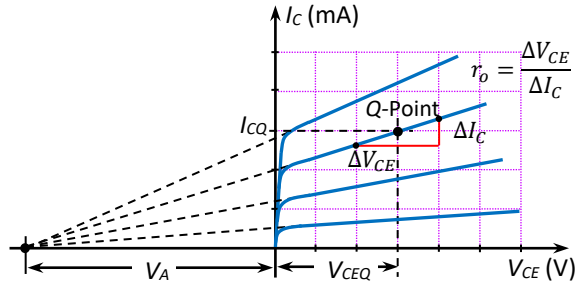


Fig.6-2: Output impedance calculation from Early voltage or slope of the output characteristic curves.

$$r_o = \frac{V_A + V_{CEQ}}{I_{CQ}} \quad (6-3)$$

If the Early voltage of the transistor is not known, the output resistance can also be calculated from the slope of the output curve at the operating point as,

$$r_o = \frac{\Delta V_{CE}}{\Delta I_C} \quad (6-4)$$

We know that the internal resistance of a current source is connected in parallel with it. So by connecting r_o is parallel with the current source, the final equivalent circuit of a transistor in CE configuration is found as shown in Fig.6-1(d).

In the same method, the r_e model of a PNP transistor in CE configuration can be drawn. The common-emitter PNP transistor and it equivalent circuits are shown in Fig.6-3.

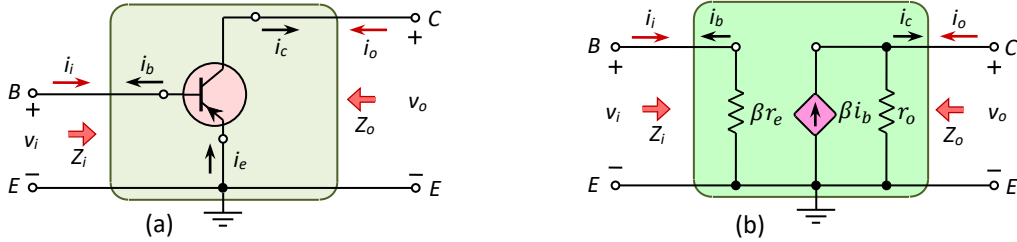


Fig.6-3: (a) PNP transistor in common-emitter configuration, and (b) r_e equivalent circuit.

6.3.2 r_e Model for CB Circuit

An NPN transistor circuit in common-base configuration is shown in Fig.6-4(a). The transistor is biased in the active region. The directions of the transistor terminal currents are as before. It is also considered as a four-terminal network, keeping the base common to both the input and output port. Thus, the emitter and the base will be in the input port, and the base and collector will be in the output port. In this circuit, the emitter current (i_e) flows through a

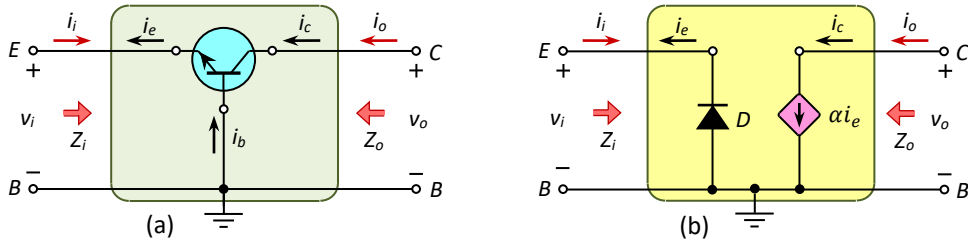


Fig.6-4: (a) Transistor in common base configuration, and (b) Equivalent circuit.

forward biased PN junction, and due to the transistor action αi_e amount of current is developed in the collector terminal. We can consider that the emitter current is flowing through a forward biased diode and the collector current is produced by a dependent current source. Thus, the CB circuit can be represented by its equivalent circuit as shown in Fig.6-4(b). As discussed earlier a forward biased diode behaves like a resistor after the knee voltage, so the diode can be replaced by a resistor which results in the circuit of Fig.6-5. This circuit is the r_e equivalent circuit of an NPN transistor in common base configuration. For all of the r_e equivalent circuits the value of r_e is calculated using the Equ.(6-1). In Chapter 3, we have found that the

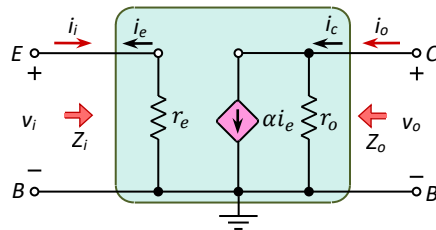


Fig.6-5: r_e model of NPN transistor in CBC.

output characteristic curves of a CB circuit are almost parallel to x -axis. So its output impedance is very high and can be neglected. However, for accurate modeling, it is connected in parallel with the output current source.

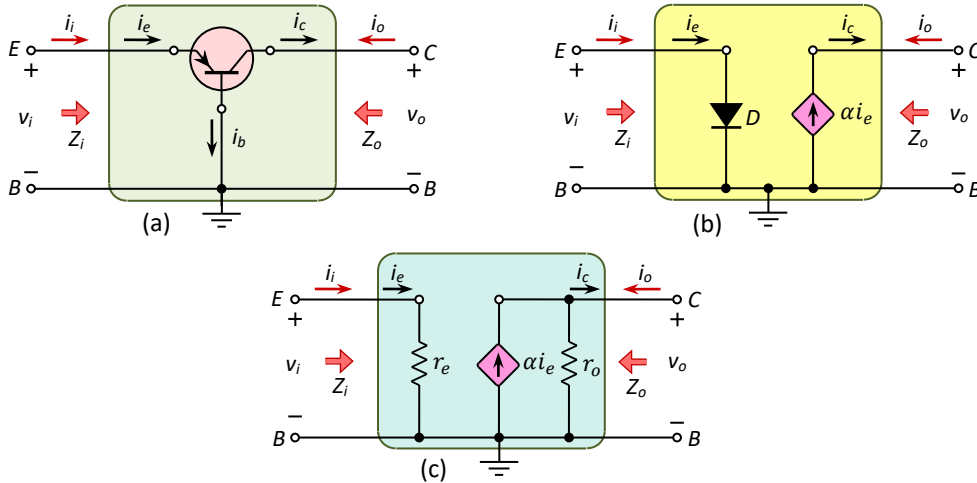


Fig.6-6: (a) PNP transistor in CBC, (b): Diode equivalent circuit, and (c) r_e model of PNP transistor in CBC

In the same method, the r_e equivalent circuit of a PNP transistor in CB configuration can be determined. These are shown in Fig.6-6.

6.3.3 r_e Model for CC Circuit

In a common-collector circuit the input signal is applied to the base and the output is taken from emitter keeping the collector common as shown in Fig.6-7(a). Here also we can consider a forward

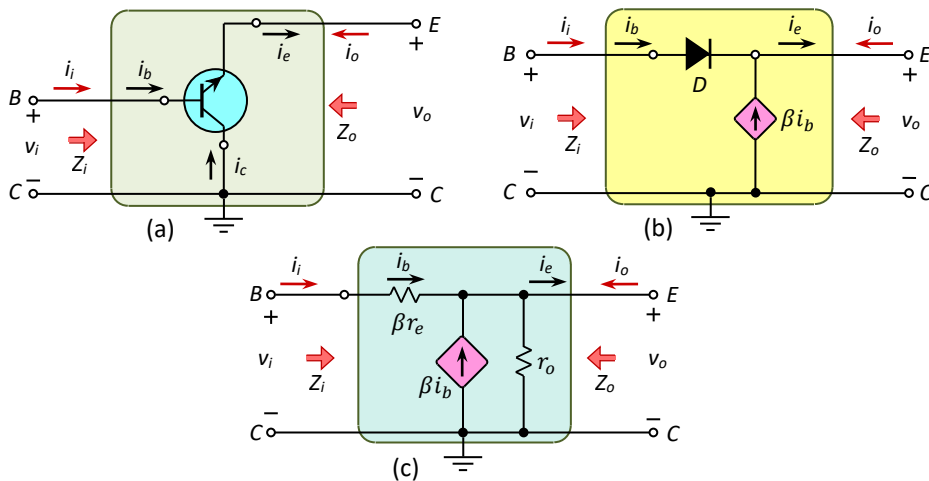


Fig.6-7: (a) Common collector circuit, (b) Alternate representation of CC circuit, and (c) r_e model of CC circuit

biased diode in the base terminal and a constant current source in the collector terminal as in Fig.6-7(b). As we have seen in Chapter 3, that the output characteristics of CC configuration is almost similar to that of CE configuration. Thus, the value of r_o will be approximately equal to that of CE circuit. Now, by replacing the diode by a resistor and connected r_o in parallel with the current source, we get the circuit of Fig.6-7(c). This is the r_e equivalent circuit of an NPN transistor in CC configuration.

6.4 Single Stage Common-Emitter Amplifier

An amplifier that amplifies the input signal only once, and produces the output is called a *single stage amplifier*. Generally, a single stage amplifier consists of a single transistor with proper biasing network. In Fig.6-8(a), a single stage common-emitter amplifier is shown. Here, the voltage divider bias is used. The input signal is applied to the base of the transistor through a capacitor C_1 and the output is taken from the collector terminal through another capacitor C_2 . These capacitors are called the

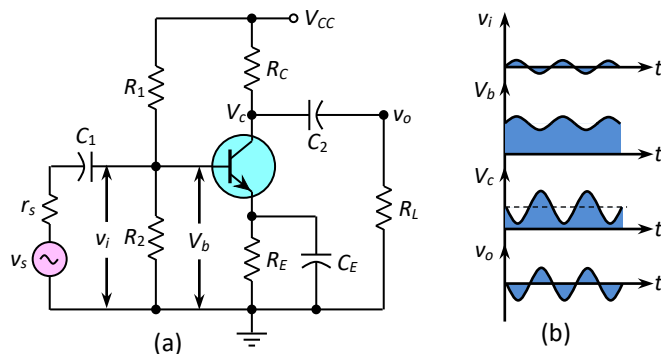


Fig.6-8: Single stage CE amplifier with voltage divider bias, and (b) Voltage waveforms of input source, base terminal, collector terminal and the load

input coupling capacitor and the *output coupling capacitor*, respectively. The functions of these capacitors are to pass the AC signal, and to block the DC signal. The input coupling capacitor (C_1) protects the source from the DC voltage V_B , while the output capacitor (C_2) protects the DC collector voltage V_C from passing to the load. Another reason is: if the source and the load were directly connected to the circuit the value of base voltage (V_B), collector current and voltage levels would have been changed. These effects are illustrated in Example 6-1. Another capacitor, used across the emitter resistor, is called the *emitter bypass capacitor* (C_E). Its function is to bypass the AC signal from the emitter resistor (R_E) and thus, increase the AC voltage gain. Without C_E the voltage gain of the amplifier will be reduced. The effect of unbypassed R_E is explained later. At the operating frequency-range of the amplifier, the reactance of all the three capacitors are so small that they can be considered as short circuited.

Example 6-1

i) Draw the load-line and operating point of the CE amplifier circuit of Fig.6-9(a). ii) Determine the DC biasing condition (operating point) if signal source is directly connected to the input, and iii) Determine the DC biasing condition if a $3\text{ k}\Omega$ load resistance is directly connected to the output. Assume that the internal resistance of the signal source is $600\ \Omega$, and the transistor has $\beta = 150$.

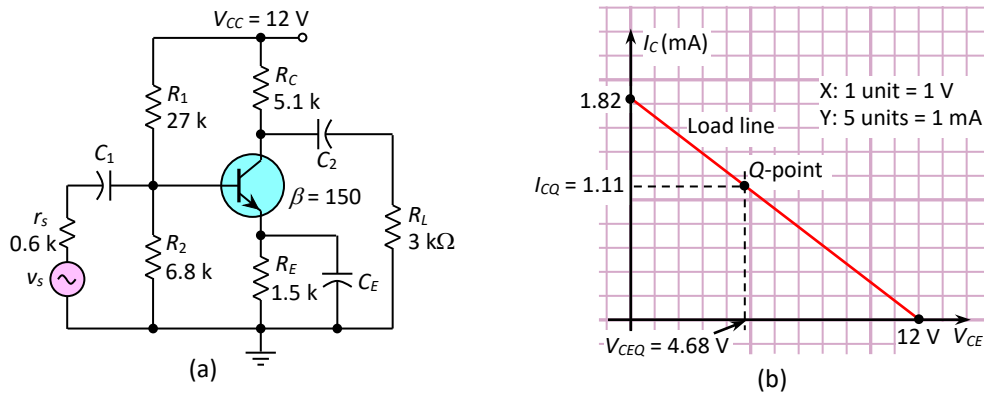


Fig.6-9: (a) CE circuit for Example 6-1, (b) Load-line and operating point of the same circuit .

Solution:

Here, the load-line is drawn by connecting two points $(V_{CC}, 0) = (12\text{ V}, 0)$ and $(0, V_{CC}/(R_L + R_C)) = (0, 12\text{ V}/6.6\text{ k}\Omega) = (0, 1.82\text{ mA})$. The DC load line is drawn on a graph paper as shown in Fig.6-9(b).

Thevenin's voltage,

$$E_{Th} = \frac{V_{CC}R_2}{R_1 + R_2} = \frac{12\text{ V} \times 6.8\text{ k}}{27\text{ k} + 6.8\text{ k}} \approx 2.414\text{ V}$$

Thevenin's resistance,

$$R_{Th} = \frac{R_1R_2}{R_1 + R_2} = \frac{27\text{ k} \times 6.8\text{ k}}{27\text{ k} + 6.8\text{ k}} \approx 5.432\text{ k}\Omega$$

DC base current,

$$I_{BQ} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.414\text{ V} - 0.7\text{ V}}{5.432\text{ k}\Omega + (150 + 1)1.5\text{ k}} \approx 7.391\text{ }\mu\text{A}$$

Quiescent collector current,

$$I_{CQ} = \beta I_{BQ} = 150 \times 7.391\text{ }\mu\text{A} \approx 1.11\text{ mA [Ans.]}$$

Quiescent C-E voltage,

$$V_{CEQ} = 12\text{ V} - 1.11\text{ mA}(5.1\text{ k} + 1.5\text{ k}) = 4.68\text{ V [Ans.]}$$

The operating point is also shown in Fig.6-9(b).

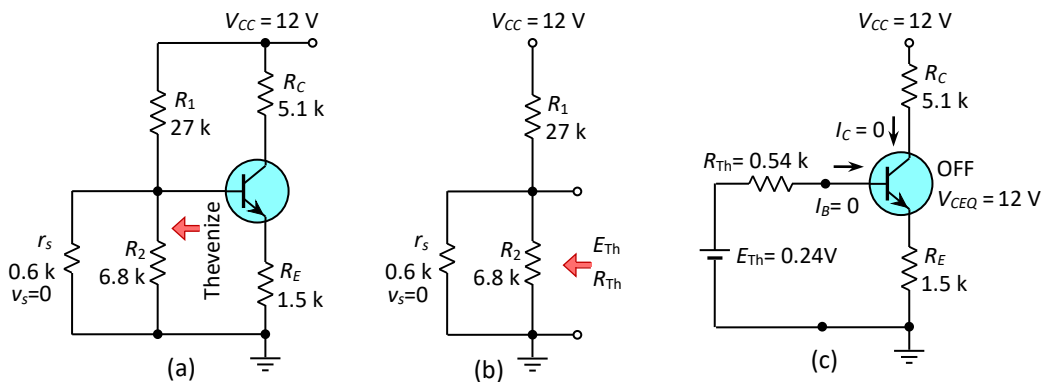


Fig.6-10: (a) Direct connected source, (b) Network to be Thevenized, and (c) Amplifier with Thevenin's equivalent circuit.

ii) For the DC analysis the signal source has to be replaced by its internal resistance as shown in Fig.6-10(a). Now, let us apply Thevenin's theorem to the biasing network.

Thevenin's voltage,
$$E_{Th} = \frac{V_{CC}(R_2 \parallel r_s)}{R_1 + (R_2 \parallel r_s)} = \frac{12 \text{ V} \times (6.8 \text{ k} \parallel 0.6 \text{ k})}{27 \text{ k} + (6.8 \text{ k} \parallel 0.6 \text{ k})} \approx 0.24 \text{ V}$$

Thevenin's resistance,
$$R_{Th} = \frac{R_1(R_2 \parallel r_s)}{R_1 + (R_2 \parallel r_s)} = \frac{27 \text{ k} \times (6.8 \text{ k} \parallel 0.6 \text{ k})}{27 \text{ k} + (6.8 \text{ k} \parallel 0.6 \text{ k})} \approx 0.54 \text{ k}\Omega$$

DC base current,
$$I_{BQ} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{0.24 \text{ V} - 0.7 \text{ V}}{5.432 \text{ k}\Omega + (150 + 1)1.5 \text{ k}} \approx -2 \mu\text{A}$$

I_{BQ} cannot be negative. Actually I_{BQ} will be zero. That is, the transistor will not conduct and its operating point will be set at the cutoff. Therefore,

$$I_{CQ} = 0 \text{ [Ans.]} \quad \text{and} \quad V_{CEQ} = 12 \text{ V [Ans.]}$$

iii) If the load is directly connected to the collector, the circuit will be as shown in Fig.6-11(a). For this circuit, when the transistor is OFF, full V_{CC} will not be applied to collector. It will be divided by R_C and R_L . Similarly, the effective collector resistance will also be changed and its value will be $R_C \parallel R_L$. This is shown in Fig.6-11(b). The new collector voltage and resistance are represented by V'_{CC} and R'_C , respectively and their values will be,

$$V'_{CC} = \frac{V_{CC}R_L}{R_C + R_L} = \frac{12 \text{ V} \times 3 \text{ k}}{5.1 \text{ k} + 3 \text{ k}} \approx 4.44 \text{ V}$$

$$R'_C = R_C \parallel R_L = \frac{R_C R_L}{R_C + R_L} = \frac{5.1 \text{ k} \times 3 \text{ k}}{5.1 \text{ k} + 3 \text{ k}} \approx 1.89 \text{ k}$$

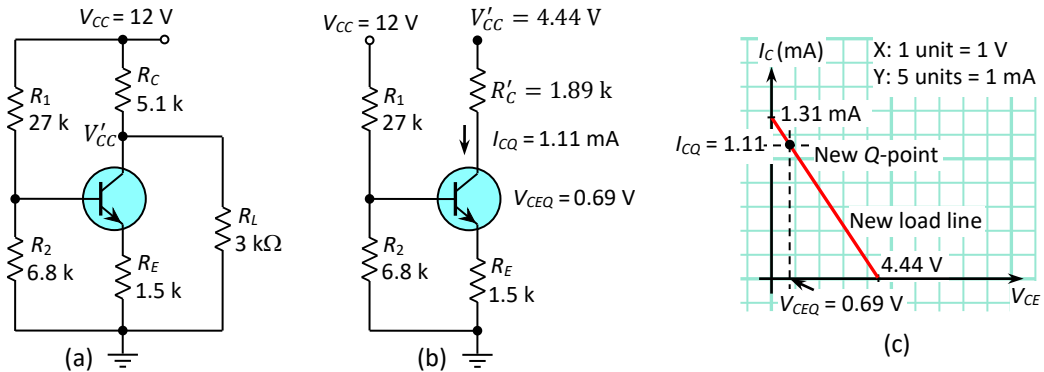


Fig.6-11: (a) Direct connected load, (b) Amplifier with effective V'_{CC} and R'_C , and (c) New load-line of the amplifier.

As the biasing network of the base side is assumed to be unchanged the value of I_{BQ} and I_{CQ} will remain same as calculated in part (i). The new load-line can be determined using the following equation,

$$V_{CE} = V'_{CC} - I_C(R'_C + R_E)$$

The new load line is shown in Fig.6-11(c). The value of I_{CQ} of the new operating point will be same as in (i), but C-E voltage will be changed. Its value can be calculated as,

$$V_{CEQ} = 4.44 \text{ V} - 1.11 \text{ mA} (1.89 \text{ k} + 1.5 \text{ k}) \approx 0.69 \text{ V [Ans.]}$$

The new operating point is also shown in Fig.6-11(c).

Comment: If the input AC source is directly connected to the base, the transistor may be cutoff. On the other hand, if the load is directly connected to the collector, the transistor may be saturated.

6.4.1 AC Load-Line

Within the operating range of frequencies of an amplifier, the reactance of the coupling and bypass capacitors are very small and they become almost short. The DC voltage source also acts as a short path because its internal resistance is very small (ideally zero). Considering these short paths the circuit of Fig.6-8(a) can be redrawn as in Fig.6-12(a). As the V_{CC} terminal is short to ground, R_1 and R_C will be connected to the ground as shown in Fig.6-12(b). We know that, the DC load-line of a CE circuit with voltage divider bias can be drawn using the following equation.

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

From this equation, we find that the DC load line starts from V_{CC} on the x -axis, and ends at $V_{CC}/(R_C + R_E)$ on the y -axis. Now from the AC equivalent circuit, it is clear that R_E is no longer effective for AC signal, and R_L works in parallel with R_C . That is, the effective AC load resistance (R'_L) will be $R_L \parallel R_C$. If the load resistance (R_L) is not connected, then $R'_L = R_C$. Using this AC load resistance (R'_L) the AC load-line can be drawn. The AC load-line is the line through which the Q -point swings when an AC signal is applied to the circuit. We know that, an AC signal must have zero values on it (points on the x -axis). For this zero values of the input signal, i.e., for no input signal, the transistor voltage and current levels are exactly same as determined by the DC Q -point. Hence, the AC load-line passes through the same DC Q -point.

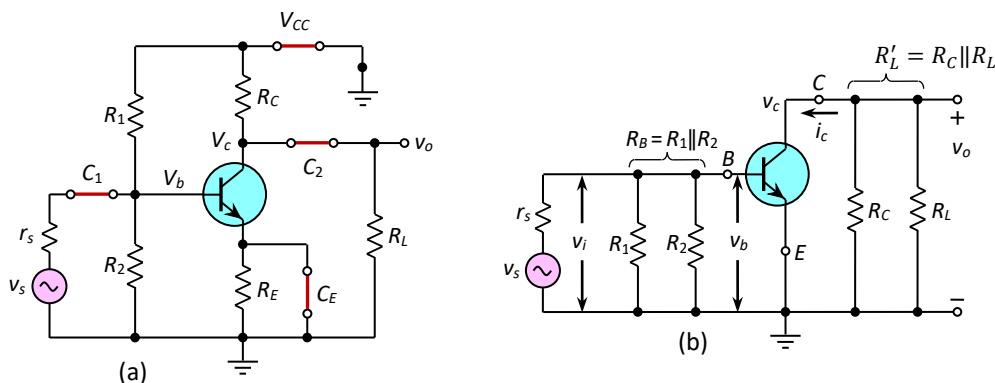


Fig.6-12: (a) AC equivalent of CE amplifier-all capacitors the voltage source is short, and
(b) Same circuit is redrawn.

Thus, we have found one point to draw the AC load-line. Starting from the Q -point, another point is found for the AC load-line by taking a convenient collector current change (usually $\Delta I_C = I_{CQ}$) and calculating the collector-emitter voltage change (ΔV_{CE}). Alternately, the AC load-line can be found by joining a point $V_{CEQ} + I_{CQ}R'_L$ on x -axis to another point $I_{CQ} + V_{CEQ}/R'_L$ on y -axis. The AC load-line

can also be found by drawing a line through Q -point with a slope of $-1/R'_L$ or with an angle of $\tan^{-1}(-1/R'_L)$. All of these methods are illustrated in Example 6-2.

Example 6-2

For the common-emitter amplifier circuit of Fig.6-13(a) determine the DC and AC load lines and operating point. Assume $\beta = 100$.

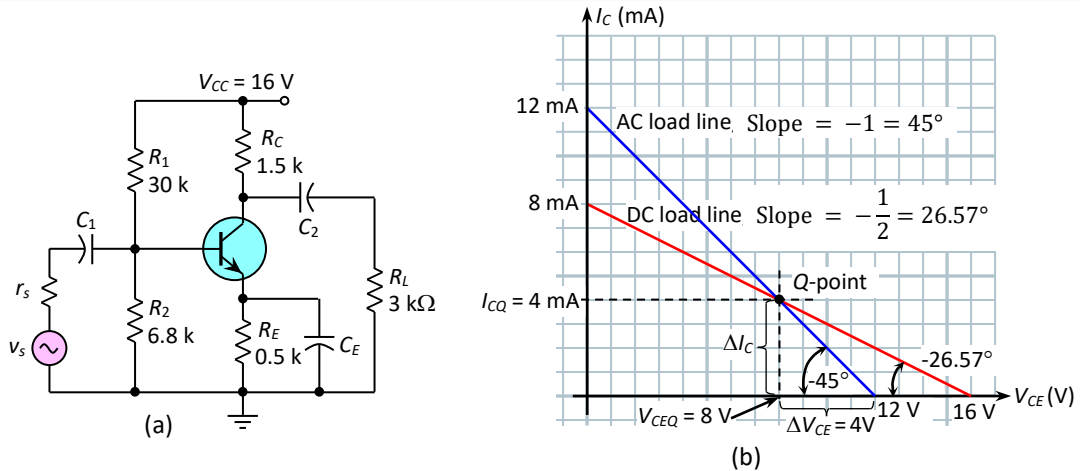


Fig.6-13: CE amplifier for Example 6-2, and (b) DC and AC load-lines and operating point.

Solution:

DC load-line and Q-point:

Here, the points to draw the DC load line will be $(V_{CC}, 0) = (16\text{ V}, 0)$ and $(0, V_{CC}/(R_L + R_C)) = (0, 16\text{ V}/2\text{ k}\Omega) = (0, 8\text{ mA})$. This DC load line is drawn on a graph paper as shown in Fig.6-13(b).

Thevenin's voltage,
$$E_{Th} = \frac{V_{CC}R_2}{R_1 + R_2} = \frac{16\text{ V} \times 6.8\text{ k}}{30\text{ k} + 6.8\text{ k}} \approx 2.957\text{ V}$$

Thevenin's resistance,
$$R_{Th} = \frac{R_1R_2}{R_1 + R_2} = \frac{30\text{ k} \times 6.8\text{ k}}{30\text{ k} + 6.8\text{ k}} \approx 5.544\text{ k}\Omega$$

DC base current,
$$I_{BQ} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.957\text{ V} - 0.7\text{ V}}{5.544\text{ k}\Omega + (100 + 1)0.5\text{ k}} \approx 40.2638\text{ }\mu\text{A}$$

Quiescent collector current,
$$I_{CQ} = \beta I_{BQ} = 100 \times 40.2638\text{ }\mu\text{A} \approx 4\text{ mA [Ans.]}$$

Quiescent C-E voltage,
$$V_{CEQ} = V_{CC} - I_{CQ}(R_C + R_E) = 16\text{ V} - 4\text{ mA}(1.5\text{ k} + 0.5\text{ k}) = 8\text{ V [Ans.]}$$

AC load-line:

The AC load resistance of the circuit is, $R'_L = R_C \parallel R_L = 1.5\text{ k} \parallel 3\text{ k} = 1\text{ k}\Omega$.

Let $\Delta I_C = I_{CQ} = 4\text{ mA}$. Therefore, $\Delta V_{CE} = \Delta I_C \times R'_L = 4\text{ mA} \times 1\text{ k}\Omega = 4\text{ V}$. Using this voltage change the AC load line is drawn through the Q -point on the same graph paper.

Alternate methods of determination of AC load-line:

Point on x -axis,
$$V_{CEQ} + I_{CQ} \times R'_L = 8 \text{ V} + 4 \text{ mA} \times 1 \text{ k} = 12 \text{ V}$$

Point on y -axis,
$$I_{CQ} + \frac{V_{CEQ}}{R'_L} = 4 \text{ mA} + \frac{8 \text{ V}}{1 \text{ k}\Omega} = 12 \text{ mA}$$

By connecting these two points, the AC load line is drawn. It passes through the DC Q -point. Even we can draw the load-lines using their slopes.

Slope of the DC load line is,
$$\tan^{-1}\left(-\frac{1}{R_C + R_E}\right) = \tan^{-1}\left(-\frac{1}{1.5 + 0.5}\right) = -26.57^\circ$$

Similarly, slope of the AC load line is,
$$\tan^{-1}\left(-\frac{1}{R'_L}\right) = \tan^{-1}\left(-\frac{1}{1}\right) = -45^\circ$$

Now we can draw the DC load line with an angle of -26.57° that intersects the x -axis at $V_{CC} = 16 \text{ V}$, and the AC load line can be found by drawing a line with slope of -45° that passes through the Q -point.

Comments: All the results found in this calculation are same that can be verified from the Fig.6-13(b). The Q -point is same for both DC and AC condition.

6.4.2 Operation of CE Amplifier

As shown in Fig.6-14, the input signal, v_{in} , has positive and negative half-cycles with very small amplitude. When this signal comes to the base through C_1 , it superimposes with the base biasing voltage V_B . For the positive half-cycles, the base current increases and consequently collector current increases too. With the increase in I_C , the voltage drop across R_C increases and hence, the collector voltage decreases. On the other hand, for the negative half-cycles the base current decreases, the collector current decreases and hence, the collector voltage increases. In this way, an amplified AC signal is developed at the collector terminal that is 180° out of phase with the input signal. Like the base signal, the amplified signal at the collector has a DC voltage level as shown in Fig.6-14. But when this signal passes through C_2 to the load resistance, the DC voltage level is blocked and a pure AC output is produced across the load.

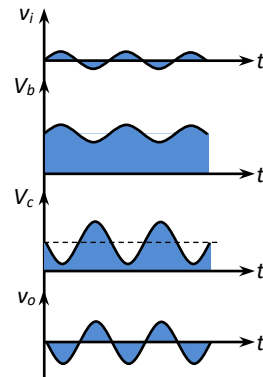


Fig.6-14: Different waveforms of CE amplifier

6.4.3 AC Analysis

The AC equivalent circuit of Fig.6-12(b) can be redrawn using r_e equivalent circuit of the transistor as shown in Fig.6-15. Here, R_B is the equivalent resistance of parallelly connected R_1 and R_2 . The directions of current and the polarities of voltages shown in this figure are those for the positive half-cycles of input

signal.

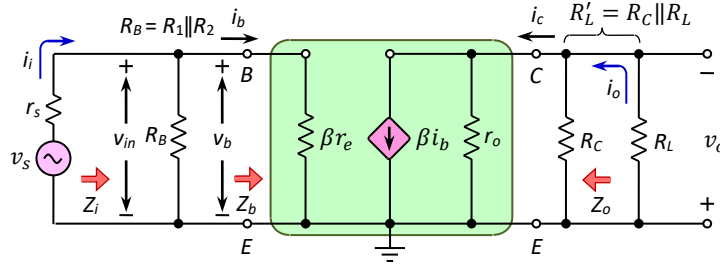


Fig.6-15: The r_e equivalent of circuit of Fig.6-12(b)

Input Impedance Z_i : From Fig.6-15, it is clear that $Z_b = \beta r_e$ and

$$Z_i = R_B \parallel Z_b = R_B \parallel \beta r_e = \frac{R_B \beta r_e}{R_B + \beta r_e} \quad (6-5)$$

If $R_B \gg \beta r_e$ $Z_i \approx \beta r_e$ (6-6)

Output Impedance Z_o : To calculate output impedance the input AC signal has to be set zero. So, i_b will be zero and output current βi_b will be zero as well. Thus, the output source can be considered open. Therefore, from Fig.6-15, the output impedance will be,

$$Z_o = R_C \parallel r_o \quad (6-7)$$

Generally, $r_o \gg R_C$ $\therefore Z_o \approx R_C$ (6-8)

Voltage Gain A_v : To determine the voltage gain, we need the base current. The base current can be calculated from the base voltage as,

$$i_b = \frac{v_b}{Z_b} = \frac{v_{in}}{Z_b} = \frac{v_{in}}{\beta r_e} \quad (6-9)$$

The output voltage is,

$$v_o = -\beta i_b (r_o \parallel R_C \parallel R_L)$$

Putting the value of i_b from Equ. (6-9) $v_o = -\beta (r_o \parallel R_C \parallel R_L) \frac{v_{in}}{\beta r_e}$

or, $A_v = \frac{v_o}{v_{in}} = -\frac{(r_o \parallel R_C \parallel R_L)}{r_e}$

Generally, r_o is very large and can be neglected. $\therefore A_v \approx -\frac{(R_C \parallel R_L)}{r_e}$ (6-10)

If R_L is not connected, i.e., considering $R_L = \infty$ $\therefore A_v \approx -\frac{R_C}{r_e}$ (6-11)

The negative sign indicates the 180° phase difference between the input and the output.

Current Gain A_i : As shown in Fig.6-15, the input current i_{in} is divided between R_B and Z_b . Using current divider rule, i_b can be determined as,

$$i_b = \frac{i_{in} R_B}{R_B + Z_b} = \frac{i_{in} R_B}{R_B + \beta r_e} \quad (6-12)$$

The current flowing through R_L is the output current. Here, current βi_b is divided among r_o , R_C and R_L . Using current divider rule, the output current can be calculated as,

$$\text{Similarly, output current is,} \quad i_o = \frac{\beta i_b (R_C \parallel r_o)}{R_L + (R_C \parallel r_o)}$$

$$\text{As } r_o \text{ is very large, it can be neglected. so} \quad i_o = \frac{\beta i_b R_C}{R_L + R_C}$$

$$\text{Putting the value of } i_b \text{ from Equ.(6-12)} \quad i_o = \left(\frac{\beta R_C}{R_L + R_C} \right) \left(\frac{i_{in} R_B}{R_B + \beta r_e} \right)$$

$$\text{Rearranging,} \quad A_i = \frac{i_o}{i_{in}} = \frac{\beta R_C R_B}{(R_L + R_C)(R_B + \beta r_e)} \quad (6-13)$$

$$\text{If } R_C \gg R_L \text{ and } R_B \gg \beta r_e \quad A_i \approx \frac{\beta R_C R_B}{(R_C)(R_B)} = \beta$$

Power Gain A_p : The power gain is not important for small signal amplifier. However, if necessary it can be calculated by multiplying the voltage gain and current gain.

$$A_p = A_v \times A_i \quad (6-14)$$

Example 6-3

For the CE amplifier of Fig.6-16, calculate Z_i , Z_o , A_v , and A_i . Assume $\beta = 150$, $r_o = \infty$, and all the capacitors are short for the operating frequency range of the amplifier.

DC analysis:

For this circuit, $\beta R_E = 150 \times (1.5 \text{ k}) = 225 \text{ k}\Omega$
and $10R_2 = 10 \times 6.8 \text{ k} = 68 \text{ k}\Omega$. Therefore, $\beta R_E > 10R_2$. So, approximate analysis can be used.

Value of E_{Th} :

$$E_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{12 \text{ V} \times 6.8 \text{ k}}{30 \text{ k} + 6.8 \text{ k}} = 2.2174 \text{ V}$$

Using Equ. (4.53)

$$I_{EQ} = \frac{E_{Th} - V_{BE}}{R_E} = \frac{2.2174 \text{ V} - 0.7 \text{ V}}{1.5 \text{ k}} \approx 1 \text{ mA}$$

AC analysis:

Using Equ.(6-1),

$$r_e = \frac{26 \text{ mV}}{I_E} + r_B \approx \frac{26 \text{ mV}}{1 \text{ mA}} \approx 26 \Omega$$

$$\therefore \beta r_e = 150 \times 26 \Omega = 3.9 \text{ k} \text{ and } R_B = R_1 \parallel R_2 = 30 \text{ k} \parallel 6.8 \text{ k} \approx 5.54 \text{ k}$$

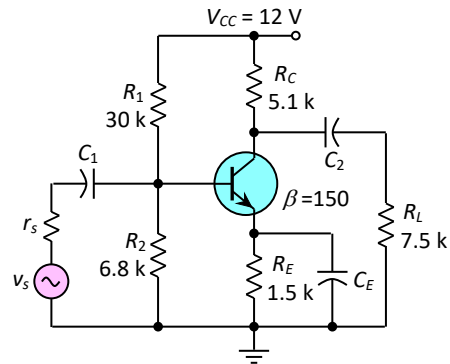


Fig.6-16: CE amplifier with voltage divider bias for Example 6-3.

Z_i : Z_i can be calculated using Equ.(6-5).

$$Z_i = \frac{R_B \beta r_e}{R_B + \beta r_e} = \frac{(5.54 \text{ k})(3.9 \text{ k})}{5.54 \text{ k} + 3.9 \text{ k}} \approx 2.3 \text{ k}\Omega$$

Z_o : Z_o can be calculated using Equ.(6-8)

$$\because r_o = \infty$$

$$\therefore Z_o = R_C = 5.1 \text{ k}\Omega$$

A_v : Using Equ.(6-10),
$$A_v \approx -\frac{(R_C \parallel R_L)}{r_e} = -\frac{5.1 \text{ k} \parallel 7.5 \text{ k}}{26 \times 10^{-3} \text{ k}} \approx -116$$

A_i : Using Equ.(6-13)
$$A_i = \frac{\beta R_C R_B}{(R_L + R_C)(R_B + \beta r_e)} = \frac{150 (5.1 \text{ k})(5.54 \text{ k})}{(7.5 \text{ k} + 5.1 \text{ k})(5.54 \text{ k} + 3.9 \text{ k})} \approx 36$$

Answers: $Z_i = 2.3 \text{ k}$, $Z_o = 5.1 \text{ k}$, $A_v = -116$, and $A_i = 36$

Comments: The transistor current gain is 150, but the circuit current gain is as low as 36.

6.5 CE Amplifier with Fixed Bias and Emitter Bias

Instead of using voltage divider bias if fixed bias or emitter bias with a bypass capacitor is used, then the AC equivalent circuit will be same as that of voltage divider bias. Only the difference is R_B . In this case R_B is a single resistor, while in case of voltage divider bias $R_B = R_1 \parallel R_2$. Fig.6-17(a) shows the CE amplifier with fixed bias, and Fig.6-17(c) is its AC equivalent circuit. Fig.6-18(a) shows the CE amplifier with emitter bias, and Fig.6-18(c) is its AC equivalent circuit. Circuits of Fig.6-17(c) and Fig.6-18(c) are similar to that of Fig.6-12(b). The AC analysis of Fig.6-12(b) has already been done. Therefore, the circuits of Fig.6-17(c) and Fig.6-18(c) can be analyzed using the same method. All of the equations from Equ.(6-5) to Equ.(6-14) are applicable for the CE amplifier with fixed bias and emitter bias.

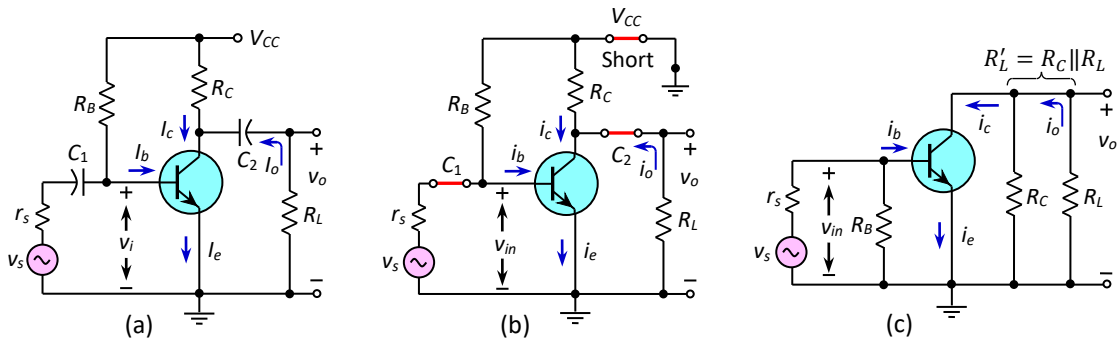


Fig.6-17: (a) CE amplifier with fixed bias, (b) AC equivalent circuit, and (c) AC equivalent circuit redrawn.

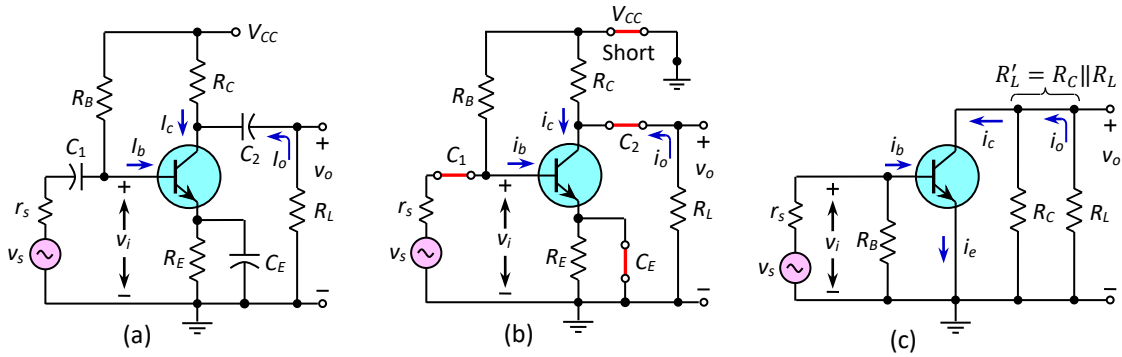


Fig.6-18: (a) CE amplifier with emitter bias, (b) AC equivalent circuit, and (c) AC equivalent circuit redrawn

6.6 CE Amplifier with Collector-to-Base Bias

The circuit diagram of a CE amplifier with collector-to-base bias is shown in Fig.6-19(a). Here, the collector resistor R_C provides the feedback to stabilize the DC operating point as does R_E resistor in emitter bias. In emitter bias, to increase the AC voltage gain, the emitter resistor R_E is bypassed using a capacitor C_E . But, in case of collector-to-base bias, R_C cannot be bypassed, as the output voltage is produced across it. If R_C is bypassed, no output will be found. For this reason, R_C produces negative feedback and reduces the voltage gain. When I_B increases, I_C also increases and V_C decreases. V_C provides the base biasing voltage. So, with the decrease in

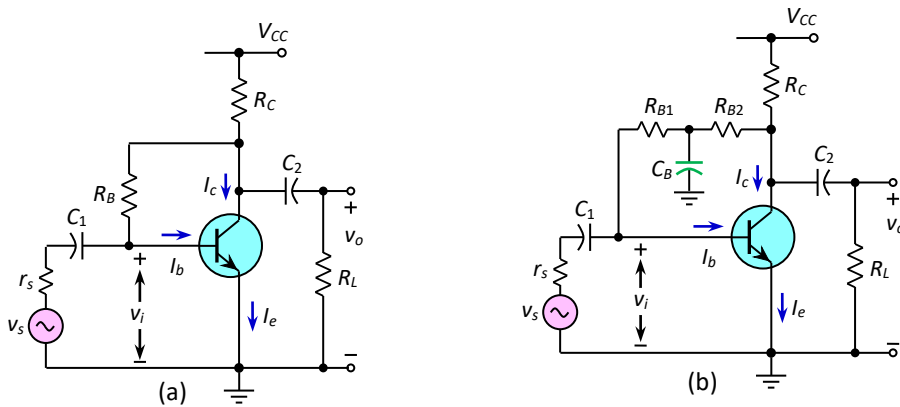


Fig.6-19: (a) CE amplifier with collector-to-base bias, and (b) Bypass capacitor C_B used to eliminate AC degeneration.

V_C , the base current I_B decreases. Thus, an attempt to increase I_B is immediately corrected by the circuit. This, of course, is good for providing stable DC operating point; however, this same process reduces the amplification of the AC signal applied to the input. This process is termed *AC degeneration*. A method to eliminate AC degeneration is shown in Fig.6-19(b). The

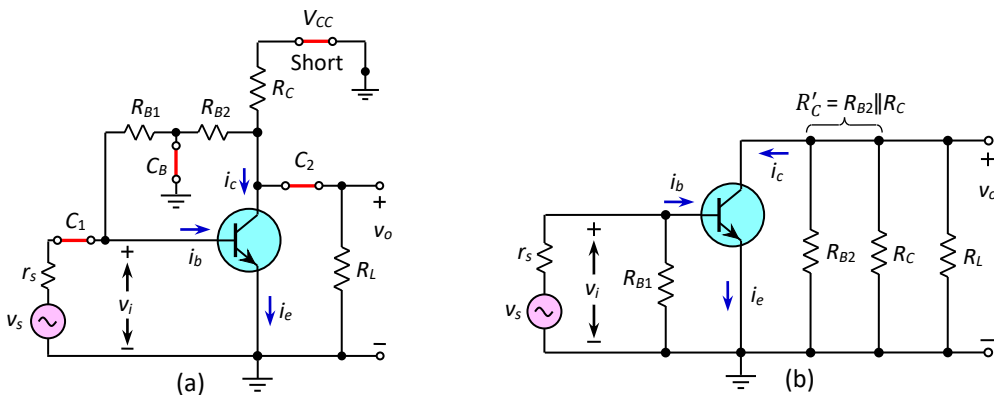


Fig.6-20: (a) AC equivalent circuit of collector-to-base bias amplifier, and (b) Alternate representation of the same circuit.

biasing resistor R_B is split into two approximately equal resistors (R_{B1} and R_{B2}). A bypass capacitor C_B is connected from the common point of R_{B1} and R_{B2} to the ground terminal as illustrated in Fig.6-19(b). For DC condition, C_B is open, and as $R_{B1} + R_{B2} \approx R_B$, the DC operating point will remain unchanged. But, C_B offers a short-circuit for AC signal, and hence, there is no feedback from the collector to base for AC voltage. Now to draw the AC equivalent circuit, all the capacitors are short-circuited as in Fig.6-20(a). This circuit can be redrawn as in Fig.6-20(b). From this figure it is clear that, R_{B1} appears in parallel to the input circuit, whereas, R_{B2} appears in parallel to the output circuit. The AC equivalent circuit of Fig.6-20(b) is same as that of CE amplifier with voltage divider bias [Fig.6-12(b)] except R_{B1} in place of R_B and $R'_C = R_{B2} \parallel R_C$ in place of a single R_C . Therefore, the AC analysis for voltage divider bias circuit can also be used for the circuit of Fig.6-19(b). All of the equations (6-5) to (6-14) can be used for this circuit only replacing R_B by R_{B1} and R_C by $R'_C (= R_{B2} \parallel R_C)$. The process is explained in the following example.

Example 6-4

For the CE amplifier of Fig.6-21, calculate Z_i , Z_o , A_v , and A_i . Assume $\beta = 145$, $r_o = \infty$, and all the capacitors are short for the operating frequency range of the amplifier.

Solution:

DC analysis:

Using Equ.(4.34),

$$I_{BQ} = \frac{12\text{ V} - 0.7\text{ V}}{400\text{ k}\Omega + (145 + 1) 3\text{ k}\Omega} = 13.485\text{ }\mu\text{A}$$

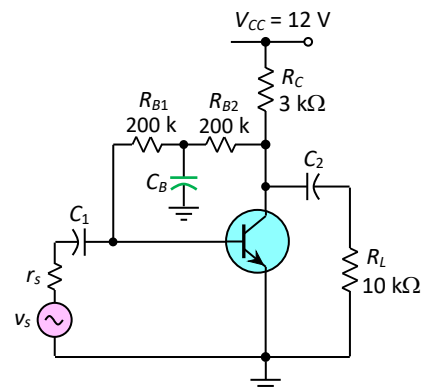


Fig.6-21: CE amplifier with collector-to-base bias for Example 6-4

$$\therefore I_{EQ} = I_{BQ}(\beta + 1) = (13.485\text{ }\mu\text{A}) \times 146 = 1.969\text{ mA}$$

AC analysis:

Using Equ.(6-1), $r_e = \frac{26 \text{ mV}}{I_E} + r_B \approx \frac{26 \text{ mV}}{1.969 \text{ mA}} \approx 13 \Omega \therefore \beta r_e = 145 \times 13 \Omega = 1.9 \text{ k}$

Z_i : Z_i can be calculated using Equ.(6-5). R_{B1} has to be used in place of R_B .

$$\therefore Z_i = \frac{R_{B1}\beta r_e}{R_{B1} + \beta r_e} = \frac{(200 \text{ k})(1.9 \text{ k})}{200 \text{ k} + 1.9 \text{ k}} \approx 1.9 \text{ k}\Omega$$

Z_o : Z_o can be calculated using Equ.(6-8) by considering R'_C in place of R_C .

$$\therefore r_o = \infty \quad \therefore Z_o = R'_C = R_{B2} \parallel R_C = \frac{R_{B2}R_C}{R_{B2} + R_C} = \frac{(200 \text{ k})(3 \text{ k})}{200 \text{ k} + 3 \text{ k}} \approx 3 \text{ k}\Omega$$

$$A_v: \text{ Using Equ.(6-10), } A_v \approx -\frac{R'_C \parallel R_L}{r_e} = -\frac{3 \text{ k} \parallel 10 \text{ k}}{13 \times 10^{-3} \text{ k}} \approx -178$$

The negative sign indicates 180° phase difference between input and output signals.

$$A_i: \text{ Using Equ.(6-13) } A_i = \frac{\beta R'_C R_{B1}}{(R_L + R'_C)(R_{B1} + \beta r_e)} = \frac{145 (3 \text{ k})(200 \text{ k})}{(10 \text{ k} + 3 \text{ k})(200 \text{ k} + 1.9 \text{ k})} \approx 33$$

Answers: $Z_i = 1.9 \text{ k}$, $Z_o = 3 \text{ k}$, $A_v = -178$, and $A_i = 33$

Comments: Same equations of voltage divider bias can be used for emitter-to-base bias.

6.7 CE Amplifier with Unbypassed Emitter Resistor

Already it is mentioned that a bypass capacitor is used across the emitter resistor (R_E) of common-emitter amplifier to increase the AC voltage gain. But, if the emitter resistor is not bypassed, i.e., if the bypass capacitor (C_E) is not used, R_E will have a significant effect on the performance of CE amplifier. Fig.6-22(a) shows a CE amplifier with unbypassed R_E . As the AC signal will pass through it, R_E will appear in the AC equivalent circuit as shown in Fig.6-22(b). Finally, using the r_e model of the transistor, the AC equivalent circuit can be drawn as shown in Fig.6-22(c). **As in the previous circuit, the current directions and voltage polarities shown in the circuit are those that occur for the positive half-cycles of input signal.**

6.7.1 AC Analysis

The AC equivalent circuit of Fig.6-22(b) can be redrawn using r_e equivalent circuit of the transistor as shown in Fig.6-22(c). Here, R_B is the equivalent resistance of parallel connected R_1 and R_2 .

Z_i : At first Z_b will be calculated using v_b and i_b . As r_o is very large, it is reasonable to assume $r_o \approx \infty$. From Fig.6-22(c), it is found that

$$v_b = i_b \beta r_e + i_e R_E = i_b \beta r_e + (\beta + 1) i_b R_E$$

$$\text{or, } Z_b = \frac{v_b}{i_b} = \beta r_e + (\beta + 1) R_E \approx \beta (r_e + R_E)$$

$$\text{Therefore, } \boxed{Z_i = R_B \parallel Z_b = R_1 \parallel R_2 \parallel \beta (r_e + R_E)} \quad (6-15)$$

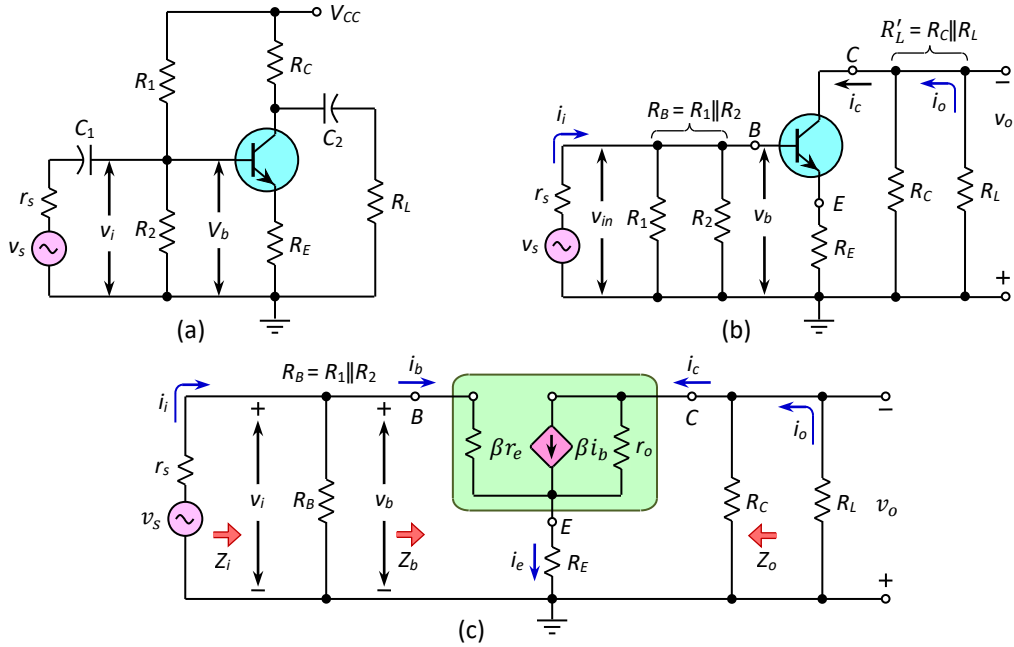


Fig.6-22: (a) CE amplifier with unbypassed R_E , (b) AC equivalent circuit, and (c) AC equivalent circuit using r_e model.

Z_o : To calculate output impedance the input source has to be removed by a short-circuit. So, $i_b = 0$ and the output current source can be considered as open. As the input is short, R_B will also be shorted. From Fig.6-22(c), the output impedance will be,

$$Z_o = R_C \parallel (r_o + \beta r_e \parallel R_E) \approx R_C \parallel r_o$$

Generally, $r_o \gg R_C$

$$\therefore \boxed{Z_o \approx R_C} \quad (6-16)$$

A_v : To determine the voltage gain, we have to calculate the base current. The base current can be calculated from the base voltage as,

$$i_b = \frac{v_b}{Z_b} = \frac{v_{in}}{Z_b} = \frac{v_{in}}{\beta(r_e + R_E)} \quad (6-17)$$

The output voltage is,

$$v_o = -\beta i_b (r_o \parallel R_C \parallel R_L)$$

Putting the value of i_b from Equ.(6-17)
$$v_o = -\beta (r_o \parallel R_C \parallel R_L) \frac{v_{in}}{\beta(r_e + R_E)}$$

or,

$$A_v = \frac{v_o}{v_{in}} = -\frac{(r_o \parallel R_C \parallel R_L)}{r_e + R_E}$$

Generally, r_o is very large and can be neglected.
$$\therefore A_v \approx -\frac{(R_C \parallel R_L)}{r_e + R_E} \quad (6-18)$$

If R_L is not connected, i.e., if $R_L = \infty$

$$\therefore \boxed{A_v \approx -\frac{R_C}{r_e + R_E}} \quad (6-19)$$

The negative sign indicates the 180° phase difference between the input and the output.

A_i: As shown in Fig.6-22(c), the input current i_{in} is divided between R_B and Z_b . Using current divider rule, i_b can be determined as,

$$i_b = \frac{i_{in} R_B}{R_B + Z_b} = \frac{i_{in} R_B}{R_B + \beta(r_e + R_E)} \quad (6-20)$$

Similarly, output current is,

$$i_o = \frac{\beta i_b [R_C \parallel (r_o + Z_b)]}{R_L + [R_C \parallel (r_o + Z_b)]} \approx \frac{\beta i_b (R_C \parallel r_o)}{R_L + R_C \parallel r_o}$$

As r_o is very large, it can be neglected. so

$$i_o = \frac{\beta i_b R_C}{R_L + R_C}$$

Putting the value of i_b from (6-20)

$$i_o = \left(\frac{\beta R_C}{R_L + R_C} \right) \left(\frac{i_{in} R_B}{R_B + \beta(r_e + R_E)} \right)$$

Rearranging,

$$A_i = \frac{i_o}{i_{in}} = \frac{\beta R_C R_B}{(R_L + R_C)[R_B + \beta(r_e + R_E)]} \quad (6-21)$$

All of these expressions derived for the unbypassed emitter resistor CE amplifier are comparable to those derived for circuit with emitter bypass resistor. In fact, if we place $R_E = 0$, these equations will immediately turn into the previous equations.

6.8 Single Stage Common-Base Amplifier

In Chapter 3, two types of biasing circuits of common-base amplifier are described. One is emitter bias and the other is voltage divider bias. Here, the circuits are reproduced in Fig.6-23(a) and Fig.6-24(a).

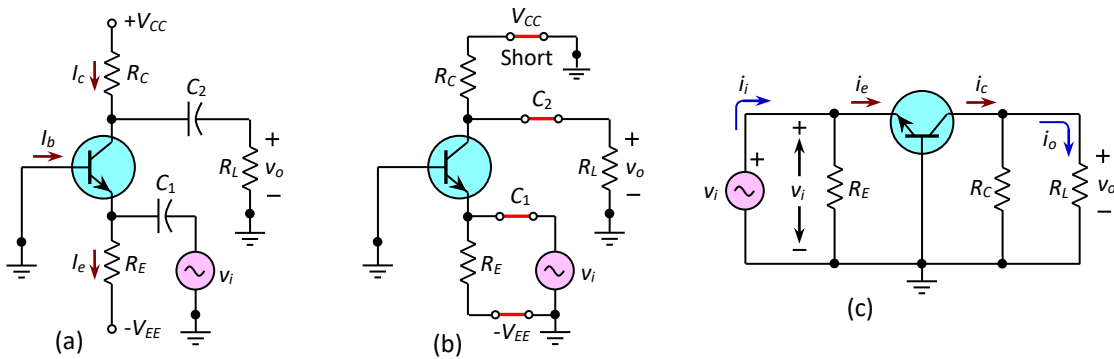


Fig.6-23: (a) CB amplifier with emitter bias, (b) AC equivalent circuit, and (c) Alternate representation of the same circuit.

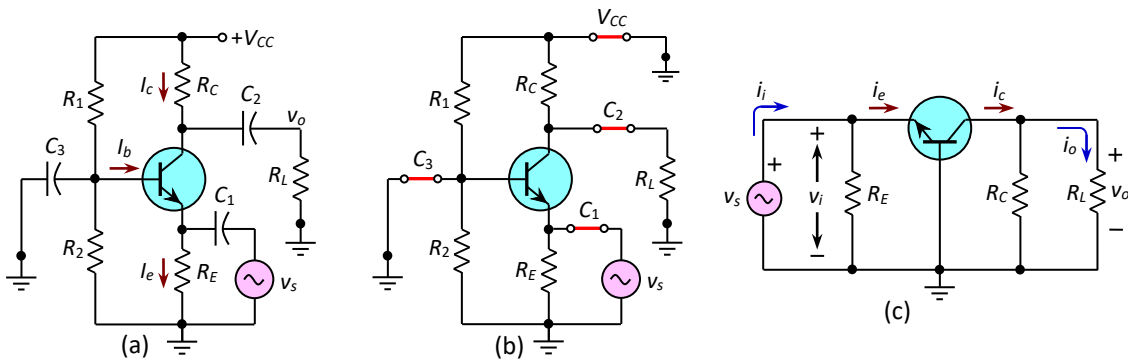


Fig.6-24: (a) CB amplifier with voltage divider bias, (b) AC equivalent circuit, and (c) Alternate representation of the same circuit.

6.8.1 Operation

As shown in Fig.6-23(a) and Fig.6-24(a), the input source is connected to the emitter with respect to ground or base. So, the positive half-cycles of the input signal will increase the emitter voltage with respect to the base, that is the forward bias of the B-E junction will decrease as shown in Fig.6-25. Therefore, for these positive half-cycles, the emitter current will decrease, and hence, the collector current will also decrease. Due to this decrease in collector current, the voltage drop across R_C will decrease, and hence, the collector voltage will increase. Here, the collector voltage is the output voltage. Thus, for a positive half-cycle of the input, a positive voltage will be developed at the output. Similarly, for the negative half-cycles of the input signal the emitter current will increase. So, the collector current will increase as well. For the increase in collector current, the voltage drop across R_C will increase and hence, the collector voltage will decrease. Thus, for the negative input voltage negative output voltage will be produced. Therefore we can say that, for common-base amplifier, the output signal is in-phase with the input. That is, there is no phase difference between the input and the output voltage.

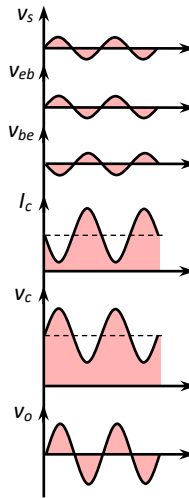


Fig.6-25: AC waveforms at different points of CB amplifier.

6.8.2 AC analysis of Common-Base Amplifier

For AC analysis of the C-B amplifier first we have to draw the AC equivalent of them. To draw the equivalent circuit, all the capacitors and voltage sources have to be short-circuited. The CB amplifiers of Fig.6-23(a) and Fig.6-24(a) are redrawn in Fig.6-23(b) and Fig.6-24 (b) with the capacitors and voltage sources shorted. Circuit of Fig.6-23(b) can easily be redrawn as in Fig.6-23(c). In Fig.6-24(b), two ends of R_1 and R_2 are connected to the ground. So these resistors will also be shorted. Therefore, the circuit of Fig.6-24(b) can also be redrawn as in Fig.6-24(c).

Now, look at the circuits of Fig.6-23(c) and Fig.6-24(c): they are exactly same. If we carefully observe these circuits, we find that the directions of current are opposite to the directions

suggested by the arrow of the NPN transistor. Actually, if a transistor is properly biased, both the positive and negative half-cycles of the input AC signal will flow through the transistor. In these figures, the current directions and voltage polarities are shown at the situation when the emitter current increases due to the input signal. Finally, by replacing the transistor with its r_e equivalent, the AC equivalent circuit of a common-base amplifier with emitter bias or voltage divider bias can be drawn as in Fig.6-26. The input impedance, output impedance, voltage gain and current gain are calculated from this circuit.

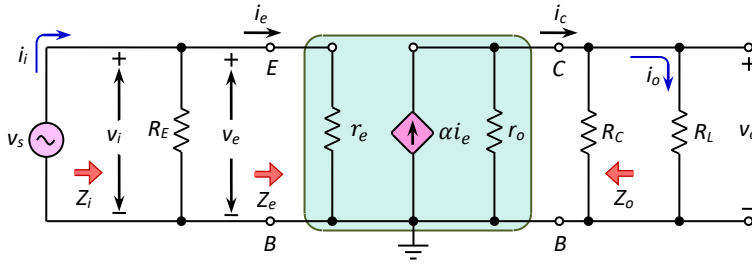


Fig.6-26: r_e equivalent circuit of common-base amplifier.

Z_i : From Fig.6-26, it is clear that

$$Z_i = R_E \parallel Z_e = R_E \parallel r_e = \frac{R_E r_e}{R_E + r_e}$$

If $R_E \gg r_e$

$$\boxed{Z_i \approx r_e} \quad (6-22)$$

Z_o : As discussed earlier, to calculate output impedance the output current source has to be considered open. So from Fig.6-26, the output impedance will be,

$$Z_o = R_C \parallel r_o \quad (6-23)$$

Generally, $r_o \gg R_C$

$$\boxed{Z_o \approx R_C} \quad (6-24)$$

A_v : To determine the voltage gain, we have to calculate the emitter current. The emitter current can be calculated from the emitter voltage as,

$$i_e = \frac{v_e}{Z_e} = \frac{v_{in}}{Z_e} = \frac{v_{in}}{r_e} \quad (6-25)$$

The output voltage is,

$$v_o = \alpha i_e (r_o \parallel R_C \parallel R_L)$$

Putting the value of i_e from (6-25),

$$v_o = \alpha (r_o \parallel R_C \parallel R_L) \frac{v_{in}}{r_e}$$

or,

$$A_v = \frac{v_o}{v_{in}} = \frac{\alpha (r_o \parallel R_C \parallel R_L)}{r_e}$$

Generally, $\alpha \approx 1$, and r_o is very large and can be neglected.

$$\therefore A_v \approx \frac{(R_C \parallel R_L)}{r_e} \quad (6-26)$$

If R_L is not connected, i.e., $R_L = \infty$ $\therefore A_v \approx \frac{R_C}{r_e}$ (6-27)

For this CB amplifier there is no negative sign in the voltage gain. It means that there no phase difference between the input and output.

A_i : As shown in Fig.6-26, the input current i_{in} is divided between R_E and Z_e . Using current divider rule, i_e can be determined as,

$$i_e = \frac{i_{in} R_E}{R_E + Z_e} = \frac{i_{in} R_E}{R_E + r_e} \quad (6-28)$$

Similarly, output current is,
$$i_o = \frac{\alpha i_e (R_C \parallel r_o)}{R_L + (R_C \parallel r_o)}$$

As r_o is very large, it can be neglected. So
$$i_o = \frac{i_e R_C}{R_L + R_C} \quad \therefore \alpha \approx 1$$

Putting the value of i_e from Equ.(6-28)
$$i_o = \left(\frac{R_C}{R_L + R_C} \right) \left(\frac{i_{in} R_E}{R_E + r_e} \right)$$

Rearranging,
$$A_i = \frac{i_o}{i_{in}} = \frac{R_C R_E}{(R_L + R_C)(R_E + r_e)} \quad (6-29)$$

If $R_C \gg R_L$ and $R_E \gg r_e$
$$A_i \approx \frac{R_C R_E}{(R_C)(R_E)} = 1$$

A_p : The power gain can be calculated by multiplying the voltage gain and current gain.

$$A_p = A_v \times 1 = A_v \quad (6-30)$$

Example 6-5

For the CB amplifier of Fig.6-27, calculate Z_i , Z_o , A_v , and A_i . Assume $\beta = 150$, $r_o = \infty$, and all the capacitors are short for the operating frequency range of the amplifier.

DC analysis:

To determine the value of I_E , first, the values of E_{Th} and R_{Th} have to be calculate.

$$\text{Value of } E_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{12 \text{ V} \times 6.8 \text{ k}}{30 \text{ k} + 6.8 \text{ k}} = 2.2174 \text{ V}$$

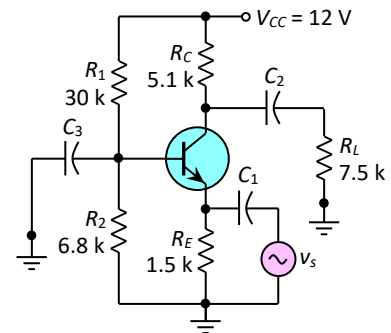


Fig.6-27: CB amplifier with voltage divider bias for Example 6-5.

Value of R_{Th} :
$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{30 \text{ k} \times 6.8 \text{ k}}{30 \text{ k} + 6.8 \text{ k}} \approx 5.54 \text{ k}$$

Using Equ.(4.48)
$$I_{BQ} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.2174 \text{ V} - 0.7 \text{ V}}{5.54 \text{ k} + (150 + 1)1.5 \text{ k}} \approx 6.54 \mu\text{A}$$

$$\therefore I_{EQ} = (\beta + 1)I_{BQ} = (150 + 1)6.54 \mu\text{A} \approx 0.988 \text{ mA}$$

AC analysis:

Using Equ.(6-1),
$$r_e = \frac{26 \text{ mV}}{I_{EQ}} + r_B \approx \frac{26 \text{ mV}}{0.988 \text{ mA}} \approx 26 \Omega$$

Z_i : Z_i can be calculated using Equ.(6-22).

$$Z_i \approx r_e = 26 \Omega$$

Z_o : Z_o can be calculated using Equ.(6-24),

$$\therefore r_o = \infty \quad \therefore Z_o = R_C = 5.1 \text{ k}\Omega$$

A_v : Using Equ.(6-10),
$$A_v \approx \frac{(R_C \parallel R_L)}{r_e} = \frac{5.1 \text{ k} \parallel 7.5 \text{ k}}{26 \times 10^{-3} \text{ k}} \approx 116$$

A_i : Using Equ.(6-29)
$$A_i = \frac{R_C R_E}{(R_L + R_C)(R_E + r_e)} = \frac{(5.1 \text{ k})(1.5 \text{ k})}{(7.5 \text{ k} + 5.1 \text{ k})(1.5 \text{ k} + 26 \Omega)} \approx 0.4$$

Answers: $Z_i = 26 \Omega$, $Z_o = 5.1 \text{ k}$, $A_v = 116$, and $A_i = 0.4$

Comments: The circuits of Example 6-3 and Example 6-5 are same. The first one is used as CE amplifier and the second one is used as CB amplifier. Comparing the results, we find that the output impedance and voltage gain are same. However, CB amplifier gives very low impedance and low current gain.

~~~~~

## 6.9 Single Stage Common-Collector Amplifier

Figures 4.67 and 4.68 of Chapter 3, were the emitter bias and voltage divider bias of a common-collector amplifier, respectively. In section 6.4, we have found that the analysis of emitter bias and voltage divider bias are exactly similar. In case of emitter bias  $R_B$  is a single resistor, whereas in case of voltage divider bias  $R_B$  is calculated from the parallel combination of  $R_1$  and  $R_2$ . For this reason, here only the CC amplifier with voltage divider bias is analyzed. For better understanding, the circuit is reproduced in Fig.6-28(a).

### 6.9.1 Operation

For the positive half-cycles of the input signal, the base current will increase, and hence, the emitter current will also increase. Due to this increase in emitter current, the voltage across  $R_E$  will increase. If external load is not connected, in CE amplifier  $R_C$  is considered as the load, and

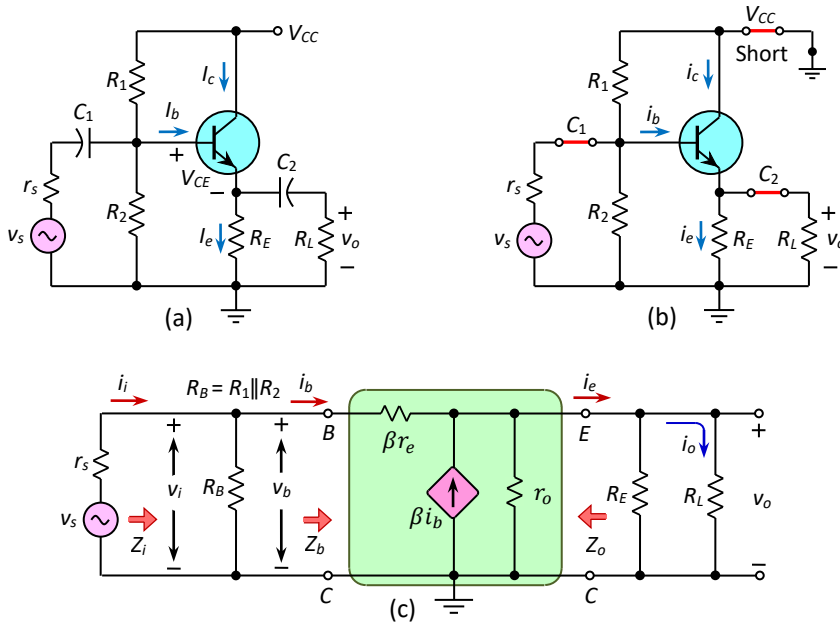


Fig.6-28: (a) CC amplifier with voltage divider bias, (b) AC equivalent circuit, and (c) AC equivalent circuit with transistor  $r_e$  model.

similarly in CC amplifier,  $R_E$  is considered as the load. Thus, we find that for positive input voltage positive output is developed across  $R_E$ . Similarly, for the negative half-cycles of the input signal the base current will decrease. So the emitter current will decrease, and the output voltage will decrease as well. Therefore, for the negative input voltage negative output voltage will be produced. Thus, we find that, for common-collector amplifier, there is no phase difference between the input and output voltage. Among the three amplifiers, only common-emitter amplifier produces  $180^\circ$  phase difference.

### 6.9.2 AC analysis of Common-Collector Amplifier

To draw the AC equivalent circuit, the coupling capacitors  $C_1$  and  $C_2$  and the supply voltage are short-circuited as shown in Fig.6-28(b). Finally the circuit is redrawn in Fig.6-28(c), replacing the transistor by its  $r_e$  equivalent model. The current directions and voltage polarities shown in this circuit are those that occur for the positive half-cycles of the input signal.

$Z_i$ : From Fig.6-28(c), the value of input impedance can be calculated as below

Input voltage 
$$v_b = i_b \beta r_e + i_e R_E \parallel R_L = i_b \beta r_e + i_b (\beta + 1) R_E \parallel R_L \quad (6-31)$$

$$Z_b = \frac{v_b}{i_b} = \beta r_e + (\beta + 1) R_E \parallel R_L$$

In CC amplifier, generally,  $R_L$  is not used. Here, the load is connected as  $R_E$ .

$\therefore$  
$$Z_b = \beta r_e + (\beta + 1) R_E \parallel R_L \approx \beta (r_e + R_E) \quad (6-32)$$

So, the input impedance

$$Z_i = R_B \parallel Z_b = (R_1 \parallel R_2) \parallel \beta(r_e + R_E) \quad (6-33)$$

**$Z_o$ :** To calculate output impedance the input signal has to be considered zero (i.e.,  $v_s = 0$ ). Thus, resistance  $r_s \parallel R_B + \beta r_e$  will come in parallel with the current source. But,  $r_s \parallel R_B + \beta r_e$  is in the base side, so it has to be transferred to emitter side. Seen from emitter, its value will be  $(r_s \parallel R_B + \beta r_e) / \beta = [(r_s \parallel R_B) / \beta + r_e]$ . With  $v_s = 0$ ,  $i_b$  will be zero. So the dependent current source can be considered as open. Therefore, the output impedance will be parallel combination of four resistors  $(r_s \parallel R_B + \beta r_e) / \beta$ ,  $r_o$ ,  $R_E$  and  $R_L$ .

$$\therefore Z_o = [(r_s \parallel R_B) / \beta + r_e] \parallel r_o \parallel R_E \parallel R_L$$

Generally,  $r_o$  is very large and  $R_L$  is not used in most of the CC circuit. So the output impedance will be:

$$\therefore Z_o = [(r_s \parallel R_B) / \beta + r_e] \parallel R_E \quad (6-34)$$

**$A_v$ :** From Fig.6-28(c), we find that  $v_{in} = v_b$

$$\text{So, using Equ.(6-31),} \quad v_b = v_{in} = i_b \beta r_e + i_b (\beta + 1) R_E \parallel R_L \quad (6-35)$$

$$\text{The output voltage is,} \quad v_o = i_e R_E \parallel R_L = (\beta + 1) i_b R_E \parallel R_L \quad (6-36)$$

$$\text{From Equ.(6-35) and (6-36)} \quad A_v = \frac{v_o}{v_{in}} = \frac{(\beta + 1) i_b R_E \parallel R_L}{i_b \beta r_e + i_b (\beta + 1) R_E \parallel R_L}$$

$$\because (\beta + 1) \approx \beta \quad A_v = \frac{R_E \parallel R_L}{r_e + R_E \parallel R_L} \quad (6-37)$$

If  $R_L$  is not used,

$$A_v = \frac{R_E}{r_e + R_E} < 1 \quad (6-38)$$

**$A_i$ :** As shown in Fig.6-28(c), the input current  $i_{in}$  is divided between  $R_B$  and  $Z_b$ . Using current divider rule,  $i_b$  can be determined as,

$$i_b = \frac{i_{in} R_B}{R_B + Z_b} = \frac{i_{in} R_B}{R_B + \beta(r_e + R_E \parallel R_L)} \quad (6-39)$$

$$\text{Similarly, output current is,} \quad i_o = \frac{i_e R_E}{R_E + R_L} = \frac{(\beta + 1) i_b R_E}{R_E + R_L}$$

$$\text{Putting the value of } i_e \text{ from (6-39)} \quad i_o = \frac{(\beta + 1) R_E}{R_E + R_L} \frac{i_{in} R_B}{R_B + \beta(r_e + R_E \parallel R_L)}$$

$$\text{Rearranging,} \quad A_i = \frac{i_o}{i_{in}} = \frac{(\beta + 1) R_B R_E}{(R_E + R_L) [R_B + \beta(r_e + R_E \parallel R_L)]} \quad (6-40)$$

Equ.(6-40) gives the exact current gain of CC amplifier. But, if  $R_L$  is not used, output current will be simply  $i_o = i_e = (\beta + 1) i_b$ . Moreover, if  $R_B$  is very large, current flowing through this resistor can be neglected. Then  $i_{in}$  will be equal to  $i_b$ , i.e.  $i_{in} = i_b$ . Dividing the output current by the input current the current gain will be simply  $(\beta + 1)$ .

**$A_p$ :** The power gain can be calculated by multiplying the voltage gain and current gain.

$$A_p = A_v \times A_i = \frac{R_E(\beta + 1)}{r_e + R_E} \quad (6-41)$$

**Example 6-6**

For the CC amplifier of Fig. 6-29, calculate  $Z_i$ ,  $Z_o$ ,  $A_v$ , and  $A_i$ . Assume  $\beta = 150$ ,  $r_o = \infty$ , and all the capacitors are short for the operating frequency range of the amplifier.

**DC analysis:**

To determine the value of  $I_E$ , first, the values of  $E_{Th}$  and  $R_{Th}$  have to be calculated.

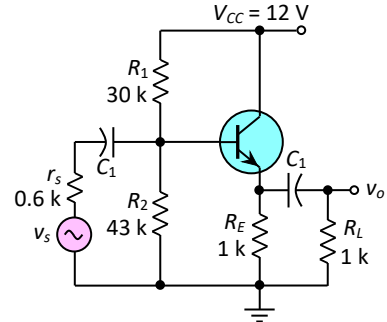


Fig.6-29: CC amplifier with voltage divider bias for Example 6-5

Value of  $E_{Th}$ : 
$$E_{Th} = \frac{V_{CC}R_2}{R_1 + R_2} = \frac{12 \text{ V} \times 43 \text{ k}}{30 \text{ k} + 43 \text{ k}} = 7.07 \text{ V}$$

Value of  $R_{Th}$ : 
$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{30 \text{ k} \times 43 \text{ k}}{30 \text{ k} + 43 \text{ k}} \approx 17.67 \text{ k} = R_B$$

Using Equ. (4.48) 
$$I_{BQ} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{7.07 \text{ V} - 0.7 \text{ V}}{17.67 \text{ k} + (150 + 1)1 \text{ k}} \approx 37.76 \mu\text{A}$$
  

$$\therefore I_{EQ} = (\beta + 1)I_{BQ} = (150 + 1)37.76 \mu\text{A} \approx 5.7 \text{ mA}$$

**AC analysis:**

Using Equ.(6-1), 
$$r_e = \frac{26 \text{ mV}}{I_{EQ}} + r_B \approx \frac{26 \text{ mV}}{5.7 \text{ mA}} \approx 4.56 \Omega$$

$Z_i$ :  $Z_i$  can be calculated using Equ.(6-33).

$$Z_i = R_B \parallel Z_b = R_{Th} \parallel \beta(r_e + R_E) = (17.67 \text{ k}) \parallel 150(4.56 \Omega + 1 \text{ k}) \approx 15.8 \text{ k}$$

$Z_o$ :  $\because r_o = \infty$ ,  $Z_o$  can be calculated using Equ.(6-34). Here  $R_L$  comes in parallel with  $R_E$ ,

$$\therefore Z_o = [(0.6 \text{ k} \parallel 17.67 \text{ k}) / 150 + 4.56 \Omega] \parallel (1 \text{ k} \parallel 1 \text{ k}) \approx 8 \Omega$$

$A_v$ : Using Equ.(6-38), 
$$A_v = A_v = \frac{R_E}{r_e + R_E} = A_v = \frac{1 \text{ k}}{4.56 \Omega + 1 \text{ k}} \approx 0.996$$

$A_i$ : Using Equ.(6-40) 
$$A_i = \frac{(\beta + 1)R_B R_E}{(R_E + R_L)[R_B + \beta(r_e + R_E \parallel R_L)]}$$
  

$$A_i = \frac{(150 + 1)(17.67 \text{ k})(1 \text{ k})}{(1 \text{ k} + 1 \text{ k})[17.67 \text{ k} + 150(4.56 \Omega + 1 \text{ k} \parallel 1 \text{ k})]} \approx 14$$

**Answers:**  $Z_i = 15.8 \text{ k} \Omega$ ,  $Z_o = 8 \Omega$ ,  $A_v = 0.996$ , and  $A_i = 14$

**Comments:** From the results, it is clear that CC amplifier has very large input impedance and current gain. Whereas, output impedance is very small and voltage gain is approximately 1. Due to its very high input impedance and very low output impedance this type of circuit is used for impedance matching.

## 6.10 Effect of $r_s$ and $R_L$ on the Amplifier

The source of the AC signal that is connected to the input terminal of an amplifier has internal resistance which is denoted by  $r_s$ . As the value of this resistance is very small, in most of the analysis, this resistance is neglected but, for some amplifier, this small source resistance may have significant effect. The main effect of  $r_s$  is to reduce the overall voltage gain. To understand this, let us represent the amplifier by a simplified block as in Fig.6-30. Every amplifier has an input impedance  $Z_i$ , an output impedance  $Z_o$  and amplifying parameter, i.e., gain  $A_v$ . The output voltage developed by the amplifier should be  $A_v$  times the source voltage but, due to the drop in  $r_s$  the value of  $v_{in}$  will be less than the source voltage  $v_s$ . Similarly, due to the voltage drop across the output impedance  $Z_o$ , the overall output voltage across  $R_L$  will also be reduced. Considering the drop in  $r_s$ , the value of  $v_{in}$  can be found using voltage divider rule as,

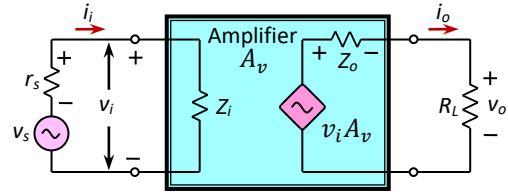


Fig.6-30: Simplified block diagram of an amplifier showing the effect of  $r_s$  and  $R_L$

Input voltage 
$$v_{in} = \frac{v_s Z_i}{r_s + Z_i} \quad \text{or,} \quad \frac{v_{in}}{v_s} = \frac{Z_i}{r_s + Z_i}$$

Voltage gain of the amplifier is 
$$A_v = \frac{v_o}{v_{in}}$$

Starting from  $v$ , the voltage gain will be 
$$A_{vs} = \frac{v_o}{v_s} = \frac{v_o}{v_{in}} \times \frac{v_{in}}{v_s} = A_v \frac{Z_i}{r_s + Z_i} \quad (6-42)$$

From Equ(6-42), it is clear that  $r_s$  decreases the voltage gain. The higher the input impedance of the amplifier the lower the effect of the source resistance  $r_s$ . When a signal source is connected, the gain of the amplifier decreases. This effect is called *input loading effect*.

Similarly, when the load resistance is connect at the output terminal, the actual voltage across  $R_L$  will be,

$$v_o = \frac{v_{in} A_v R_L}{Z_o + R_L} = \quad \text{or,} \quad \frac{v_o}{v_{in}} = A_{vL} = A_v \frac{R_L}{Z_o + R_L} \quad (6-43)$$

From Equ.(6-43), it is clear that the voltage gain has been reduced due to the connection of  $R_L$ . This phenomenon is called *output loading effect*. The lower the output impedance of the amplifier the lower the effect of the load resistance  $R_L$ .

Now, if the effect of  $r_s$  and  $R_L$  is considered simultaneously, the net voltage gain will be,

$$A_{v(Net)} = A_v \times \frac{R_L}{Z_o + R_L} \times \frac{Z_i}{r_s + Z_i} \quad (6-44)$$

**Example 6-7**

The values of voltage gain, input impedance and output impedance of a common-base amplifier are 200, 1 k $\Omega$ , and 5 k $\Omega$ , respectively. If the internal source resistance is 600  $\Omega$  and the load resistance is 1 k $\Omega$ , calculate the net voltage gain of the amplifier.

**Solution:**

As the source and the load are simultaneously connected to the amplifier, we have to consider their effect at a time. Therefore using Equ.(6-44), the net voltage gain of the amplifier will be,

$$A_{v(Net)} = A_v \times \frac{R_L}{Z_o + R_L} \times \frac{Z_i}{r_s + Z_i}$$

Putting the values  $A_{v(Net)} = 200 \times \frac{1 \text{ k}\Omega}{5 \text{ k}\Omega + 1 \text{ k}\Omega} \times \frac{1 \text{ k}\Omega}{0.6 \text{ k}\Omega + 1 \text{ k}\Omega} \approx 20 \text{ [Ans.]}$

**Comments:** Loading effect of  $R_L$  is more severe than the source resistance. But if the input resistance of the amplifier is very low then, the loading effect of source resistance will also be severe.

## 6.11 Hybrid Equivalent Circuit ( $h$ -parameters)

So far, we have analyzed the single-stage amplifier using  $r_e$  equivalent models of transistors. There is another transistor equivalent model which is called **hybrid model** or **hybrid equivalent circuit** of transistors. The hybrid equivalent model was widely used before the popularity of  $r_e$  model. Both models have advantages and disadvantages of their own. Presently, both of them are used for transistor circuit analysis, depending on the required level of accuracy and direction of investigation of the circuit. The main advantage of  $r_e$  model is that, it is simple and its parameters are defined by the actual operating current of the transistor but the main disadvantage of this model is that, it does not consider the feedback effect, that is, the effect of output to input. On the other hand, hybrid model considers the feedback effect but the parameters of hybrid model cannot be determined from the operating current and voltage of the amplifier. The values of the parameters are taken from the data-sheets. In this section, we will discuss the hybrid model of a transistor.

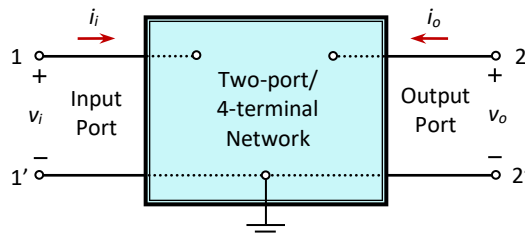


Fig.6-31: Four variables of a two-port network

To define the parameters of hybrid model, let us start with a two-port network. We have already discussed about a two-port network. As shown in Fig.6-31, there are four variables related with a

two-port network. In the input side, the variables are input voltage ( $v_i$ ) and input current ( $i_i$ ), and in the output side, the variables are output voltage ( $v_o$ ) and output current ( $i_o$ ). The values of these variables will be determined by the internal characteristics (e.g., impedance, reactance, feedback, gain etc.) of the network. As long as the network is linear, the internal characteristics will remain constant for any values of the variables. We can relate the variables in different ways. For example, considering the impedance of the network the variables can be related using the following two equations:

$$\begin{aligned}v_i &= i_i Z_{11} + i_o Z_{12} \\v_o &= i_i Z_{21} + i_o Z_{22}\end{aligned}$$

Here, the parameters relating the input and the output variables are called the **Z-parameters** or **impedance parameters**. Similarly, the four variables can be related using the following equations:

$$\begin{aligned}i_i &= v_i Y_{11} + v_o Y_{12} \\i_o &= v_i Y_{21} + v_o Y_{22}\end{aligned}$$

Here, the parameters are called the **Y-parameters** or **admittance parameters**.

As the name suggests, hybrid parameters are the combination of Z-parameters and Y-parameters and the four variables are related using the following two equations:

$$v_i = i_i h_{11} + v_o h_{12} \quad (6-45)$$

$$i_o = i_i h_{21} + v_o h_{22} \quad (6-46)$$

The unit of the variable on the left-hand side of Equ.(6-45) is Volt. So the units of two terms on the right-hand side of this equation must be Volt. To match the units of both sides,  $h_{11}$  must be an impedance parameter and  $h_{12}$  must be a unitless constant. Similarly in Equ.(6-46), the two terms on right-hand side should be current. Therefore, the parameter  $h_{21}$  must be a unitless constant and  $h_{22}$  must be an admittance parameter. Now, we will determine the actual value of each parameter one by one. **Here, the subscript 1 with  $h$  indicates input terminal and the subscript 2 with  $h$  indicates output terminal. The first subscripts indicate the positions where the parameters are measured and the second subscripts indicate where they come from.** Thus,  $h_{12}$  means that the parameter is measured in the input side, but it comes from output side.

**$h_{11}$ :** If we arbitrarily set  $v_o = 0$ , that is, if we short the output terminals and solve Equ.(6-45), we get,

$$h_{11} = \left. \frac{v_i}{i_i} \right|_{v_o=0} \quad \text{Ohms } (\Omega) \quad (6-47)$$

From Equ.(6-47), we find that  $h_{11}$  is the ratio of input voltage and input current. As  $v/i$  gives impedance, and  $h_{11}$  is measured with the output terminal shorted, so it is called **short-circuit input-impedance parameter**. We know that the unit of impedance is Ohms. Hence, the unit of  $h_{11}$  will also be Ohms ( $\Omega$ ). The subscripts 11 of  $h_{11}$  refer to the fact that it is calculated by the ration of two input quantities.

**$h_{12}$ :** To calculate the value of  $h_{12}$ , we have to set  $i_i = 0$ , by keeping the input terminal open. Then, from Equ.(6-45) we find,

$$h_{12} = \left. \frac{v_i}{v_o} \right|_{i_i=0} \quad \text{unitless} \quad (6-48)$$

Equ.(6-48) shows that  $h_{12}$  is the ratio of input voltage to output voltage with input terminal open and it indicates reverse flow of output voltage to input side. Therefore, it is called **open-circuit reverse transfer voltage ratio parameter**. Being the ratio of two same quantities, it is unitless. The subscript 12 of  $h_{12}$  indicates that it is a reverse transfer parameter determined by the ratio of input (1) to output (2) quantities.

**$h_{21}$ :** To calculate the value of  $h_{21}$ , we have to set  $v_o = 0$ , by shorting the output terminals. From Equ.(6-46) we find,

$$h_{21} = \left. \frac{i_o}{i_i} \right|_{v_o=0} \quad \text{unitless} \quad (6-49)$$

Equ.(6-49) shows that  $h_{21}$  is the ratio of output current to input current with output terminals shorted and the parameter indicates the forward flow of input current to output side. Therefore, it is called **short-circuit forward transfer current ratio parameter**. Being the ratio of two same quantities, it is unitless. The meaning of the subscripts 21 of  $h_{12}$  is same as before, that is,  $h_{12}$  is a forward parameter determined by the ratio of output (2) to input (1) quantities.

**$h_{22}$ :** Again, if we set  $i_i = 0$ , that is if we open the input terminals and solve Equ.(6-46), we get

$$h_{22} = \left. \frac{i_o}{v_o} \right|_{i_i=0} \quad \text{siemens or mho (}\mathcal{U}\text{)} \quad (6-50)$$

From Equ.(6-50) we find that  $h_{22}$  is the ratio of output voltage and output current. As  $i/v$  gives admittance, and  $h_{22}$  is measured with the input terminals shorted, it is called **open-circuit output-admittance parameter**. We know that the unit of admittance is siemens or mho. Hence, the unit of  $h_{22}$  will also be siemens ( $\mathcal{U}$ ). The subscripts 22 of  $h_{22}$  refer to the fact that it is measured by the ratio of two output quantities.

Using KVL to any circuit, we can find an equation. Reversely, we can draw a circuit for a given equation. If we represent Equ.(6-45) by a circuit, it will be as in Fig.6-32(a). Similarly, Equ.(6-46) can be represented by a circuit of Fig.6-32(b). Here, the value of the output resistance is  $1/h_{22}$  as the reciprocal of admittance gives impedance. The first one is the input circuit and the second one is the output circuit. As the lower terminals of the circuits are reference terminals, they will be connected to ground. Therefore, the input and output circuits can be combined by connecting the same ground (lower) terminals as in Fig.6-32(c). In this circuit, the hybrid parameters are represented by letter subscripts instead of number subscripts. The choice of the letters is from their short names as given below:

$h_{11} \rightarrow$  input resistance  $\rightarrow h_i$

$h_{12} \rightarrow$  reverse transfer voltage ratio  $\rightarrow h_r$

$h_{21} \rightarrow$  forward transfer current ratio  $\rightarrow h_f$

$h_{22} \rightarrow$  output conductance  $\rightarrow h_o$

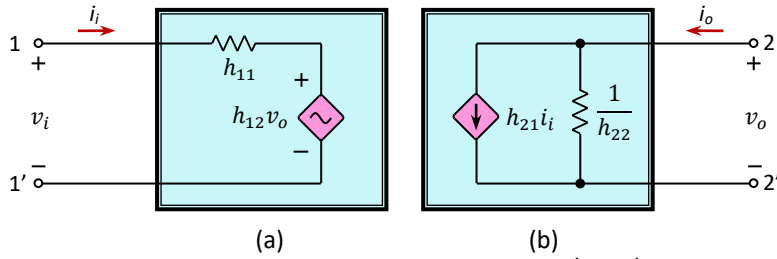


Fig.6-32: Hybrid equivalent circuit (a) Input side indicates Equ.(6-45), and (b) Output side indicates Equ.(6-46)

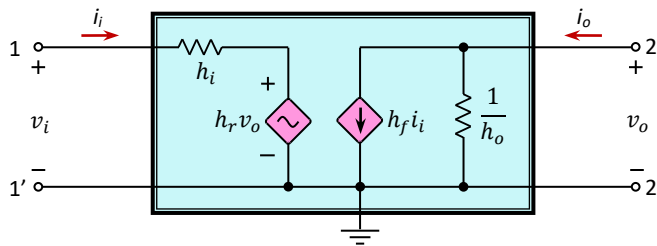


Fig.6-33: (c) Combined hybrid equivalent circuit of two-port

We have already discussed that a transistor circuit is also a two-port network. The hybrid parameters, derived here, can be used for transistor circuits. Depending on the configurations of transistors, a second subscript letter is added with the hybrid parameters.

### 6.11.1 Hybrid Equivalent Circuit of CE Configuration

A transistor in common emitter configuration is shown in Fig.6-(a) and its hybrid equivalent is given in Fig.6-(b). To represent CE configuration, the letter 'e' is added with the four hybrid parameters. The input voltage and input current are  $v_{be}$  and  $i_b$  similarly the output voltage and current are  $v_{ce}$  and  $i_c$ , respectively. The voltage polarities and current directions shown in these circuits are those developed during the positive half-cycles of the AC input signal. Parameter  $h_{fe}$  is the transistor current gain  $\beta$ ,  $h_{ie}$  is the emitter resistance measured at the base terminal, that is,  $\beta r_e$ , and  $1/h_{oe}$  is equal to  $r_o$ .

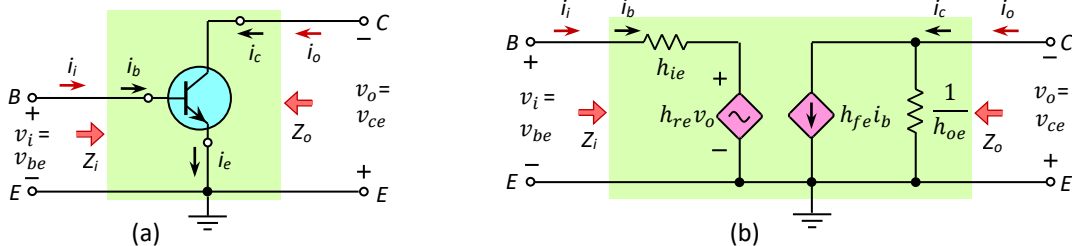
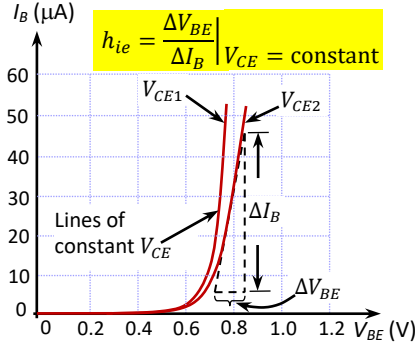
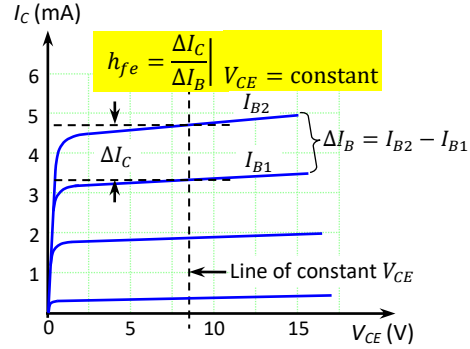
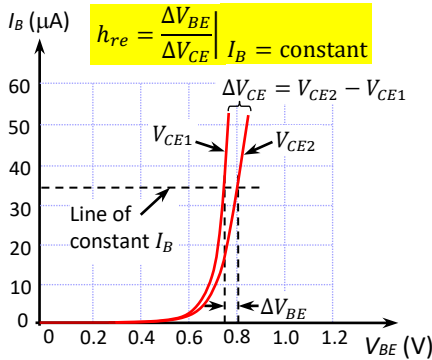
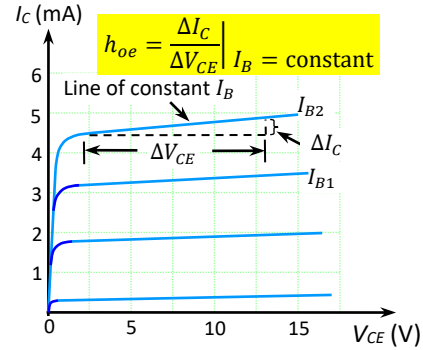


Fig.6-33: (a) Transistor in CE configuration, and (b) Hybrid equivalent circuit of a transistor in CE configuration

The four  $h$ -parameters represent the characteristics of the transistor. Therefore, the  $h$ -parameters of CE configuration can be calculated from the CE input and output characteristic curves as shown in Fig.6-34(a) through Fig.6-34(d).

Fig. 5.34 (a):  $h_{ie}$  from input characteristicsFig. 5.34 (b):  $h_{fe}$  from output characteristicsFig. 5.34 (c):  $h_{re}$  from input characteristicsFig. 5.34 (d):  $h_{oe}$  from output characteristics

### 6.11.2 Hybrid Equivalent Circuit of CB Configuration

A transistor in common base configuration is shown in Fig.6-35. Here, the letter ' $b$ ' with the four hybrid parameters represents common-base configuration. For this circuit, input voltage and input current are  $v_{be}$  and  $i_e$ , similarly the output voltage and current are  $v_{cb}$  and  $i_c$ , respectively. The voltage polarities and current directions shown in these circuits are those produced for the positive half-cycles of the input signal. In this equivalent circuit, parameter  $h_{fb}$  is the transistor current gain  $\alpha$ ,  $h_{ib}$  is the emitter resistance, that is,  $r_e$ , and  $1/h_{ob}$  is equal to  $r_o$ .

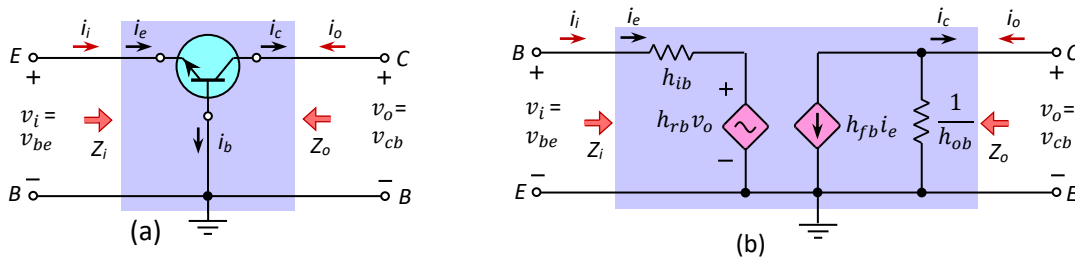


Fig.6-34: (a) Transistor in CB configuration, and (b) Hybrid equivalent circuit of a transistor in CB configuration.

### 6.11.3 Hybrid Equivalent Circuit of CC Configuration

Fig.6-36 shows a transistor in common-collector configuration as well as its hybrid equivalent circuit. Here, the letter 'c' is attached with the hybrid parameters to indicate common-collector configuration. For this circuit, input voltage and input current are  $v_{cb}$  and  $i_b$ , and the output voltage and current are  $v_{ce}$  and  $i_e$ , respectively. The voltage polarities and current directions given in these circuits are those developed during the positive half-cycles of the AC input signal. Here, parameter  $h_{fc}$  is the transistor current gain  $\gamma$ ,  $h_{ic}$  is the emitter resistance, that is,  $\beta r_e$ , and  $1/h_{oc}$  is equal to  $r_o$ .

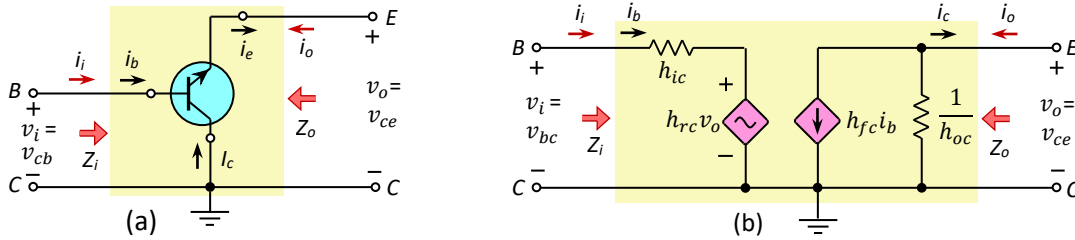


Fig.6-35: (a) Transistor in CC configuration, and (b) Hybrid equivalent circuit of a transistor in CC configuration.

### 6.11.4 Conversion of $h$ -parameters

Nowadays, most of the datasheets of transistor do not mention the values of  $h$ -parameters, except  $h_{fe}$ . In that case, we can analyze the amplifier circuit using  $r_e$  model. Some datasheets mention the values of  $h$ -parameters either for CB configuration or for CE configuration. However, it is possible to convert the values of  $h$ -parameters from one configuration to other configurations. The conversion formulae are given in Table 5.1.

Table 5.1: Formulae for  $h$ -parameters conversion

| From CB to CE                 | From CE to CB                        | From CE to CC     |
|-------------------------------|--------------------------------------|-------------------|
| $h_{ie} = h_{ib}(1 + h_{fb})$ | $h_{ib} = \frac{h_{ie}}{1 + h_{fe}}$ | $h_{ic} = h_{ie}$ |

|                                                     |                                                     |                                 |
|-----------------------------------------------------|-----------------------------------------------------|---------------------------------|
| $h_{re} = \frac{h_{ib}h_{ob}}{1 + h_{fb}} - h_{rb}$ | $h_{rb} = \frac{h_{ie}h_{oe}}{1 + h_{fe}} - h_{re}$ | $h_{rc} = 1 - h_{re} \approx 1$ |
| $h_{fe} = \frac{-h_{fb}}{1 + h_{fb}}$               | $h_{fb} = \frac{-h_{fe}}{1 + h_{fe}}$               | $h_{fc} = -(1 + h_{fe})$        |
| $h_{oe} = \frac{h_{ob}}{1 + h_{fb}}$                | $h_{ob} = \frac{h_{oe}}{1 + h_{fe}}$                | $h_{oc} = h_{oe}$               |

### 6.11.5 Typical Values of h-parameters

Like other parameters, the values of  $h$ -parameters change from transistor to transistor. However, to get some idea about the values of  $h$ -parameters, their typical values for different transistor configurations are given in Table 5.2.

Table 5.2: Typical values of  $h$ -parameters for different configurations

| Parameters | CB Configuration       | CE Configuration      | CC Configuration       |
|------------|------------------------|-----------------------|------------------------|
| $h_i$      | 25 $\Omega$            | 1 k $\Omega$          | 1 k $\Omega$           |
| $h_r$      | $3 \times 10^{-4}$     | $2.5 \times 10^{-4}$  | $\approx 1$            |
| $h_f$      | -0.98                  | 100                   | -100                   |
| $h_o$      | $0.5 \times 10^{-6}$ S | $25 \times 10^{-6}$ S | $255 \times 10^{-6}$ S |

## 6.12 AC Analysis of Transistor Circuits using Hybrid Parameters

In previous sections, amplifiers circuits have been analyzed using  $r_e$  models of transistors. In this sections, the same analysis will be done using hybrid models of transistors.

### 6.12.1 Common-Emitter Amplifier

A single-stage common-emitter amplifier is shown in Fig.6-37(a). Already we have discussed the operation and the AC load-line of this circuit. Here also the AC equivalent circuit of the amplifier is drawn by shorting the capacitors and DC voltage sources which is given in Fig.6-37(b). Finally, replacing the transistor by its hybrid equivalent model, the AC equivalent circuit of is found, which is shown in Fig.6-37(c).

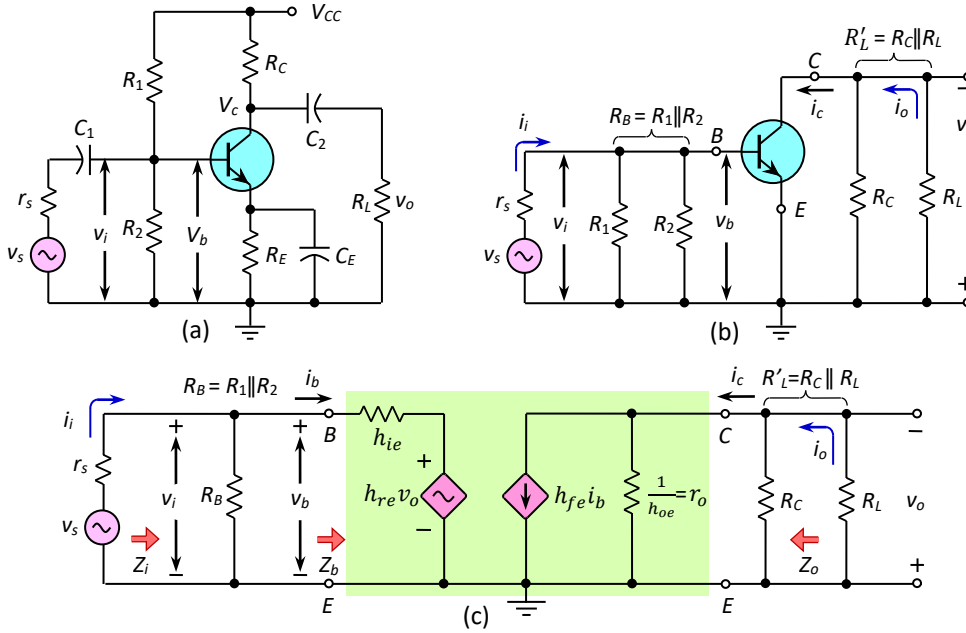


Fig.6-36: (a) Common-emitter amplifier, (b) AC equivalent circuit, and (c) AC equivalent circuit using hybrid model of transistor.

$A_v$ : The output voltage of the amplifier can be calculated as,

$$v_o = -i_o R_L = -i_c (R_C \parallel R_L)$$

$$\text{Let } R_C \parallel R_L = R'_L, \quad \therefore v_o = -i_c R'_L = -h_{fe} i_b (r_o \parallel R'_L) \quad (6-51)$$

The input voltage of the amplifier can be calculated as,

$$v_{in} = v_b = i_b h_{ie} + h_{re} v_o \quad (6-52)$$

Putting the value of  $v_o$  from Equ.(6-51) into Equ.(6-52),

$$v_{in} = v_b = i_b h_{ie} - h_{re} h_{fe} i_b (r_o \parallel R'_L) \quad (6-53)$$

Therefore, the voltage gain of the amplifier can be found by dividing Equ.(6-51) by Equ.(6-53).

$$A_v = \frac{v_o}{v_{in}} = - \frac{h_{fe} (r_o \parallel R'_L)}{h_{ie} - h_{re} h_{fe} (r_o \parallel R'_L)} \quad (6-54)$$

The second term in the denominator of Equ.(6-54) is due to the feedback effect of output voltage into input side. The reverse transfer factor relating the feedback effect is so small that it can be neglected. Thus considering  $h_{re} = 0$ ,

$$A_v \approx - \frac{h_{fe} (r_o \parallel R'_L)}{h_{ie}} \quad (6-55)$$

Again, the value of output impedance of the transistor,  $r_o$ , is very large and can be considered as  $r_o = \infty$ .

$$\therefore \boxed{A_v \approx -\frac{h_{fe}R'_L}{h_{ie}} = -\frac{h_{fe}}{h_{ie}} \left( \frac{R_C R_L}{R_C + R_L} \right)} \quad (6-56)$$

$Z_i$ : From Equ.(6-53) the input impedance at the base terminal can be calculated as,

$$Z_b = \frac{v_b}{i_b} = h_{ie} - h_{re}h_{fe}(r_o \parallel R'_L) \quad (6-57)$$

Here again, considering  $h_{re} = 0$ , we can write,

$$Z_b \approx h_{ie} \quad (6-58)$$

Now, looking from the input terminal, the input impedance will be,

$$\boxed{Z_i = R_B \parallel Z_b = R_B \parallel h_{ie}} \quad (6-59)$$

$$\text{If } R_B \gg h_{ie}, \quad \boxed{Z_i \approx h_{ie}} \quad (6-60)$$

$Z_o$ : To calculate the value of output impedance, we have to consider that the input source is short. That is,  $v_s = 0$ . As there is no AC input voltage, AC input current  $i_b$  will also be zero. Therefore, we can consider that the value of the dependent current source at output is zero. That is, the current source can be considered open. Then, the value of output impedance will be,

$$Z_o = r_o \parallel R_C$$

$$\text{As } r_o \text{ is very large,} \quad \boxed{Z_o \approx R_C} \quad (6-61)$$

$$\text{The output impedance with } R_L, \quad \boxed{Z_o = R_C \parallel R_L} \quad (6-62)$$

$A_i$ : The input current will be divided between  $R_B$  and  $Z_b$ . Using voltage divider rule,

$$i_b = \frac{i_{in}R_B}{R_B + Z_b} \quad \text{or,} \quad \frac{i_b}{i_{in}} = \frac{R_B}{R_B + Z_b} \quad (6-63)$$

Similarly, current  $h_{fe}i_b$  will be divided among  $r_o$ ,  $R_C$ , and  $R_L$ . The current flowing through  $R_L$  is considered as output current. Now neglecting  $r_o$  as it is very large, the output current can be calculated as,

$$i_o = \frac{h_{fe}i_b R_C}{R_C + R_L} \quad \text{or,} \quad \frac{i_o}{i_b} = \frac{h_{fe}R_C}{R_C + R_L} \quad (6-64)$$

The current gain can be calculated as,

$$A_i = \frac{i_o}{i_{in}} = \frac{i_b}{i_{in}} \times \frac{i_o}{i_b}$$

$$\text{Using Equ.(6-63) and (6-64),} \quad \boxed{A_i = \frac{R_B}{R_B + Z_b} \times \frac{h_{fe}R_C}{R_C + R_L}} \quad (6-65)$$

If  $R_B \gg Z_b$  and  $R_L \gg R_C$ , Equ. reduces to,

$$\boxed{A_i \approx h_{fe}} \quad (6-66)$$

**Example 6-8**

Fig.6-37(a) shows a CE amplifier with voltage divider bias. The transistor, used in this circuit, has  $h_{ie} = 3 \text{ k}\Omega$ ,  $h_{fe} = 150$ ,  $h_{re} = 2 \times 10^{-6}$ , and  $h_{oe} = 10 \text{ }\mu\text{S}$ . Assuming  $V_{CC} = 12 \text{ V}$ ,  $R_1 = 30 \text{ k}$ ,  $R_2 = 6.8 \text{ k}$ ,  $R_C = 5.1 \text{ k}$ ,  $R_E = 1.5 \text{ k}$ , and  $R_L = 7.5 \text{ k}$  calculate  $A_v$ ,  $A_i$ ,  $Z_i$ , and  $Z_o$ .

**Solution:**

Value of bias resistance,  $R_B = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} = \frac{30 \text{ k} \times 6.8 \text{ k}}{30 \text{ k} + 6.8 \text{ k}} \approx 5.54 \text{ k}$

Value of AC load resistance,  $R'_L = R_C \parallel R_L = \frac{R_C R_L}{R_C + R_L} = \frac{5.1 \text{ k} \times 7.5 \text{ k}}{5.1 \text{ k} + 7.5 \text{ k}} \approx 3 \text{ k}$

Value of  $r_o$ ,  $r_o = \frac{1}{h_{oe}} = \frac{1}{10 \times 10^{-6} \text{ S}} = 100 \text{ k}\Omega$

Using Equ.(6-55),  $A_v \approx \frac{150(100 \text{ k}\Omega \parallel 3 \text{ k}\Omega)}{3 \text{ k}\Omega} \approx 146$

Using Equ.(6-56),  $A_v \approx -\frac{h_{fe} R'_L}{h_{ie}} = \frac{150 \times 3 \text{ k}}{3 \text{ k}} = 150$

Using Equ.(6-65),  $A_i = \frac{R_B}{R_B + Z_b} \times \frac{h_{fe} R_C}{R_C + R_L} = \frac{R_B}{R_B + h_{ie}} \times \frac{h_{fe} R_C}{R_C + R_L}$

Therefore,  $A_i = \frac{5.54 \text{ k}}{5.54 \text{ k} + 3 \text{ k}} \times \frac{150 \times 5.1 \text{ k}}{5.1 \text{ k} + 7.5 \text{ k}} \approx 39$

Using Equ.(6-59),  $Z_i = R_B \parallel h_{ie} = 5.54 \text{ k} \parallel 3 \text{ k} = \frac{5.54 \text{ k} \times 3 \text{ k}}{5.54 \text{ k} + 3 \text{ k}} \approx 2 \text{ k}\Omega$

Using Equ.(6-62),  $Z_o = r_o \parallel R_C \parallel R_L = r_o \parallel R'_L = \frac{100 \text{ k} \times 3 \text{ k}}{100 \text{ k} + 3 \text{ k}} \approx 2.9 \text{ k}\Omega$

**Results:**  $A_v = 146$ ,  $A_i = 39$ ,  $Z_i = 2 \text{ k}\Omega$ , and  $Z_o = 2.9 \text{ k}\Omega$ .

**Comments:** The exact voltage gain is 146, whereas, approximate voltage gain is 150. On the other hand, although the device current gain is  $\beta = 150$ , but the circuit current gain is only 39.

### 6.12.2 Common-Emitter Amplifier with Unbypassed Emitter Resistor

A single-stage common-emitter amplifier is shown in Fig.6-38(a). Here, no capacitor is used across the emitter resistor. Already, we have discussed in  $r_e$  modeling, the unbypassed emitter

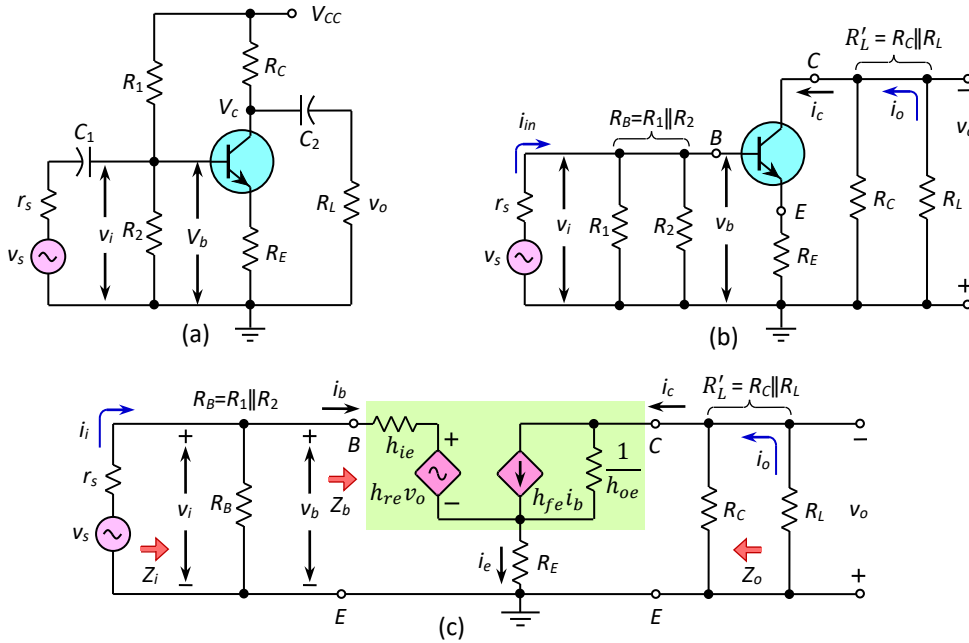


Fig.6-37: (a) Common-emitter amplifier, (b) AC equivalent circuit, and (c) AC equivalent circuit using hybrid model of transistor.

resistor has a significant effect on the performance of amplifiers. It increases the output impedance and decreases the voltage gain. It also reduces the current gain and increases the input impedance. Here also, the AC equivalent circuit of the amplifier is drawn by shorting the capacitors and DC voltage sources, which is given in Fig.6-38(b). Replacing the transistor by its hybrid equivalent model, the final AC equivalent circuit is found which is shown in Fig.6-38(c).

$A_v$ : The output voltage of the amplifier can be calculated as

$$v_o = -i_o R_L = -i_c (R_C \parallel R_L)$$

$$\text{Let } R_C \parallel R_L = R'_L \quad \therefore v_o = -i_c R'_L = -h_{fe} i_b (r_o \parallel R'_L) \quad (6-67)$$

The input voltage of the amplifier can be calculated as

$$\begin{aligned} v_{in} = v_b &= i_b h_{ie} + h_{re} v_o + i_e R_E \\ \therefore i_e &= (h_{fe} + 1) i_b \quad \therefore v_{in} = i_b h_{ie} + h_{re} v_o + (h_{fe} + 1) i_b R_E \end{aligned} \quad (6-68)$$

Putting the value of  $v_o$  from Equ.(6-67) into Equ.(6-68)

$$v_{in} = v_b = i_b h_{ie} + (h_{fe} + 1) i_b R_E - h_{re} h_{fe} i_b (r_o \parallel R'_L) \quad (6-69)$$

Now, the voltage gain of the amplifier will be found by dividing Equ.(6-67) by Equ.(6-69).

$$A_v = \frac{v_o}{v_{in}} = - \frac{h_{fe} (r_o \parallel R'_L)}{h_{ie} + (h_{fe} + 1) i_b R_E - h_{re} h_{fe} (r_o \parallel R'_L)} \quad (6-70)$$

Considering  $h_{re} = 0$ ,

$$A_v \approx -\frac{h_{fe}(r_o \parallel R'_L)}{h_{ie} + (h_{fe} + 1)R_E} \quad (6-71)$$

Again, the value of output impedance,  $r_o$ , of transistor is very large and can be considered  $r_o = \infty$ .

$\therefore$

$$A_v \approx -\frac{h_{fe}R'_L}{h_{ie} + (h_{fe} + 1)R_E} \quad (6-72)$$

**$Z_i$ :** From Equ.(6-69) the input impedance at the base terminal can be calculated as,

$$Z_b = \frac{v_b}{i_b} = h_{ie} + (h_{fe} + 1)R_E - h_{re}h_{fe}(r_o \parallel R'_L) \quad (6-73)$$

Here again, considering  $h_{re} = 0$ , we can write,

$$Z_b \approx h_{ie} + (h_{fe} + 1)R_E \quad (6-74)$$

Now, looking at the input terminal, the input impedance will be,

$$Z_i = R_B \parallel Z_b = R_B \parallel [h_{ie} + (h_{fe} + 1)R_E] \quad (6-75)$$

**$Z_o$ :** To calculate the value of output impedance, we have to set  $v_{in} = 0$ . Therefore,  $i_b$  will also be zero. The value of the dependent current source at the output will be zero as well. That is, this current source can be considered as open circuit. Then the value of the output impedance will be,

$$Z_o = (r_o + R_E \parallel h_{ie}) \parallel R_C \parallel R_L$$

As  $(r_o + R_E \parallel h_{ie}) \gg R_C$ ,

$$\therefore Z_o \approx R_C \parallel R_L \quad (6-76)$$

The output impedance without  $R_L$ ,

$$Z_o = R_C \parallel R_L \quad (6-77)$$

**$A_i$ :** The input current will be divided between  $R_B$  and  $Z_b$ . Using voltage divider rule,

$$i_b = \frac{i_{in}R_B}{R_B + Z_b} \quad \text{or,} \quad \frac{i_b}{i_{in}} = \frac{R_B}{R_B + Z_b} \quad (6-78)$$

The current  $h_{fe}i_b$  will also be divided among  $r_o$ ,  $R_C$ , and  $R_L$ . The current flowing through  $R_L$  is considered as the output current. Generally,  $r_o$  is so large that its current can be neglected. Thus, the output current will be,

$$i_o = \frac{h_{fe}i_bR_C}{R_C + R_L} \quad \text{or,} \quad \frac{i_o}{i_b} = \frac{h_{fe}R_C}{R_C + R_L} \quad (6-79)$$

Now, the current gain of the circuit can be calculated as,

$$A_i = \frac{i_o}{i_{in}} = \frac{i_b}{i_{in}} \times \frac{i_o}{i_b}$$

Using Equ.(6-78) and (6-79),

$$A_i = \frac{R_B}{R_B + Z_b} \times \frac{h_{fe}R_C}{R_C + R_L} \quad (6-80)$$

Where,  $Z_b$  is given by Equ.(6-73).

### Example 6-9

Fig.6-38(a) shows a CE amplifier with voltage divider bias. The transistor, used in this circuit, has  $h_{ie} = 3 \text{ k}\Omega$ ,  $h_{fe} = 150$ ,  $h_{re} = 2 \times 10^{-6}$ , and  $h_{oe} = 10 \text{ }\mu\text{S}$ . Assuming  $V_{CC} = 12 \text{ V}$ ,  $R_1 = 30 \text{ k}$ ,  $R_2 = 6.8 \text{ k}$ ,  $R_C = 5.1 \text{ k}$ ,  $R_E = 1.5 \text{ k}$ , and  $R_L = 7.5 \text{ k}$  calculate  $A_v$ ,  $A_i$ ,  $Z_i$ , and  $Z_o$ .

**Solution:**

Value of bias resistance,  $R_B = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} = \frac{30 \text{ k} \times 6.8 \text{ k}}{30 \text{ k} + 6.8 \text{ k}} \approx 5.54 \text{ k}$

Value of AC load resistance,  $R'_L = R_C \parallel R_L = \frac{R_C R_L}{R_C + R_L} = \frac{5.1 \text{ k} \times 7.5 \text{ k}}{5.1 \text{ k} + 7.5 \text{ k}} \approx 3 \text{ k}$

Using Equ.(6-72),  $A_v \approx -\frac{h_{fe} R'_L}{h_{ie} + (h_{fe} + 1) R_E} = \frac{150 \times 3 \text{ k}}{3 \text{ k} + (150 + 1) 1.5 \text{ k}} \approx 2$

Using Equ.(6-74), the value of  $Z_b$  can be calculated as,

$$Z_b \approx h_{ie} + (h_{fe} + 1) R_E = 3 \text{ k} + (150 + 1) 1.5 \text{ k} = 229.5 \text{ k}$$

Now, using Equ.(6-80),  $A_i = \frac{5.54 \text{ k}}{5.54 \text{ k} + 229.5 \text{ k}} \times \frac{150 \times 5.1 \text{ k}}{5.1 \text{ k} + 7.5 \text{ k}} \approx 1.43$

Using Equ.(6-75),  $Z_i = R_B \parallel Z_b = 5.54 \text{ k} \parallel 229.5 \text{ k} \approx 5.4 \text{ k}$

Using Equ.(6-77),  $Z_o = R_C \parallel R_L = R'_L = 3 \text{ k}$

**Results:**  $A_v = 2$ ,  $A_i = 1.43$ ,  $Z_i = 1.43 \text{ k}\Omega$ , and  $Z_o = 3 \text{ k}\Omega$ .

**Comments:** Although, the circuit components are same as in Example 6-5, the voltage gain, current gain and input impedance are comparatively small due to the effect of unbypassed emitter resistor. However, the output impedance remains almost unchanged.

### 6.13 Analysis of Common-Base Amplifier using Hybrid Equivalent Circuit

Common-base amplifier has already been analyzed using  $r_e$  equivalent circuit. Here, the same thing is done using hybrid equivalent circuit. As the emitter bias and the voltage divider bias result in the same AC equivalent circuit, here the voltage divider bias is considered only. The CB amplifier with voltage divider bias is redrawn in Fig.6-39(a) and its AC equivalent circuit in Fig.6-39(b). The final equivalent circuit is drawn in Fig.6-39(c) by replacing the transistor with its hybrid model.

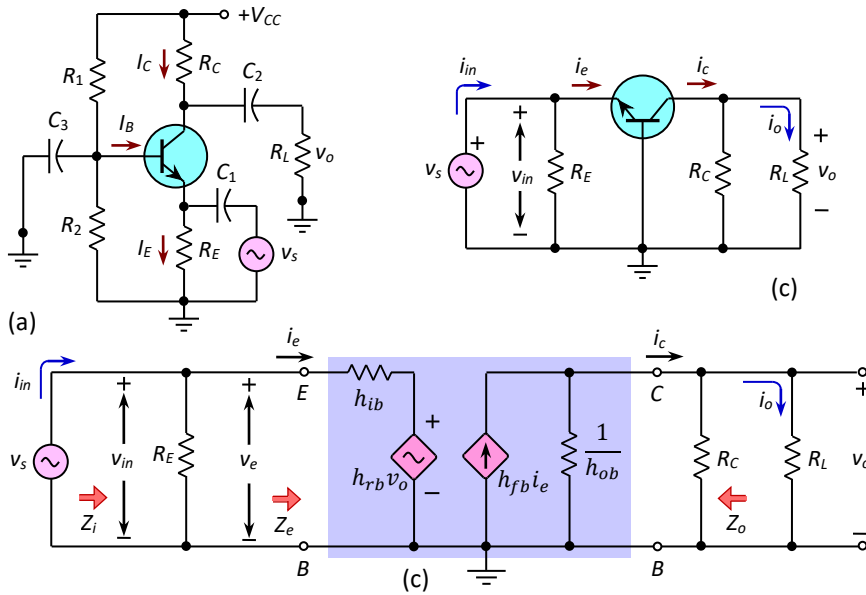


Fig.6-38: (a) CB amplifier with emitter bias, (b) AC equivalent circuit, and (c) Alternate representation of the same circuit.

The input impedance, output impedance, voltage gain and current gain are calculated from this circuit.

$A_v$ : The output voltage of the amplifier can be calculated as

$$v_o = i_o R_L = i_c (R_C \parallel R_L)$$

$$\text{Let } R_C \parallel R_L = R'_L \text{ and } r_o = \frac{1}{h_{ob}} \quad \therefore v_o = i_c R'_L = h_{fb} i_e (r_o \parallel R'_L) \quad (6-81)$$

The input voltage of the amplifier can be calculated as

$$v_{in} = v_e = i_e h_{ib} + h_{rb} v_o \quad (6-82)$$

Putting the value of  $v_o$  from Equ.(6-81) into Equ.(6-82),

$$v_{in} = v_e = i_e h_{ib} + h_{rb} h_{fb} i_e (r_o \parallel R'_L) \quad (6-83)$$

Therefore, the voltage gain of the amplifier can be found dividing Equ.(6-81) by Equ.(6-83).

$$\therefore A_v = \frac{v_o}{v_{in}} = \frac{h_{fb} (r_o \parallel R'_L)}{h_{ib} + h_{rb} h_{fb} (r_o \parallel R'_L)} \quad (6-84)$$

The second term in the denominator of Equ.(6-84) is due to the effect of output voltage into the input side. The reverse transfer factor,  $h_{rb}$ , relating this feedback effect, is very small and can be neglected. Thus, considering  $h_{rb} = 0$ ,

$$A_v \approx \frac{h_{fb}(r_o \parallel R'_L)}{h_{ib}} \quad (6-85)$$

Again, the value of output impedance,  $r_o$ , of the transistor is very large and comes in parallel with  $R'_L$ . So neglecting  $r_o$ , the voltage gain will be,

$$A_v \approx \frac{h_{fb}R'_L}{h_{ib}} = \frac{h_{fb}}{h_{ib}} \left( \frac{R_C R_L}{R_C + R_L} \right) \quad (6-86)$$

**$Z_i$ :** From Equ.(6-83) the input impedance at the emitter terminal can be calculated as,

$$Z_e = \frac{v_e}{i_e} = h_{ib} + h_{rb}h_{fb}(r_o \parallel R'_L) \quad (6-87)$$

Here again, considering  $h_{rb} = 0$ , we can write,

$$Z_e \approx h_{ib} \quad (6-88)$$

Now, looking from the input terminal, the input impedance will be,

$$Z_i = R_E \parallel Z_e = R_E \parallel h_{ib} \quad (6-89)$$

If  $R_E \gg Z_e$ ,

$$Z_i \approx h_{ib} \quad (6-90)$$

**$Z_o$ :** To calculate the output impedance, we have to short the input source. That is,  $v_{in} = 0$  and hence  $i_e$  will also be zero. Therefore, we can consider the output current source open. Then the value of the output impedance will be,

$$Z_o = r_o \parallel R_C \left[ \text{where, } r_o = \frac{1}{h_{ob}} \right] \quad (6-91)$$

As  $r_o \gg R_C$ ,

$$Z_o \approx R_C \quad (6-92)$$

The output impedance with  $R_L$ ,

$$Z_o = R_C \parallel R_L$$

**$A_i$ :** The input current will be divided between  $R_E$  and  $Z_e$ . Using voltage divider rule,

$$i_e = \frac{i_{in}R_E}{R_E + Z_e} \quad \text{or,} \quad \frac{i_e}{i_{in}} = \frac{R_E}{R_E + Z_e} \quad (6-93)$$

The current  $h_{fb}i_e$  will be divided among  $r_o$ ,  $R_C$ , and  $R_L$ . The current flowing through  $R_L$  is considered as the output current. Neglecting  $r_o$ , as it is very large, the output current can be calculated as,

$$i_o = \frac{h_{fb}i_e R_C}{R_C + R_L} \quad \text{or,} \quad \frac{i_o}{i_e} = \frac{h_{fb}R_C}{R_C + R_L} \quad (6-94)$$

The current gain can be calculated as,

$$A_i = \frac{i_o}{i_{in}} = \frac{i_e}{i_{in}} \times \frac{i_o}{i_e}$$

Using Equ.(6-93) and Equ.(6-94)

$$A_i = \frac{R_E}{R_E + Z_e} \times \frac{h_{fb}R_C}{R_C + R_L} \quad (6-95)$$

For common-base amplifier  $R_E \gg Z_e$

$$\therefore A_i \approx \frac{h_{fb} R_C}{R_C + R_L} = h_{fb} R'_L \quad (6-96)$$

Again if  $R_C \gg R_L$ , Equ.(6-96) reduces to

$$A_i = h_{fb} \approx 1 \quad (6-97)$$

### Example 6-10

For the CB amplifier of Fig.6-40, calculate  $A_v$ ,  $Z_i$ ,  $Z_o$ , and  $A_i$ . The transistor used in this circuit has  $h_{ib} = 20 \Omega$ ,  $h_{fb} = 0.995$ ,  $h_{rb} = 3 \times 10^{-5}$ , and  $h_{ob} = 0.5 \mu S$ . Assume all the capacitors are short for the operating frequency range of the amplifier.

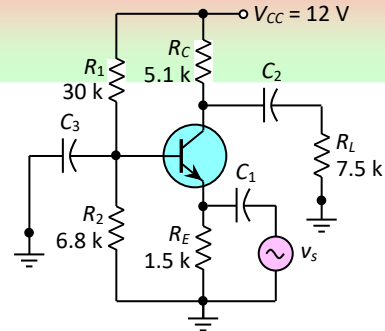


Fig.6-39: CB amplifier with voltage divider bias for Example 6-7.

#### Solution:

The value of AC load resistance and output impedance  $r_o$  will be,

$$R'_L = R_C \parallel R_L = 5.1 \text{ k} \parallel 7.5 \text{ k} \approx 3 \text{ k}$$

$$r_o = \frac{1}{h_{ob}} = \frac{1}{0.5 \times 10^{-6} \text{ S}} = 2 \text{ M}\Omega$$

As  $r_o$  is very large, it will be neglected in this calculation.

$$\text{Using Equ.(6-86),} \quad A_v \approx \frac{h_{fb}}{h_{ib}} \left( \frac{R_C R_L}{R_C + R_L} \right) = \frac{0.998}{20 \times 10^{-3} \text{ k}\Omega} \left( \frac{5.1 \text{ k}\Omega \times 7.5 \text{ k}\Omega}{5.1 \text{ k}\Omega + 7.5 \text{ k}\Omega} \right) \approx 151$$

$$\text{Using Equ.(6-90),} \quad Z_i \approx h_{ib} = 20 \Omega$$

$$\text{The output impedance with } R_L \text{ is,} \quad Z_o = R_C \parallel R_L = R'_L = 3 \text{ k}\Omega$$

$$\text{Using Equ.(6-96),} \quad A_i \approx \frac{h_{fb} R_C}{R_C + R_L} = \frac{0.995 \times 5.1 \text{ k}\Omega}{5.1 \text{ k}\Omega + 7.5 \text{ k}\Omega} \approx 0.41$$

**Results:**  $A_v = 151$ ,  $Z_i = 20 \Omega$ ,  $Z_o = 3 \text{ k}\Omega$  and  $A_i = 0.41$

**Comments:** Circuit current gain is much less than the device current gain  $h_{ib}$  or  $\alpha$ .

## 6.14 Analysis of Common-Base Amplifier with unbypassed Base Resistor

In the common-base amplifier with voltage divider bias, a capacitor is connected from the base terminal to the ground. If this capacitor is not used as shown in Fig.6-41(a), the bias resistor

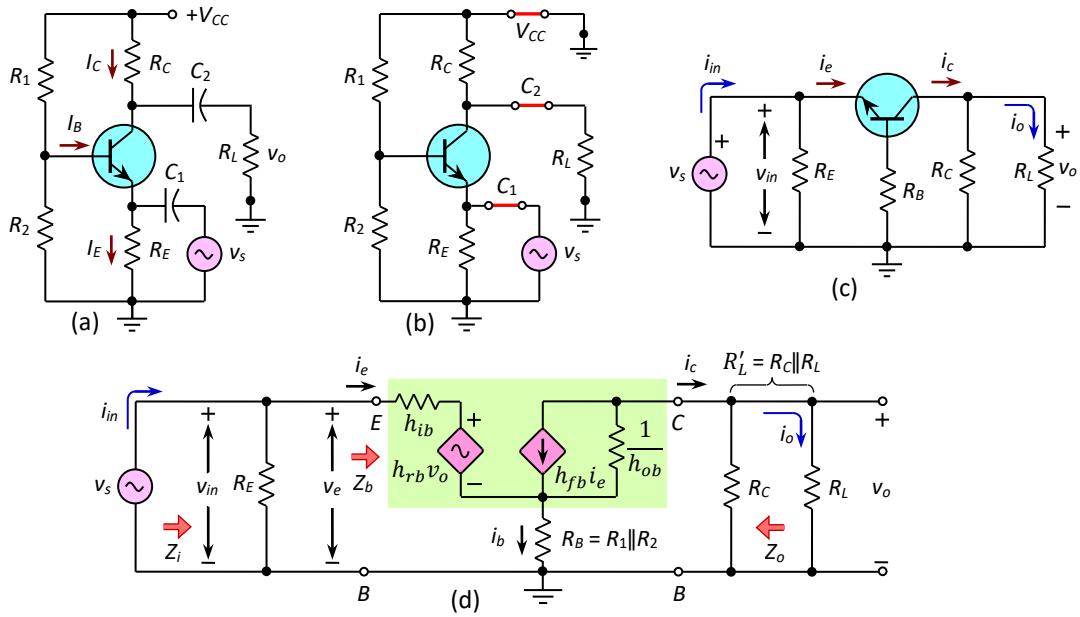


Fig.6-40: (a) CB amplifier with unbypassed base resistor, (b) AC equivalent circuit, (c) Alternate representation of the same circuit, and (d) Hybrid equivalent circuit.

$R_B = R_1 \parallel R_2$  will come in series at the base terminal which is shown in Fig.6-41(b) and (c). This unbypassed resistor will affect the performance of the amplifier. The hybrid equivalent circuit for this amplifier is shown in Fig.6-41(d).

**$A_v$ :** The output voltage of the amplifier can be calculated as

$$v_o = i_o R_L = i_c (R_C \parallel R_L)$$

$$\text{Let } R_C \parallel R_L = R'_L \text{ and } r_o = \frac{1}{h_{ob}} \quad \therefore v_o = i_c R'_L = h_{fb} i_e (r_o \parallel R'_L) \quad (6-98)$$

$$\text{We know that, for CB amplifier, } h_{fb} = \frac{i_c}{i_e} \quad \text{or, } i_c = h_{fb} i_e$$

$$\text{From } i_e = i_c + i_b \quad i_b = i_e - i_c = i_e - h_{fb} i_e = i_e (1 - h_{fb}) \quad (6-99)$$

Now, the input voltage of the amplifier can be calculated as

$$v_{in} = v_e = i_e h_{ib} + h_{rb} v_o + R_B i_b$$

Putting the value of  $v_o$  and  $i_b$  from Equ.(6-98) and (6-99)

$$v_{in} = v_e = i_e h_{ib} + h_{rb} h_{fb} i_e (r_o \parallel R'_L) + R_B i_e (1 - h_{fb}) \quad (6-100)$$

Therefore, the voltage gain of the amplifier can be found dividing Equ.(6-98) by Equ.(6-100).

$$A_v = \frac{v_o}{v_{in}} = \frac{h_{fb} (r_o \parallel R'_L)}{h_{ib} + h_{rb} h_{fb} (r_o \parallel R'_L) + R_B (1 - h_{fb})} \quad (6-101)$$

As the value of reverse transfer factor,  $h_{rb}$ , is very small, it can be neglected. Thus considering  $h_{rb} = 0$ ,

$$A_v \approx \frac{h_{fb}(r_o \parallel R'_L)}{h_{ib} + R_B(1 - h_{fb})} \quad (6-102)$$

Again the value of output impedance of the transistor,  $r_o$  is very large and comes in parallel with  $R'_L$ . So neglecting  $r_o$ , the voltage gain will be

$$\therefore \quad \boxed{A_v \approx \frac{h_{fb}R'_L}{h_{ib} + R_B(1 - h_{fb})}} \quad (6-103)$$

**$Z_i$ :** From Equ.(6-100) the input impedance at the emitter terminal can be calculated as,

$$Z_e = \frac{v_e}{i_e} = h_{ib} + h_{rb}h_{fb}(r_o \parallel R'_L) + R_B(1 - h_{fb})$$

Here again, considering  $h_{rb} = 0$ , we can write

$$Z_e \approx h_{ib} + R_B(1 - h_{fb}) \quad (6-104)$$

Now, looking from the input terminal, the input impedance will be

$$Z_i = R_E \parallel Z_e = R_E \parallel [h_{ib} + R_B(1 - h_{fb})] \quad (6-105)$$

If  $R_E \gg Z_e$

$$\boxed{Z_i \approx h_{ib} + R_B(1 - h_{fb})} \quad (6-106)$$

**$Z_o$ :** To calculate the output impedance, we have to short the input source. That is  $v_{in} = 0$  and hence  $i_e$  will also be zero. Therefore, we can consider the output current source open. As,  $r_o$  is very large it can also be considered open. Thus, the value of output impedance will be,

$$\boxed{Z_o \approx R_C} \quad (6-107)$$

The output impedance with  $R_L$

$$\boxed{Z_o \approx R_C \parallel R_L = R'_L} \quad (6-108)$$

**$A_i$ :** The emitter current can be calculated using voltage divider rule,

$$i_e = \frac{i_{in}R_E}{R_E + Z_e} \quad \text{or,} \quad \frac{i_e}{i_{in}} = \frac{R_E}{R_E + Z_e} \quad (6-109)$$

The current of the dependent current source,  $h_{fb}i_e$ , will be divided among  $r_o$ ,  $R_C$ , and  $R_L$ . Now neglecting  $r_o$  (as it is very large), the output current can be calculated as,

$$i_o = \frac{h_{fb}i_eR_C}{R_C + R_L} \quad \text{or,} \quad \frac{i_o}{i_e} = \frac{h_{fb}R_C}{R_C + R_L} \quad (6-110)$$

The current gain can be calculated as,

$$A_i = \frac{i_o}{i_{in}} = \frac{i_e}{i_{in}} \times \frac{i_o}{i_e}$$

Using Equ.(6-109) and Equ.(6-110),

$$A_i = \frac{R_E}{R_E + Z_e} \times \frac{h_{fb}R_C}{R_C + R_L} \quad (6-111)$$

If  $R_E \gg Z_e$

$$\therefore \quad A_i \approx \frac{h_{fb} R_C}{R_C + R_L} \quad (6-112)$$

Again if  $R_L \gg R_C$ , Equ.(6-96) reduces to

$$\text{or,} \quad A_i = h_{fb} \approx 1 \quad (6-113)$$

### Example 6-11

For the CB amplifier of Fig.6-42, calculate  $A_v$ ,  $Z_i$ ,  $Z_o$ , and  $A_i$ . The transistor used in this circuit has  $h_{ib} = 20 \Omega$ ,  $h_{fb} = 0.995$ ,  $h_{rb} = 3 \times 10^{-5}$ , and  $h_{ob} = 0.5 \mu S$ . Assume all the capacitors are short for the operating frequency range of the amplifier.

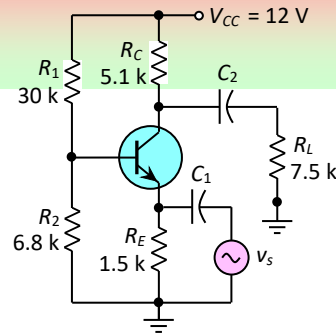


Fig.6-41: CB amplifier with voltage divider bias for Example 6-8.

### Solution:

The value of bias resistance is,

$$R_B = R_1 \parallel R_2 = 30 \text{ k} \parallel 6.8 \text{ k} \approx 5.5 \text{ k}$$

The value of AC load resistance is,  $R'_L = R_C \parallel R_L = 5.1 \text{ k} \parallel 7.5 \text{ k} \approx 3 \text{ k}$

and the value of output resistance is,  $r_o = \frac{1}{h_{ob}} = \frac{1}{0.5 \times 10^{-6} \text{ S}} = 2 \text{ M}\Omega$

As  $r_o$  is very large, it will be neglected in this calculation. Therefore,

$$\text{Using Equ.(6-103),} \quad A_v \approx \frac{h_{fb} R'_L}{h_{ib} + R_B(1 - h_{fb})} = \frac{0.995 \times 3 \text{ k}}{20 \times 10^{-3} + 5.5 \text{ k}(1 - 0.995)} \approx 62$$

$$\text{Using Equ.(6-104),} \quad Z_e = h_{ib} + R_B(1 - h_{fb}) = 20 \times 10^{-3} + 5.5 \text{ k}(1 - 0.995) \approx 102 \Omega$$

$$\text{Using Equ.(6-105),} \quad Z_i = R_E \parallel Z_e \approx Z_e = 102 \Omega$$

$$\text{The output impedance with } R_L \quad Z_o = R_C \parallel R_L = R'_L = 3 \text{ k}\Omega \quad [\text{from Equ.(6-108)}]$$

$$\text{From Equ.(6-112),} \quad A_i \approx \frac{h_{fb} R_C}{R_C + R_L} = h_{fb} R'_L = \frac{0.985 \times 5.1 \text{ k}}{5.1 \text{ k} + 7.5 \text{ k}} \approx 0.41$$

**Results:**  $A_v = 62$ ,  $Z_i = 102 \Omega$ ,  $Z_o = 3 \text{ k}\Omega$  and  $A_i = 0.41$

**Comments:** Due to the unbypassed base resistance, the voltage gain of the amplifier has decreased and the input impedance has increased. But the output impedance and current gain remain same as the circuit with no resistor in the base.

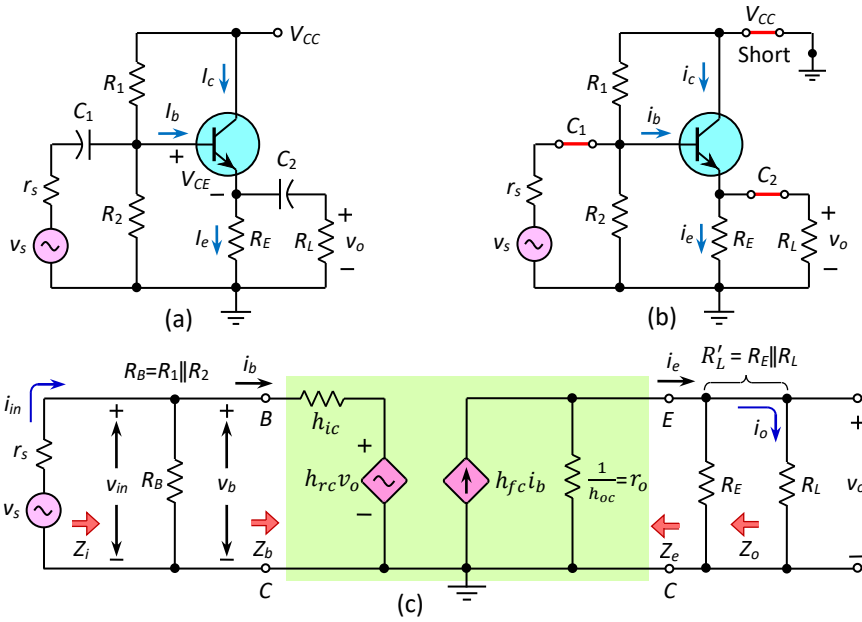


Fig.6-42: (a) CC amplifier with voltage divider bias, (b) AC equivalent circuit, and (c) AC equivalent circuit with transistor hybrid model.

### 6.15 Common-Collector Amplifier

A single-stage common-collector amplifier is shown in Fig.6-43(a). Already we have discussed the operation of this circuit. Here, also, the AC equivalent circuit of the amplifier is drawn by shorting the capacitors and DC voltage sources which is given in Fig.6-43(b). Finally replacing the transistor by its hybrid equivalent model, the AC equivalent circuit of Fig.6-43(c) is found.

$A_v$ : The output voltage of the amplifier can be calculated as

$$v_o = i_o R_L = i_e (R_E \parallel R_L)$$

$$\text{Let } R_E \parallel R_L = R'_L \text{ and } r_o = \frac{1}{h_{oc}} \quad \therefore v_o = i_c R'_L = h_{fc} i_b (r_o \parallel R'_L) \quad (6-114)$$

The input voltage of the amplifier can be determined as

$$v_{in} = v_b = i_b h_{ic} + h_{rc} v_o \quad (6-115)$$

Putting the value of  $v_o$  from Equ.(6-114) into Equ.(6-115),

$$v_{in} = v_b = i_b h_{ic} + h_{rc} h_{fc} i_b (r_o \parallel R'_L) \quad (6-116)$$

The voltage gain of the amplifier will be found by dividing Equ.(6-114) by Equ.(6-116).

$$A_v = \frac{v_o}{v_{in}} = \frac{h_{fc} (r_o \parallel R'_L)}{h_{ic} + h_{rc} h_{fc} (r_o \parallel R'_L)}$$

The second term in the denominator of the above equation is due to the effect of output voltage to the input side. The value of the reverse transfer factor ( $h_{rc}$ ) is approximately 1.

$$\therefore A_v \approx \frac{h_{fc}(r_o \parallel R'_L)}{h_{ic} + h_{fc}(r_o \parallel R'_L)} \quad (6-117)$$

Again the value of output impedance of the transistor,  $r_o$  is very large and comes in parallel with  $R'_L$ . So neglecting  $r_o$ , the voltage gain will be,

$$\therefore A_v \approx \frac{h_{fc}R'_L}{h_{ic} + h_{fc}R'_L} \quad (6-118)$$

If the external load  $R_L$  is not used, the voltage gain will be,

$$A_v = \frac{h_{fc}R_E}{h_{ic} + h_{fc}R_E} \approx 1 \quad [\because h_{ic} \ll h_{fc}R_E] \quad (6-119)$$

$Z_i$ : From Equ.(6-116), the input impedance at the base terminal can be calculated as

$$Z_b = \frac{v_b}{i_b} = h_{ic} + h_{rc}h_{fc}(r_o \parallel R'_L)$$

Here again, considering  $h_{rc} = 1$ , and  $r_o = \infty$ , we can write

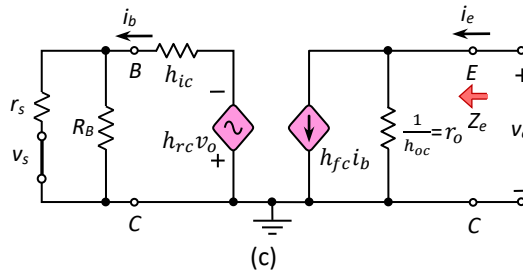
$$Z_b \approx h_{ic} + h_{fc}R'_L \quad (6-120)$$

Now, looking from the input terminal, the input impedance will be,

$$Z_i = R_B \parallel Z_b = R_B \parallel (h_{ic} + h_{fc}R'_L) \quad (6-121)$$

$$\text{If } h_{ic} \ll h_{fc}R'_L, \text{ we can neglect } h_{ic}. \quad \therefore Z_i = R_B \parallel h_{fc}R'_L \quad (6-122)$$

$Z_o$ : To calculate the output impedance, we have to short the input source, that is  $v_s = 0$ . Then we will apply a voltage, say  $v_o$  to the output terminal and calculate the output current. By dividing the applied voltage by the current the impedance is calculated. The situation is shown in Fig. As shown in this figure, due to the feedback of the output voltage, an input current  $i_b$  will be produced.



Applying KVL in the input circuit, we get,

$$r_s \parallel R_B i_b + h_{ic} i_b - h_{rc} v_o = 0$$

$$\text{or,} \quad i_b = \frac{h_{rc} v_o}{r_s \parallel R_B + h_{ic}}$$

Now applying KCL at point E of the output circuit we get,

$$i_e = \frac{v_o}{r_o} + h_{fc} i_b$$

Putting the value of  $i_b$ ,

$$i_e = \frac{v_o}{r_o} + \frac{h_{fc} h_{rc} v_o}{r_s \parallel R_B + h_{ic}}$$

or,

$$i_e = v_o \left( \frac{1}{r_o} + \frac{h_{fc} h_{rc}}{r_s \parallel R_B + h_{ic}} \right)$$

$$\therefore Z_e = \frac{v_o}{i_e} = \frac{1}{1/r_o + h_{fc} h_{rc} / (r_s \parallel R_B + h_{ic})}$$

For common collector amplifier  $h_{rc} \approx 1$ , and  $1/r_o \approx 0$  as  $r_o$  is very large. Therefore, Equ. can be approximately written as,

$$Z_e = \frac{v_o}{i_e} \approx \frac{1}{h_{fc} / (r_s \parallel R_B + h_{ic})} = \frac{r_s \parallel R_B + h_{ic}}{h_{fc}}$$

Therefore, the output impedance of the amplifier will be,

$$Z_o = Z_e \parallel R_E = \left( \frac{r_s \parallel R_B + h_{ic}}{h_{fc}} \right) \parallel R_E \quad (6-123)$$

$A_i$ : The input current will be divided between  $R_B$  and  $Z_b$ . Using voltage divider rule, we get

$$i_b = \frac{i_{in} R_B}{R_B + Z_b} \quad \text{or,} \quad \frac{i_b}{i_{in}} = \frac{R_B}{R_B + Z_b} \quad (6-124)$$

The current  $h_{fc} i_b$  will be divided among  $r_o$ ,  $R_E$ , and  $R_L$ . The current flowing through  $R_L$  is considered as the output current. Now neglecting  $r_o$  as it is very large, the output current can be calculated using current divider rule.

$$i_o = \frac{h_{fc} i_b R_E}{R_E + R_L} \quad \text{or,} \quad \frac{i_o}{i_b} = \frac{h_{fc} R_E}{R_E + R_L} \quad (6-125)$$

The current gain of the amplifier can be calculated as,

$$A_i = \frac{i_o}{i_{in}} = \frac{i_b}{i_{in}} \times \frac{i_o}{i_b}$$

Using Equ.(6-124) and Equ.(6-125),

$$A_i = \frac{R_B}{R_B + Z_b} \times \frac{h_{fc} R_E}{R_E + R_L} \quad (6-126)$$

If  $R_B \gg Z_b$ , then  $Z_b$  can be neglected.

$$\therefore A_i \approx \frac{h_{fc} R_E}{R_E + R_L} \quad (6-127)$$

Again if  $R_E \gg R_L$ , or if  $R_L$  is not connected, Equ.(6-127) reduces to

$$A_i = h_{fc} \quad (6-128)$$

**Example 6-12**

For the CC amplifier of Fig. 6-43, calculate  $A_v$ ,  $Z_i$ ,  $Z_o$ , and  $A_i$ . The transistor used in this circuit has  $h_{ic} = 3 \text{ k}\Omega$ ,  $h_{fc} = 150$ ,  $h_{rc} = 1$ , and  $h_{oc} = 10 \text{ }\mu\text{S}$ . Assume all the capacitors are short for the operating frequency range of the amplifier.

**Solution:**

Value of bias resistance is,

$$R_B = R_1 \parallel R_2 = (30 \text{ k}) \parallel (43 \text{ k}) = 17.7 \text{ k}$$

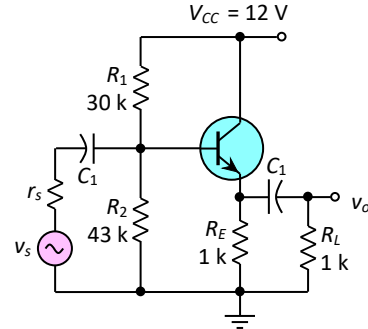


Fig.6-43: CC amplifier with voltage divider bias for Example 6-9.

Value of AC load resistance is,

$$R'_L = R_E \parallel R_L = 1 \text{ k} \parallel 1 \text{ k} = 0.5 \text{ k}$$

$$\text{Using Equ.(6-118), } A_v \approx \frac{h_{fc} R'_L}{h_{ic} + h_{fc} R'_L} = \frac{150 \times 0.5 \text{ k}}{3 \text{ k} + 150 \times 0.5 \text{ k}} \approx 0.96$$

$$\text{Using Equ.(6-121), } Z_i = R_B \parallel (h_{ic} + h_{fc} R'_L) = (17.7 \text{ k}) \parallel (3 \text{ k} + 150 \times 0.5 \text{ k}) \approx 2.98 \text{ k}$$

As  $r_o$  is very large, we can neglect it. So

$$Z_o \approx R'_L = 0.5 \text{ k}$$

[from Equ. Error!  
Reference source not  
found.]

$$\text{Using Equ.(6-127), } A_i \approx \frac{h_{fc} R_E}{R_E + R_L} = \frac{150 \times 1 \text{ k}}{1 \text{ k} + 1 \text{ k}} = 75$$

**Results:**  $A_v = 0.96$ ,  $Z_i = 2.98 \text{ k}\Omega$ ,  $Z_o = 0.5 \text{ k}\Omega$  and  $A_i = 75$

**Comments:** Common-collector amplifier provides high current gain but unity voltage gain, high input impedance and low output impedance.

## 6.16 Common-collector Amplifier with Darlington Pair

In Chapter 5, we have discussed that a Darlington pair has a very large current gain. Darlington transistor is generally used as common-collector amplifier. Fig.6-45(a) shows such a common-

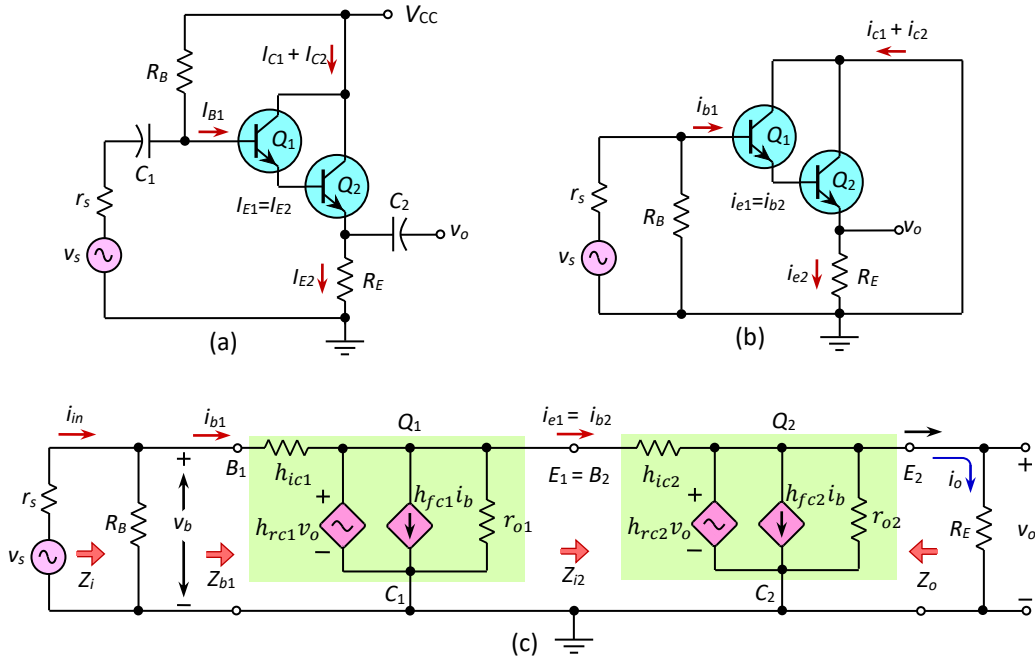


Fig.6-44: (a) Common-collector amplifier with Darlington pair transistor, (b) AC equivalent circuit, and (c) AC equivalent circuit with hybrid model of the transistor.

collector amplifier with Darlington transistor. The hybrid equivalent current is shown in Fig.6-45(c). Two transistors constitute two common-emitter amplifiers. The analysis of the 2<sup>nd</sup> stage CC amplifier is exactly same as that of single stage CC amplifier. So the voltage gain, input impedance, output impedance, and current gain of this stage will be found from the previous analysis.

Using Equ.(6-119),

$$A_{v2} = \frac{h_{fc2}R_E}{h_{ic2} + h_{fc2}R_E} \quad (6-129)$$

In the previous topic, we have found the input impedance of CC amplifier is given by

$$Z_i = R_B || h_{fc}R'_L$$

But, in the 2<sup>nd</sup> stage there is no biasing resistor, and no load resistor. So  $R_B = \infty$  and  $R'_L = R_E$

$$\therefore Z_{i2} \approx h_{fc2}R_E \quad (6-130)$$

The output impedance of this stage will be the output impedance of the whole amplifier. As no load resistor is connected, here  $R_E$  will work as the load. So the output impedance can be calculated using Equ. **Error! Reference source not found.**,

$$Z_{o2} = R_E = Z_o \quad (6-131)$$

Current gain of a single stage CC amplifier is given by Equ.(6-126) which is,

$$A_i = \frac{R_B}{R_B + Z_b} \times \frac{h_{fc}R_E}{R_E + R_L}$$

For the 2<sup>nd</sup> stage, there is no  $R_B$  and there is no  $R_L$ . Considering  $R_B = \infty$  and  $R_L = \infty$  the above equation reduces to

$$A_{i2} = h_{fc2} \quad (6-132)$$

The input impedance of 2<sup>nd</sup> stage,  $Z_{i2}$ , will work as the load for the 1<sup>st</sup> stage. Moreover, for the 1<sup>st</sup> stage we have to consider the bias resistor  $R_B$ . Therefore, the voltage gain, input impedance, output impedance, and current gain of this stage will be:

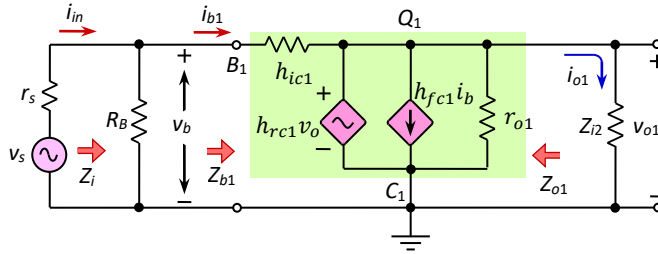


Fig.6-45:  $Z_{i2}$  of second stage works as the load of the first stage

$$A_{v1} = \frac{h_{fc1}Z_{i2}}{h_{ic1} + h_{fc1}Z_{i2}} = \frac{h_{fc1}h_{fc2}R_E}{h_{ic1} + h_{fc1}h_{fc2}R_E} \quad [\because Z_{i2} \approx h_{fc2}R_E]$$

The input impedance of the 1<sup>st</sup> stage will be that of the overall circuit. Its value will be,

$$\therefore Z_{i1} = R_B \parallel h_{fc1}Z_{i2} = R_B \parallel h_{fc1}h_{fc2}R_E \quad (6-133)$$

The output impedance of 1<sup>st</sup> stage will be

$$Z_{o1} = r_{o1}$$

The current gain of a CC amplifier is given by

$$A_i = \frac{R_B}{R_B + Z_{b1}} \times \frac{h_{fc}R_E}{R_E + R_L}$$

For the 1<sup>st</sup> stage, there is no extra  $R_L$ . So, the current gain will be,

$$A_{i1} = \frac{h_{fc1}R_B}{R_B + Z_{b1}} \quad (6-134)$$

The value of  $Z_{b1}$  will be  $Z_{b1} = h_{ic1} + h_{fc1}Z_{i2} = h_{ic1} + h_{fc1}h_{fc2}R_E \approx h_{fc1}h_{fc2}R_E$

Putting the value of  $Z_{b1}$  into Equ.(6-134),  $A_{i1} = \frac{h_{fc1}R_B}{R_B + h_{fc1}h_{fc2}R_E} \quad (6-135)$

Now combining the voltage and current amplification of two stages, the overall gain will be calculated.

**Voltage gain:** 
$$A_v = A_{v1} \times A_{v2} = \frac{h_{fc1}h_{fc2}R_E}{h_{ic1} + h_{fc1}h_{fc2}R_E} \times \frac{h_{fc2}R_E}{h_{ic2} + h_{fc2}R_E} \quad (6-136)$$

As  $h_{fc1}h_{fc2}R_E \gg h_{ic1}$  and  $h_{fc2}R_E \gg h_{ic2}$ , overall voltage gain of the amplifier will be,

$$A_v \approx \frac{h_{fc1}h_{fc2}R_E}{h_{fc1}h_{fc2}R_E} \times \frac{h_{fc2}R_E}{h_{fc2}R_E} = 1 \quad (6-137)$$

**Input impedance:**  $\therefore Z_i = Z_{i1} = R_B \parallel h_{fc1}Z_{i2} = R_B \parallel h_{fc1}h_{fc2}R_E$  (6-138)

**Output impedance:**  $Z_o = Z_{o2} = R_E$  (6-139)

**Current gain:**  $A_i = A_{i1} \times A_{i2} = \frac{h_{fc1}h_{fc2}R_B}{R_B + h_{fc1}h_{fc2}R_E}$  (6-140)

**Example 6-13**

For the CC amplifier of Fig. 6-46, calculate  $A_v$ ,  $Z_i$ ,  $Z_o$ , and  $A_i$ . The transistors used in this circuit are identical and has  $h_{ic} = 1 \text{ k}\Omega$ ,  $h_{fc} = 25$ ,  $h_{rc} = 1$ , and  $h_{oc} = 10 \mu\text{S}$ . Assume all the capacitors are short for the operating frequency range of the amplifier.

**Solution:**

Using Equ.(6-136), the voltage gain is,

$$A_v = \frac{h_{fc1}h_{fc2}R_E}{h_{ic1} + h_{fc1}h_{fc2}R_E} \times \frac{h_{fc2}R_E}{h_{ic2} + h_{fc2}R_E}$$

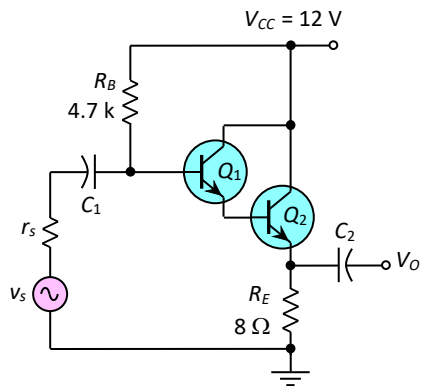


Fig.6-46: CC amplifier with voltage divider bias for Example 6-10.

$$A_v = \frac{25 \times 25 \times 8 \times 10^{-3}\text{k}}{1 \text{ k} + 25 \times 25 \times 8 \times 10^{-3}\text{k}} \times \frac{25 \times 8 \times 10^{-3}\text{k}}{1 \text{ k} + 25 \times 8 \times 10^{-3}\text{k}} \approx 0.14$$

Using Equ.(6-138),  $Z_i = Z_{i1} = R_B \parallel h_{fc1}h_{fc2}R_E$

$$= 4.7\text{k} \parallel (25 \times 25 \times 8 \times 10^{-3}\text{k}) \approx 2.4 \text{ k}$$

Using Equ.(6-139),  $Z_o = R_E = 8 \Omega$

Current gain is calculated using Equ.(6-140).

$$A_i = \frac{h_{fc1}h_{fc2}R_B}{R_B + h_{fc1}h_{fc2}R_E} = \frac{25 \times 25 \times 4.7 \text{ k}}{4.7 \text{ k} + 25 \times 25 \times (8 \times 10^{-3})\text{k}} \approx 302$$

**Results:**  $A_v = 0.14$ ,  $Z_i = 2.4 \text{ k}\Omega$ ,  $Z_o = 8 \Omega$  and  $A_i = 302$

**Comments:** As the load resistance is very small, the current gain of the circuit is very large and the voltage gain is very small.

## 6.17 The Miller Theorem and Miller Effect

Before going to the hybrid- $\pi$  model, we will discuss the *Miller theorem* and the *Miller effect*, because this theorem will be used in this model. The name of this theorem is according to **J. J. Miller** who introduced the theorem in 1969.

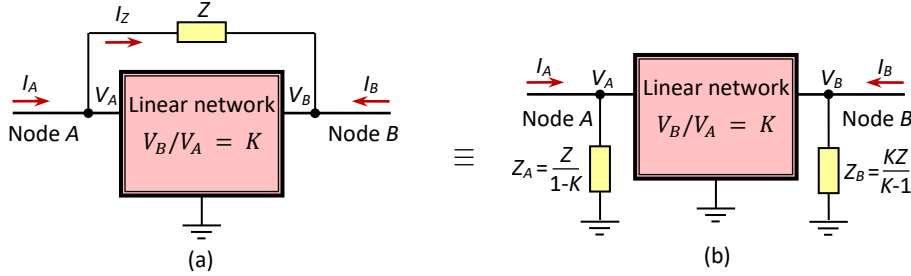


Fig.6-48: Miller Theorem: (a) A linear network with a floating impedance,  $Z$  (b) Impedance  $Z$  is split with Miller theorem

### 6.17.1 The Miller Theorem

The Miller theorem states that any floating impedance,  $Z$ , connected between two nodes (node  $A$  and  $B$  of Fig.6-) of a linear circuit, can be split into two grounded impedances. If the voltages of the nodes are  $V_A$  and  $V_B$ , respectively, then the impedance seen at node  $A$  will be  $Z/(1 - K)$  and the impedance seen at node  $B$  will be  $KZ/(K - 1)$ , where  $K = V_B/V_A$ .

**Proof:**

The value of current flowing through  $Z$  from node  $A$  to node  $B$  will be,

$$I_Z = \frac{V_A - V_B}{Z}$$

After splitting the impedance, the value of current flowing through  $Z_A$  from node  $A$  to ground is,

$$I_{ZA} = \frac{V_A}{Z_A}$$

At node  $A$ , the effect of  $Z_A$  will be same if the values of two currents  $I_Z$  and  $I_{ZA}$  are same. That is,

$$\frac{V_A - V_B}{Z} = \frac{V_A}{Z_A} \quad \text{or,} \quad Z_A = \frac{ZV_A}{V_A - V_B}$$

$$\therefore \quad Z_A = \frac{Z}{1 - \frac{V_B}{V_A}} = \frac{Z}{1 - K} \quad (6-141)$$

Similarly, the value of current flowing through  $Z$  from node  $B$  to node  $A$  will be,

$$I_Z = \frac{V_B - V_A}{Z}$$

The value of current flowing through  $Z_B$  from node  $B$  to ground is,

$$I_{ZB} = \frac{V_B}{Z_B}$$

To get the same effect before and after splitting  $Z$  will be,

$$\frac{V_B - V_A}{Z} = \frac{V_B}{Z_B} \quad \text{or,} \quad Z_B = \frac{ZV_B}{V_B - V_A}$$

$\therefore$

$$Z_B = \frac{Z}{1 - \frac{V_A}{V_B}} = \frac{Z}{1 - 1/K} = \frac{KZ}{K - 1} \quad (6-142)$$

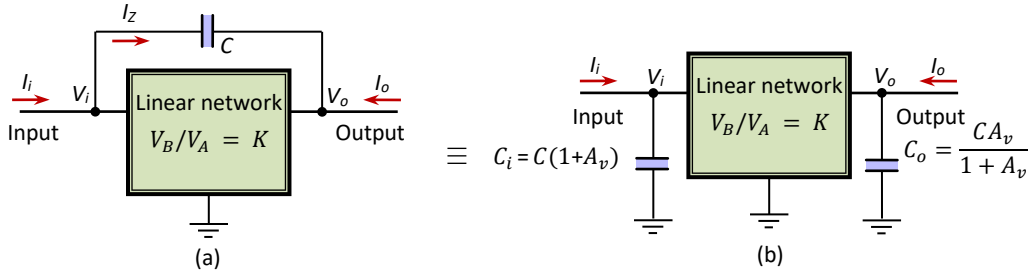


Fig.6-49: Miller Effect: (a) A linear network with a floating impedance,  $Z$  (b) Impedance  $Z$  is split with Miller theorem

### 6.17.2 The Miller Effect

The **Miller effect** accounts for the increase in the **equivalent input capacitance** of an inverting voltage **amplifier** due to amplification of the effect of capacitance between the input and output terminals. The virtually increased input capacitance due to the Miller effect is given by  $C(1 + A_v)$ , where,  $C$  is the capacitance connected between the input and output terminal of an amplifier with gain  $-A_v$ . To prove this, let us apply the Miller theorem in the amplifier of Fig.6-49. The value of  $C_i$  can be calculated using Equ.(6-141), as,

$$\frac{1}{j\omega C_i} = \frac{1/j\omega C}{1 - (-A_v)} \quad [\because K = -A_v]$$

or,

$$\frac{1}{j\omega C_i} = \frac{1}{j\omega C(1 + A_v)}$$

$\therefore$

$$C_i = C(1 + A_v) \quad (6-143)$$

Therefore, according to the Miller effect, in a voltage amplifier with  $180^\circ$  phase difference, any capacitance connected between the input and output will appear as a very large capacitance at the input. The multiplying factor,  $(1 + A_v)$ , is called **Miller multiplier**. This capacitance is also called the **Miller capacitance** and is denoted as  $C_M$ .

Similarly, the value of capacitance seen at the output terminal can be calculated using Equ.(6-142) as,

$$\frac{1}{j\omega C_o} = \frac{-A_v(1/j\omega C)}{-A_v - 1} \quad [\because K = -A_v]$$

$$\text{or,} \quad \frac{1}{j\omega C_o} = \frac{A_v}{j\omega C(1 + A_v)}$$

$$\therefore \quad \boxed{C_o = \frac{C(1 + A_v)}{A_v} \approx C} \quad (6-144)$$

Therefore, when a floating capacitor is seen from the output side, its value will remain almost same.

### 6.17.3 Physical Significance of Miller Effect

As shown in Fig.6-1, with an increase in input voltage say  $\Delta v_i$  the output voltage will be decreased by  $\Delta v_i A_v$ . Thus, the resultant voltage change across the capacitor,  $C$ , will be  $(\Delta v_i + \Delta v_i A_v)$ . Due to this voltage change, the charge drawn by the capacitor from input terminal will be,

$$Q_C = C(\Delta v_i + \Delta v_i A_v) = C\Delta v_i (1 + A_v) \quad (6-145)$$

But, if the capacitor were connected to ground from the input terminal, the voltage change would have been only,  $\Delta v_i$ . Then the charge taken by the capacitor would have been only,

$$Q_{Ci} = C\Delta v_i \quad (6-146)$$

By comparing Equ.(6-145) and Equ.(6-146), we get the multiplying factor  $(1 + A_v)$ . Actually, the capacitance of  $C$  appears larger at input terminal due to the negative voltage gain of the amplifier.

#### Example 6-14

Fig.6-37 shows the block diagram of a CE amplifier with the base-collector capacitance,  $C_{bc} = 10$  pF. Considering the voltage gain of the amplifier  $A_v = -150$ , split the  $C_{bc}$  using the Miller effect.

**Solution:**

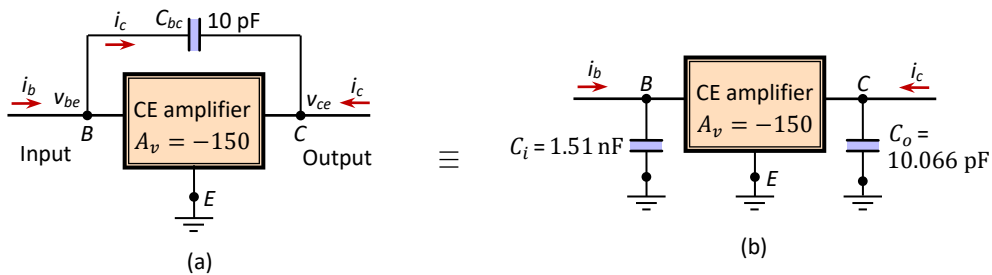


Fig.6-50: (a) CE amplifier with base-collector capacitance, and (b)  $C_{bc}$  is split using Miller theorem

Using Equ.(6-143),  $C_i = C(1 + A_v) = 10 \text{ pF}(1 + 150) = 1.51 \text{ nF}$  [Ans.]

Using Equ.(6-144),  $C_o = \frac{C(1 + A_v)}{A_v} = \frac{10 \text{ pF}(1 + 150)}{150} = 10.066 \text{ pF}$  [Ans.]

**Comments:** The parasitic capacitance ( $C_{bc}$ ) increases when transferred to input side but the output capacitance is approximately equal to  $C_{bc}$ .

#### 6.17.4 The Miller effect is not always a nuisance

As we will discuss later, Miller effect reduces the bandwidth of amplifiers. But in some cases, we can use Miller effect purposefully. It may be exploited to synthesize larger capacitors from smaller ones. One such example is in the stabilization of feedback amplifiers, where stability is achieved by adding capacitance. The required capacitance may be too large to include directly in the circuit, so a smaller capacitance is added that is made larger by the Miller effect. This approach is particularly important in integrated circuits, where large capacitors consume significant area, increasing costs.

#### 6.17.5 Miller Approximation

Because the Miller capacitance (MC) depends upon  $A_v$ , and  $A_v$  depends on frequency (near the cutoff regions). Thus, the Miller capacitance is also frequency dependent. Due to its frequency dependency, it becomes difficult to calculate the MC. However, in many cases, frequency dependence of  $A_v$  arises only at frequencies higher than the corner frequency caused by the Miller effect, so for frequencies up to the corner, roll-off of the gain does not occur, and  $A_v$  is approximated adequately by its low-frequency value. This approximation, the determination of MC using  $A_v$  evaluated at low frequencies, is the so-called Miller approximation. *So long as the Miller approximation is accurate, CM is frequency independent.*

### 6.18 Hybrid- $\pi$ Model of Transistor

So far, we have discussed two types of transistor models ( $r_e$  model and hybrid model). Both of them are low frequency small signal models. These models can be used for low frequencies—approximately below 1 MHz. For frequencies greater than 1 MHz the response of the transistor will be limited by its internal and external parasitic capacitances. So, in order to estimate the gain and switching (ON/OFF) times of BJT at higher frequencies, the high frequency model of BJT has to be used to get reasonably accurate results. The high frequency *hybrid- $\pi$  model* is such a model that considers transistor parasitic capacitances and provides reasonably accurate results (though not exact) at higher frequencies. If we want to get exact results, the circuit will be complicated and its analysis will be very difficult. Although designed for high frequency, hybrid- $\pi$  model can also be used for low frequencies. This model is also called *Giacoletto model* named after L. J. Giacioletto who introduced it in 1969. Hybrid- $\pi$  model is so called, because the circuit resemblances the form of  $\pi$  (pi) and its parameters used hybrid units like  $\Omega$  and Siemens.

Hybrid- $\pi$  model considers the junction capacitances and the bulk resistances of transistors. For better understanding let us consider a point  $B'$  at the middle of base material as shown in Fig.6-50(a). This point is called *internal base* or *virtual base* terminal as it is not physically accessible. Fig.6-50(a) shows the resistances between this internal base terminal and the other three terminals. Fig.6-50(b) shows different parasitic inter-electrode capacitances.

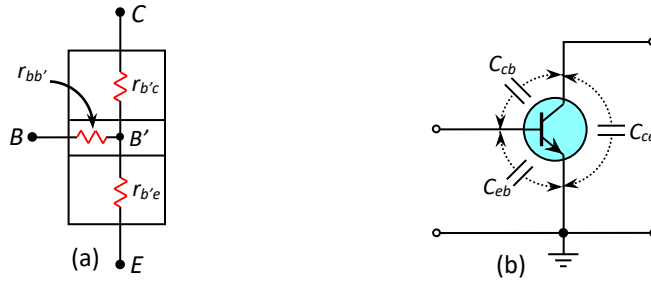


Fig.6-50: Parasitic components of a BJT: (a) Internal resistance, and (b) Inter-electrode capacitances

They include junction diffusion capacitance ( $C_D$ ), junction transition capacitance ( $C_T$ ) and capacitance between contact electrodes. However, most of the datasheets mention the values of  $C_{ibo}$ , and  $C_{obo}$ . The values of  $C_{ibo}$ , and  $C_{obo}$  are measured as shown in Fig.6-51 and Fig.6-52.

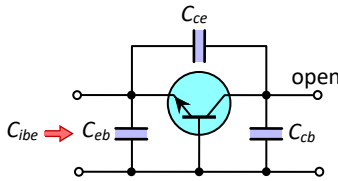


Fig.6-51: Equivalent circuit to measure

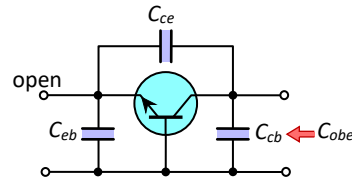


Fig.6-52: Equivalent circuit to measure

Here,  $C_{ibo}$  is called **Capacitance, Input common-Base with output Open**. The value of this capacitance can be calculated from the Fig.6-51 as,

$$C_{ibo} = C_{eb} + \frac{C_{ce} \times C_{cb}}{C_{ce} + C_{cb}} \quad (6-147)$$

This capacitance is also known as  $C_{ib}$ ,  $C_{ebo}$ , or simply  $C_e$ .

Here,  $C_{obo}$  is called **Capacitance, Output common-Base with input Open**. The value of this capacitance can be calculated from the Fig.6-52 as,

$$C_{obo} = C_{cb} + \frac{C_{ce} \times C_{eb}}{C_{ce} + C_{eb}} \quad (6-148)$$

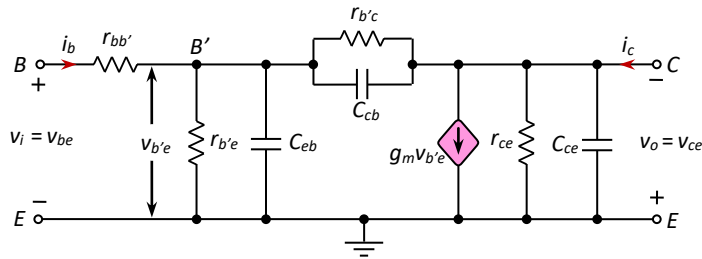
This capacitance is also known as  $C_{ob}$ ,  $C_{cbo}$ , or simply  $C_c$ .

The datasheets also mention the value of **unity gain cut-off frequency**,  $f_T$ .  $f_T$  is related with  $C_{eb}$  as,

$$C_{eb} = \frac{g_m}{2\pi f_T} - C_{ce} \approx \frac{g_m}{2\pi f_T} \quad (6-149)$$

Using the equations (6-) to (6-), the values of  $C_{eb}$ ,  $C_{cb}$ , and  $C_{ce}$  can be calculated.

With the help of Fig.6-50, and Fig.6- the high frequency equivalent circuit of a transistor in CE configuration can be drawn as shown in Fig.6-53.

Fig.6-53: Hybrid- $\pi$  model of a common-emitter configuration transistor

Where,

- $r_{bb'}$  = resistance between internal base terminal ( $B'$ ) and the external base terminal ( $B$ ). It includes the base contact, and base spreading (bulk) resistance. Its value is  $100\ \Omega$  or less. In the original hybrid- $\pi$  model, this resistance is also represented as  $r_x$ .
- $r_{b'e}$  = resistance between virtual base  $B'$  and emitter terminal,  $E$ . This is the resistance that a forward biased PN junction shows for AC signals. Its value is  $1000\ \Omega$  or less. In the original hybrid- $\pi$  model, this resistance is denoted as  $r_\pi$ .
- $r_{b'c}$  = resistance between  $B'$  and  $C$ . This resistance is due to the Early-effect as described before. With the increase of collector-emitter voltage ( $V_{CE}$ ), base width decreases and hence base current decreases and vice versa. Thus, the output voltage has an effect on the input current. This feedback effect is represented by this resistance. Its value is very high, in the range of few to few hundred  $M\Omega$ . In the hybrid- $\pi$  model, this resistance is represented as  $r_\mu$ .
- $r_{ce}$  = resistance between  $C$  and  $E$ . This is also because of the Early-effect. Due to this effect the base current decreases with the increase of output voltage. As a result the current increases. Due to this change in output current a resistance is used in parallel with the current source. Actually, it is the output resistance ( $r_o$ ) of CE amplifier and its value is  $50\ k\Omega$  to  $200\ k\Omega$ .
- $C_{eb}$  = capacitance of the forward biased base emitter junction. It includes both diffusion capacitance ( $C_D$ ) and transition capacitance ( $C_T$ ). But  $C_D$  is very larger than  $C_T$ . The value of this capacitance is  $1000\ pf$  or less. In the original hybrid- $\pi$  model, this capacitance is denoted as  $C_\pi$ .
- $C_{cb}$  = capacitance of the reverse biased base collector junction. It is mainly transition capacitance and its value is in the order of  $1\ pf$  to  $10\ pf$ . In the original hybrid- $\pi$  model, this capacitance is represented as  $C_\mu$ .
- $C_{ce}$  = the interelectrode ( $C$  and  $E$ ) capacitance. Its value is very small, so in the original hybrid- $\pi$  model this capacitance has not been included.
- $g_m$  = transconductance calculated as  $g_m = \Delta i_c / \Delta v_{b'e}$  at constant  $v_{ce}$ . Its value depends on the collector current and ranges from  $50\ mA/V$  to  $200\ mA/V$ .
- $v_{b'e}$  = small change in voltage across base emitter junction. Portion of the input AC signal that appears across the  $B$ - $E$  junction.

The values of some of the components described above can be calculated using  $h$ -parameters. The formulae are given in Table- (Notations of original hybrid- $\pi$  model are used):

| Parameter | Exact value                                        | Approximate value                                        |
|-----------|----------------------------------------------------|----------------------------------------------------------|
| $g_m$     | $g_m = \frac{I_C - I_{CO}}{V_T}$                   | $g_m \approx \frac{I_C}{V_T}$ or $\approx \frac{1}{r_e}$ |
| $r_\pi$   | $r_\pi = \frac{h_{fe}}{g_m}$                       | $r_\pi \approx \beta r_e$                                |
| $r_x$     | $r_x = h_{ie} - r_{b'e}$                           | $r_x \approx \text{few } \Omega$                         |
| $r_\mu$   | $r_\mu = \frac{r_\pi}{h_{re}}$                     | $r_\mu \approx 10^4 \times r_{b'e}$ $1/h_{oe} = r_o$     |
| $r_{ce}$  | $r_{ce} = \frac{1}{h_{oe} - h_{fe}h_{re}/r_{b'e}}$ | $r_{ce} \approx 1/h_{oe} = r_o$                          |
| $C_\pi$   | $C_\pi = \frac{g_m}{2\pi f_T} - C_{ce}$            | $C_\pi \approx \frac{g_m}{2\pi f_T}$                     |

The diagram of the original hybrid- $\pi$  model is shown in Fig.6-54. Note that, being very small valued capacitance,  $C_{ce}$  was not present in the original model. But here we will consider this capacitance.

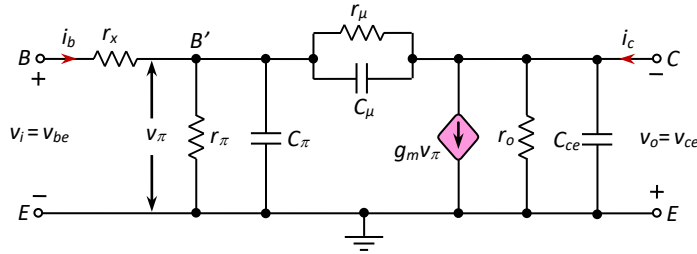


Fig.6-54: Original hybrid- $\pi$  model of a common-emitter configuration

The value of  $r_\mu$  is very large and the value of  $r_x$  is very small, so these resistances can also be neglected (short). Thus, the high frequency model of Fig.6-54 can be simplified as in Fig.6-55.

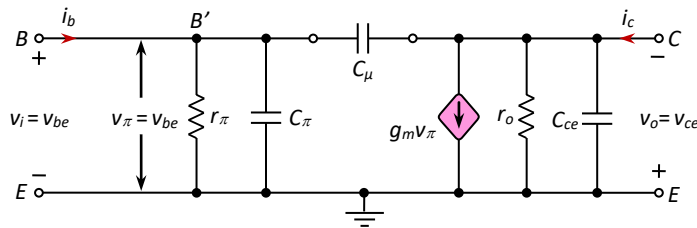


Fig.6-55: Simplified hybrid- $\pi$  model of a common-emitter configuration

For this simplified configuration, the input voltage is  $v_i = v_{be} = v_\pi$  and the output voltage is  $-g_m v_\pi r_o$ . Thus the gain of this amplifier will be

$$A_v = \frac{v_o}{v_i} = \frac{-g_m v_\pi r_o}{v_\pi} = -g_m r_o \quad (6-150)$$

Now, using the Miller theorem, capacitance  $C_\mu$  can be split into two components. The value of the Miller capacitance at the input will be  $C_{Mi} = C_\mu(1 + g_m r_o) = C_{cb}(1 + g_m r_o)$  and the value of the Miller capacitance at the output will be  $C_{Mo} \approx C_\mu = C_{cb}$ . Thus, the circuit of Fig.6-55 can be redrawn as in Fig.6-56.

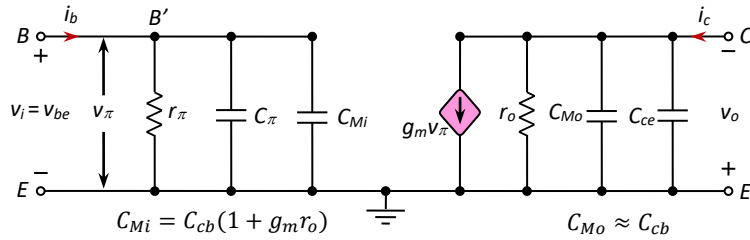


Fig.6-56: Hybrid- $\pi$  model simplified using Miller theorem

The circuit of Fig.6-56 can be simplified further as in Fig.6-57.

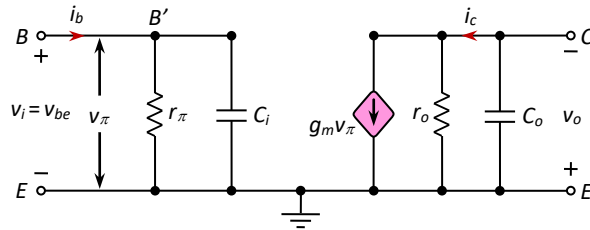


Fig.6-57: Simplified hybrid- $\pi$  model of a common-emitter

Here,

$$C_i = C_{eb} + C_{Mi} = C_{eb} + C_{cb}(1 + g_m r_o) \quad (6-151)$$

and,

$$C_o = C_{ce} + C_{Mo} = C_{ce} + C_{cb} \quad (6-152)$$

Using this model the high frequency response of CE amplifier is described in Chapter 10.

### Example 6-15

Assume that a small signal BJT has  $r_{bb'} = 50 \Omega$ ,  $r_{b'e} = r_\mu = 800 \Omega$ ,  $r_{b'c} = 2 \text{ M}\Omega$ ,  $r_{ce} = 120 \text{ k}\Omega$ ,  $r_o = 115 \text{ k}\Omega$ ,  $C_{eb} = 65 \text{ pF}$ ,  $C_{ce} = 3 \text{ pF}$ ,  $C_{cb} = C_\mu = 6 \text{ pF}$ , and  $g_m = 75 \text{ mA/V}$ . Determine and draw the hybrid- $\pi$  equivalent circuit.

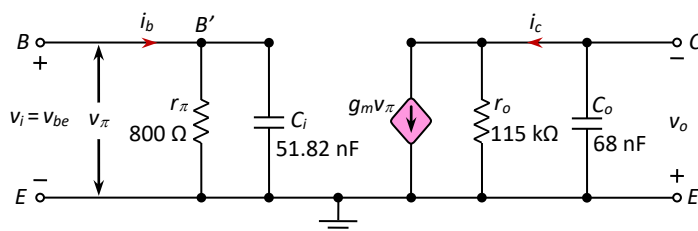
### Solution:

Using Equ.(6-150),  $A_v = -g_m r_o = -(75 \text{ mA/V})(115 \text{ k}\Omega) = -8,625$

Using Equ.(6-151),  $C_i = C_{eb} + C_{cb}(1 + g_m r_o) = 65 \text{ pF} + 6 \text{ pF}(1 + 75 \times 115)$

or,  $C_i \approx 51.82 \text{ nF}$

Using Equ.(6-152),  $C_o = C_{ce} + C_{cb} = 3 \text{ pF} + 65 \text{ pF} = 68 \text{ pF}$

Fig.6-58: Simplified hybrid- $\pi$  model of Example 6-

**Comments:** The calculated equivalent current is shown in Fig.4-. Due to the Miller effect, the input capacitance,  $C_i$  in common emitter amplifier is very large.

## 6.19 Questions

1. What do you mean by modeling? Why we need to determine the equivalent circuit of a transistor?
2. What do you mean by small signal low frequency model of a transistor?
3. Give a comparison between  $r_e$  model and hybrid ( $h$ ) model for a BJT.
4. What are the advantages and disadvantages of  $r_e$  equivalent circuit over  $h$ -equivalent circuit?
5. What are the advantages and disadvantages of  $h$  equivalent circuit over  $r_e$ -equivalent circuit?
6. Draw the  $r_e$ -equivalent circuit of transistor on CE configuration and describe.
7. How the value of  $r_e$  can be calculated?
8. What is Early Effect and Early Voltage? How the output impedance of a transistor can be calculated from this phenomenon?
9. Draw the  $r_e$ -equivalent circuit of a transistor in CB configuration and describe.
10. Draw the  $r_e$ -equivalent circuit of transistor on CC configuration and describe.
11. Draw the circuit diagram of single stage common-emitter amplifier and describe its operation. Draw the waveforms of AC signals at different points of the circuit.
12. Why we use the coupling capacitors in this amplifier?
13. What do you mean by DC and AC load line? Explain a method to draw AC load line.
14. Why the AC load-line passes through the DC operating point? – Explain.
15. Why the slope of the AC load-line is higher than the DC load-line?
16. Draw the single stage common-emitter amplifier and its AC equivalent circuit.
17. Draw the  $r_e$  equivalent circuit of a CE amplifier and determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
18. Draw the  $r_e$  equivalent circuit of a CE amplifier with fixed bias and determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
19. Draw the  $r_e$  equivalent circuit of a CE amplifier with emitter feedback bias and determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
20. Draw the  $r_e$  equivalent circuit of a CE amplifier with collector feedback bias and determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).

21. Draw the  $r_e$  equivalent circuit of a CE amplifier without emitter bypass capacitor and determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
22. Draw the single stage common-base amplifier and describe its operation.
23. Draw the single stage common-base amplifier and its AC equivalent circuit.
24. Draw the single stage common-base amplifier and its AC equivalent circuit capacitor and determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
25. Draw the single stage common-collector amplifier and describe its operation.
26. Draw the single stage common-collector amplifier and its AC equivalent circuit.
27. Draw the single stage common-collector amplifier and its AC equivalent circuit capacitor and determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
28. Give a comparison among common-emitter, common-base, and common-collector amplifiers.
29. Explain why the gain of a common-collector amplifier is less than unity.
30. What do you mean by the loading effect of an amplifier?
31. Describe the effect of source resistance ( $r_s$ ) and load resistance ( $R_L$ ) on the performance of an amplifier.
32. Define the hybrid parameters? Why the name hybrid?
33. Draw a two-port network and find out the expression for the hybrid parameters.
34. Draw the combined hybrid equivalent circuit of a two-port network and describe effect of the  $h$ -parameters on the performance of the amplifier.
35. Draw the hybrid equivalent circuit of a CE amplifier and describe
36. the effect of the parameters on the characteristics of a transistor in CEC.
37. Draw the hybrid equivalent circuit of a transistor in common-base configuration.
38. Draw the hybrid equivalent circuit of a transistor in common-collector configuration.
39. Write the expressions for converting the  $h$ -parameters of CBC into CEC.
40. Write the expressions for converting the  $h$ -parameters of CEC into CBC.
41. Write the expressions for converting the  $h$ -parameters of CEC into CCC.
42. Writes the typical values of four  $h$ -parameters of a transistor in three different configurations.
43. Draw the circuit diagram of a CE amplifier and using the hybrid equivalent circuit determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
44. Draw the circuit diagram of a CE amplifier with unbypassed emitter resistor and using the hybrid equivalent circuit determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
45. Draw the circuit diagram of a CB amplifier and using the hybrid equivalent circuit analyze it to determine voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
46. Draw the circuit diagram of a CB amplifier with unbypassed emitter resistor and using the hybrid equivalent circuit determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
47. Draw the circuit diagram of a CC amplifier and using the hybrid equivalent circuit analyze it to determine voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
48. Why we use Darlington Pair? Draw the circuit diagram of a CE amplifier with Darlington Pair and using the hybrid equivalent circuit determine its voltage gain ( $A_v$ ), input impedance ( $Z_i$ ), and output impedance ( $Z_o$ ).
49. State and explain the Miller Theorem and the Miller Effect.

50. Is there any use of the Miller Effect? Explain.
51. Describe the parasitic capacitance and resistance of a bipolar junction transistor.
52. What are  $C_{ibo}$  and  $C_{obo}$ ? Explain how these values can be determined.
53. Draw the hybrid- $\pi$  model of a transistor in common-emitter configuration and simplify it.

## 6.20 Exercise

1. i) Draw the load-line and operating point of the CE amplifier circuits of Fig.6-960.  
 ii) Determine the DC biasing condition (operating point) if the signal source is directly connected to the input (that is without using the input coupling capacitor,  $C_1$ ), and  
 iii) Determine the DC biasing condition if the  $4.7\text{ k}\Omega$  load resistance is directly connected to the output (that is without using the output coupling capacitor,  $C_2$ ).

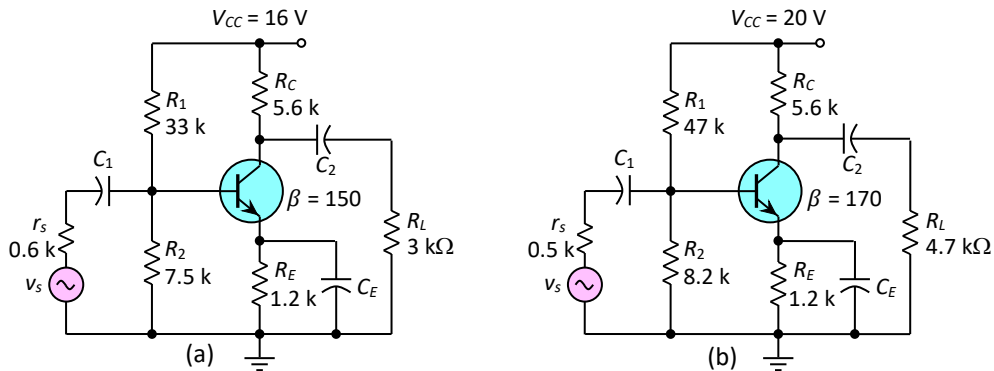


Fig.6-60: CE amplifier circuits for Exercise 1

2. For the common-emitter amplifier circuit of Fig.6-1360 determine the DC and AC load lines and operating points. Assume  $\beta = 120$  in place of the values given in the circuits.
3. For the CE amplifier of Fig.6-161, calculate  $Z_i$ ,  $Z_o$ ,  $A_v$ , and  $A_i$  using  $r_e$  model of transistor. Assume  $\beta = 150$ ,  $r_o = \infty$ , and all the capacitors are short for the operating frequency range of the amplifier.

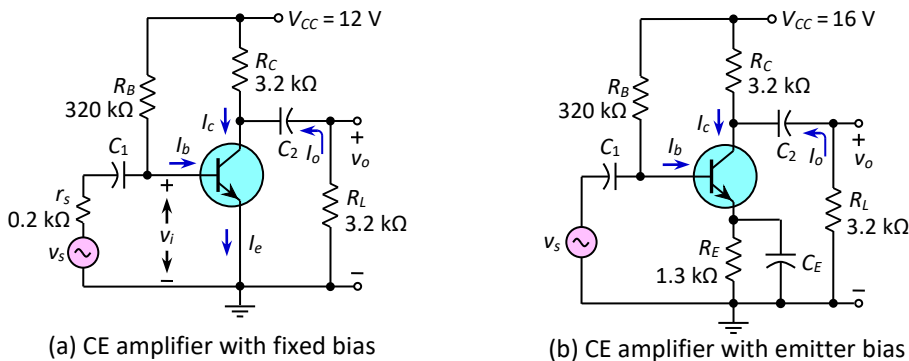
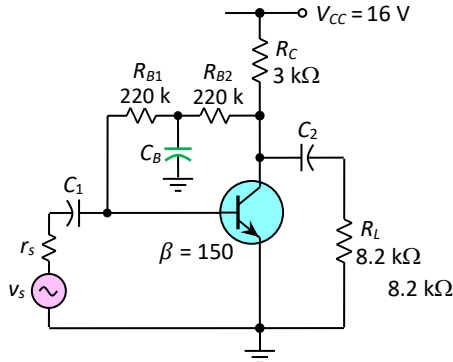
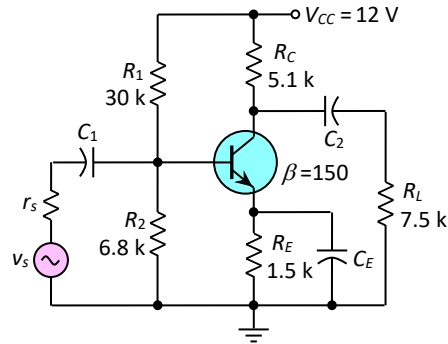


Fig.6-61: CE amplifiers for Exercise 3

4. For the CE amplifier of Fig.6-16, calculate  $Z_i$ ,  $Z_o$ ,  $A_v$ , and  $A_i$  using  $r_e$  equivalent circuit of the transistor. Assume  $\beta = 150$ ,  $r_o = \infty$ , and all the capacitors are short for the operating frequency range of the amplifier.



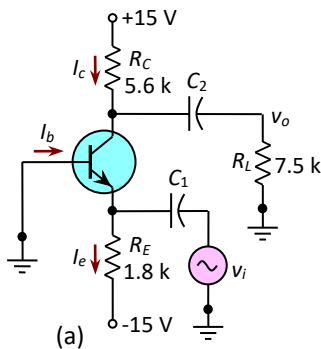
(a) CE amplifier with collector feedback bias



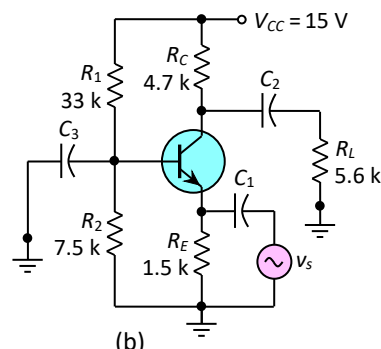
(b) CE amplifier with voltage divider bias

Fig.6-62: CE amplifiers for Exercise 4

5. For the CE amplifier of Fig.6-21, calculate  $Z_i$ ,  $Z_o$ ,  $A_v$ , and  $A_i$  using  $r_e$  model of transistor. Assume  $\beta = 145$ ,  $r_o = \infty$ , and all the capacitors are short for the operating frequency range of the amplifier.
6. For the CE amplifier of Fig.6-21(b), calculate  $Z_i$ ,  $Z_o$ ,  $A_v$ , and  $A_i$  without the emitter bypass capacitor. Assume  $\beta = 145$ ,  $r_o = \infty$ , and all the capacitors are short for the operating frequency range of the amplifier.
7. For the CB amplifier of Fig.6-27, calculate  $Z_i$ ,  $Z_o$ ,  $A_v$ , and  $A_i$ . Assume  $\beta = 150$ ,  $r_o = \infty$ , and all the capacitors are short for the operating frequency range of the amplifier.



(a)



(b)

Fig.6-63: CB amplifier with voltage divider bias for Exercise no. 7

8. For the CC amplifier of Fig.6-64, calculate  $Z_i$ ,  $Z_o$ ,  $A_v$ , and  $A_i$  using  $r_e$  model. Assume  $\beta = 150$ ,  $r_o = \infty$ , and all the capacitors are short for the operating frequency range of the amplifier.

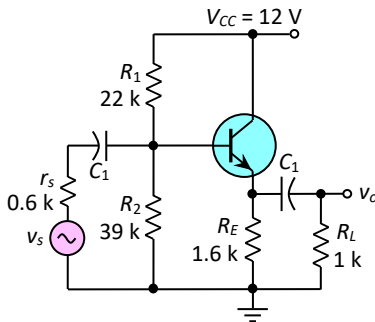


Fig.6-648: CE amplifier with voltage divider bias for Example 6-

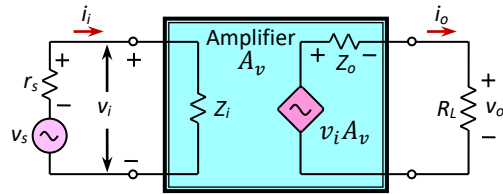


Fig.6-648: Simplified block diagram of an amplifier showing the effect of  $r_s$  and  $R_L$

9. The values of voltage gain, input impedance and output impedance of a common-base amplifier are 500, 1.2 k $\Omega$ , and 4.7 k $\Omega$ , respectively. If the internal source resistance is 600  $\Omega$  and the load resistance is 1.5 k $\Omega$ , calculate the net voltage gain of the amplifier.
10. Repeat problem no. 3 using hybrid equivalent circuit.
11. Repeat problem no. 4 using hybrid equivalent circuit.
12. Repeat problem no. 5 using hybrid equivalent circuit.
13. Repeat problem no. 6 using hybrid equivalent circuit.
14. Repeat problem no. 7 using hybrid equivalent circuit.
15. Repeat problem no. 8 using hybrid equivalent circuit

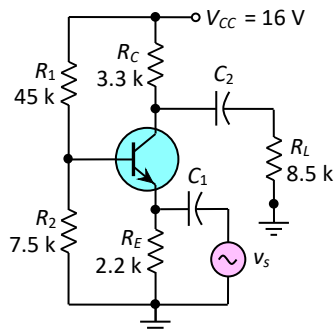


Fig.6-66: CB amplifier with voltage divider bias for Example 6-8.

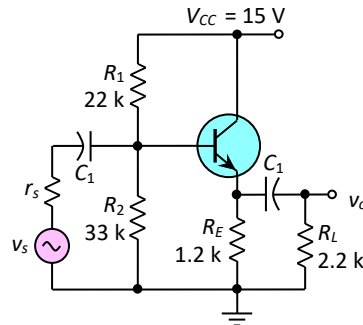


Fig.6-67: CE amplifier with voltage divider bias for Example 6-9.

16. For the CB amplifier of Fig.6-42, calculate  $A_v$ ,  $Z_i$ ,  $Z_o$ , and  $A_i$ . The transistor used in this circuit has  $h_{ib} = 20 \Omega$ ,  $h_{fb} = 0.995$ ,  $h_{rb} = 3 \times 10^{-5}$ , and  $h_{ob} = 0.5 \mu\text{S}$ . Assume all the capacitors are short for the operating frequency range of the amplifier.
17. For the CE amplifier of Fig.6-68, calculate  $A_v$ ,  $Z_i$ ,  $Z_o$ , and  $A_i$ . The transistors used in this circuit are identical and has  $h_{ic} = 1 \text{ k}\Omega$ ,  $h_{fc} = 25$ ,  $h_{rc} = 1$ , and  $h_{oc} = 10 \mu\text{S}$ . Assume all the capacitors are short for the operating frequency range of the amplifier.

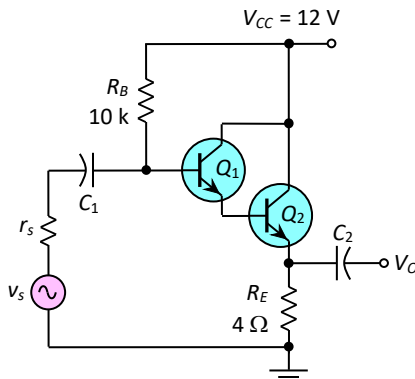


Fig.6-68: CB amplifier with voltage divider bias for Example 6-10

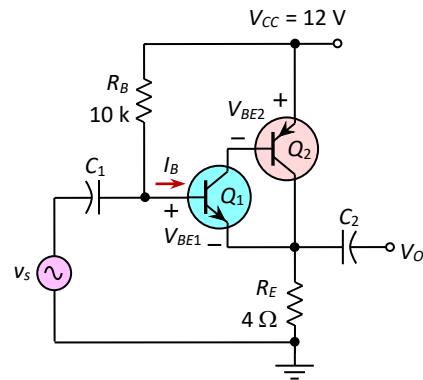


Fig.6-69: CB amplifier with voltage divider bias for Example 6-10

18. A CC amplifier is designed using a Sziklai-pair of  $\beta_1 = 100$  and  $\beta_2 = 75$ . The transistors used in this circuit are identical and has  $h_{ic} = 1.2 \text{ k}\Omega$ ,  $h_{fc} = 25$ ,  $h_{rc} = 1$ , and  $h_{oc} = 12 \mu\text{S}$ . Assume all the capacitors are short for the operating frequency range of the amplifier.
19. Draw the load-line and operating point of the CE amplifier circuits of Fig.6-969, using the slope and the y-intercept.

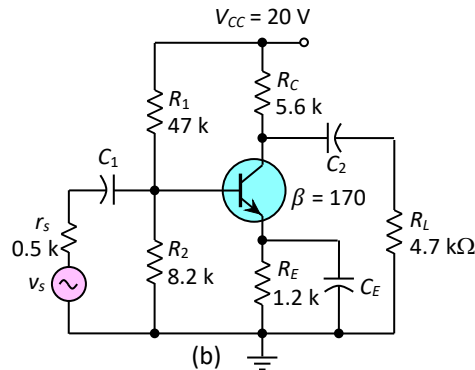
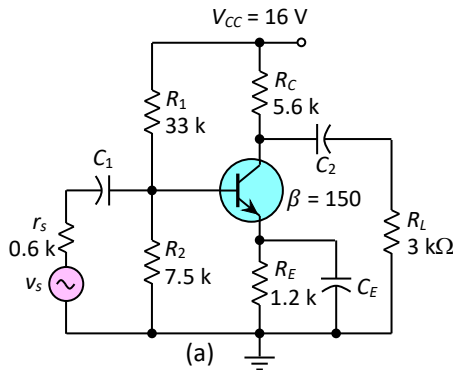


Fig.6-70: CE amplifier circuits for Exercise 14

20. Assume that a small signal BJT has  $r_{bb'} = 32 \Omega$ ,  $r_{b'e} = r_{\mu} = 600 \Omega$ ,  $r_{b'c} = 2.2 \text{ M}\Omega$ ,  $r_{ce} = 140 \text{ k}\Omega$ ,  $r_o = 150 \text{ k}\Omega$ ,  $C_{eb} = 60 \text{ pF}$ ,  $C_{ce} = 3.5 \text{ pF}$ ,  $C_{cb} = C_{\mu} = 6.5 \text{ pF}$ , and  $g_m = 80 \text{ mA/V}$ . Determine and draw the hybrid- $\pi$  equivalent circuit.